

American Board of Family Medicine



2025 IN-TRAINING EXAMINATION

CRITIQUE BOOK

This book contains the answers to each question in the In-Training Examination, as well as a rationale that provides an explanation for the correct answer. Bibliographic references are included at the end of each critique to facilitate any further study you may wish to do in a particular area.

Item 1

ANSWER: B

Quality improvement is a key area for clinical practice, but initiating these projects can be challenging. Many different methodologies have been used, including the Plan-Do-Study-Act cycle or Plan-Do-Check-Act cycle, but clear identification of the current issues is critical in any framework. In order to plan appropriate interventions, it must first be determined what barriers are responsible for hindering performance in a certain area.

If evaluation shows that there is a systemic issue in capturing data, rather than a clinician issue, no other interventions will be beneficial. When used as part of an organizational plan for staffing and clinical procedures, patient-specific reminders and use of the entire health care team are more likely to improve screening uptake than other methods, but the initial step should be delineating any current issues prior to implementing a change. Educating patients about preventive health care and screenings is essential and should be part of routine care. However, general materials such as clinic posters are less likely to increase screening rates than direct physician recommendations or conversations. Sending generalized reminders for preventive care may miss the patients in most critical need of services. General reminders that rely on clinician evaluation of every individual visit may be less effective than patient-specific reminders that can be addressed at the point of care.

References

Stone EG, Morton SC, Hulscher ME, et al. Interventions that increase use of adult immunization and cancer screening services: a meta-analysis. *Ann Intern Med*. 2002;136(9):641-651.

Alam Z, Cairns JM, Scott M, Dean JA, Janda M. Interventions to increase cervical screening uptake among immigrant women: a systematic review and meta-analysis. *PLoS One*. 2023;18(6):e0281976.

Item 2

ANSWER: A

Given the growing prevalence of diabetes mellitus, clinicians should be familiar with effective prevention strategies. Intensive lifestyle and behavior modification programs, such as the National Diabetes Prevention Program, are low-cost and can reduce the development of diabetes, with one U.S. study demonstrating a 27% reduction over 15 years. These programs help patients set weight loss goals $\geq 7\%$ of their baseline, which is thought to be a key component of diabetes prevention. This is typically achieved through a caloric reduction of 500–1000 kcal per day, with activity goals ≥ 150 minutes per week, including up to 75 minutes per week of strength training. While a variety of dietary patterns are thought to be effective for diabetes prevention, including low intake of carbohydrates, the evidence for changes in dietary patterns is not as strong as that of intensive lifestyle programs.

References

American Diabetes Association Professional Practice Committee. 3. Prevention or delay of diabetes and associated comorbidities: standards of care in diabetes—2025. *Diabetes Care*. 2025;48(Supplement_1):S50-S58.

Item 3

ANSWER: A

This patient's imaging reveals a fifth metatarsal shaft fracture, which should be managed with a boot or hard-soled shoe for 3–6 weeks with weight-bearing. Foot fractures are typically caused by an axial or twisting force on the foot or a crush injury and account for about one-third of lower extremity fractures in adults. Patients present with swelling and bony point tenderness of the affected area, as seen in this patient. The ability to bear weight varies based on the patient's pain tolerance and the extent of the fracture. Complications of foot fractures include infection, arthritis, nonunion, malunion, and compartment syndrome. A short leg cast is not appropriate because it is used to treat proximal fifth metatarsal and tarsal bone fractures. Physical therapy is not first-line management for this patient's injury. If reduction is needed or if nonunion is present after 6 months, then referral to an orthopedic surgeon would be indicated.

References

Silver S, Williams E, Plunkett ML. Common foot fractures. *Am Fam Physician*. 2024;109(2):119-129.

Item 4

ANSWER: C

This patient is in septic shock secondary to pneumonia. Surviving Sepsis Campaign guidelines recommend early goal-directed therapy (EGDT) consisting of initial fluid resuscitation with 30 mL/kg of crystalloid solution within the first 3 hours and appropriate antibiotics within 1 hour of evaluation, which have already been done for this patient. The guidelines also recommend initiating vasopressor therapy with norepinephrine as the first-line agent if the mean arterial pressure remains <65 mm Hg after fluid resuscitation. Norepinephrine is a more potent vasoconstrictor than dopamine. Additional fluid therapy is controversial and patient-specific, with some experts recommending discrete 500-mL boluses with hemodynamic monitoring of response, and others recommending balanced crystalloids such as lactate Ringer solution over normal saline. Vasopressin should be added for adults with septic shock who are receiving norepinephrine but continue to have inadequate mean arterial pressure. Vasopressor therapy should not be delayed for placement of a central line.

References

Evans L, Rhodes A, Alhazzani W, et al. Surviving Sepsis Campaign: international guidelines for management of sepsis and septic shock 2021. *Critical Care Medicine*. 2021;49(11):e1063-e1143.

Item 5

ANSWER: C

Thyroid storm is a medical emergency that must be treated promptly. The initial treatment for thyroid storm is propylthiouracil (PTU). While PTU has been shown to decrease circulating T_3 hormone levels more quickly than methimazole, they have both been shown to be equally effective clinically. Glucocorticoids can also be used to block the conversion of T_4 to T_3 . Hydrocortisone and dexamethasone, not prednisone, are the preferred glucocorticoids if used in addition to PTU. While iodine is used to block the synthesis of new thyroid hormone, it should be started at least 1 hour after the initiation of PTU. Plasmapheresis is reserved for patients who do not respond to pharmacologic treatment.

References

De Leo S, Lee SY, Braverman LE. Hyperthyroidism. *Lancet*. 2016;388(10047):906-918.

Ross DS, Burch HB, Cooper DS, et al. 2016 American Thyroid Association guidelines for diagnosis and management of hyperthyroidism and other causes of thyrotoxicosis. *Thyroid*. 2016;26(10):1343-1421.

Lee SY, Modzelewski KL, Law AC, Walkey AJ, Pearce EN, Bosch NA. Comparison of propylthiouracil vs methimazole for thyroid storm in critically ill patients. *JAMA Netw Open*. 2023;6(4):e238655.

Item 6

ANSWER: E

According to the 2022 American Gastroenterological Association guidelines for diarrhea-predominant irritable bowel syndrome (IBS-D), the nonabsorbable antibiotic rifaximin has the best evidence for controlling diarrhea, discomfort, bloating, and abdominal pain. It has a broad spectrum with activity against gram-positive, gram-negative (including *Clostridioides difficile*), and anaerobic bacteria. It is thought to work by reducing bacterial byproducts and altering the gut flora. It is prescribed as a 14-day course for acute and recurrent symptoms. Rifaximin does not have any significant adverse effects, but it is a high-cost medication.

Low-dose amitriptyline and dicyclomine are often effective in reducing symptoms, but the evidence is not as strong as that of rifaximin. Linaclotide is used for constipation-predominant IBS (IBS-C), but its chief side effect of diarrhea results in many patients discontinuing the medication. Lubiprostone is another medication used for IBS-C and for opioid-induced constipation, not IBS-D.

References

Karuppiyah S, Pomianowski K. Rifaximin (Xifaxan) for irritable bowel syndrome. *Am Fam Physician*. 2017;95(4):258-259.

Chang L, Sultan S, Lembo A, Verne GN, Smalley W, Heidelbaugh JJ. AGA clinical practice guideline on the pharmacologic management of irritable bowel syndrome with constipation. *Gastroenterology*. 2022;163(1):118-136.

Arnold MJ. Medications for irritable bowel syndrome: guidelines from the AGA. *Am Fam Physician*. 2023;108(5):527-529.

Lembo A, Sultan S, Chang L, Heidelbaugh JJ, Smalley W, Verne GN. AGA clinical practice guideline on the pharmacological management of irritable bowel syndrome with diarrhea. *Gastroenterology*. 2022;163(1):137-151.

Item 7

ANSWER: C

First-line treatments for the relief of painful diabetic neuropathy include amitriptyline, duloxetine, gabapentin, and pregabalin. Second-line treatments include topical lidocaine and capsaicin, with less evidence of benefit. Opioids such as hydrocodone should be avoided in the treatment of neuropathic pain. Studies of peripheral transcutaneous electrical nerve stimulation have shown modest benefit; however, it is not a first-line treatment.

References

Bragg S, Marrison ST, Haley S. Diabetic peripheral neuropathy: prevention and treatment. *Am Fam Physician*. 2024;109(3):226-232.

Item 8

ANSWER: B

Urinary incontinence is common among women and can substantially diminish quality of life while contributing to increased financial strain. Risk factors include obesity, parity, increasing age, and a history of hysterectomy. The initial evaluation includes a thorough history and should establish whether the incontinence is transient or chronic. It should also identify the incontinence subtype as well as any red flags that would warrant subspecialist referral, such as pelvic organ prolapse or fistula. Urinalysis should be ordered for all patients with urinary incontinence. An incontinence screening questionnaire, cough stress test (SOR C), postvoid residual measurement, and a 3-day voiding diary are helpful tools. Renal and bladder ultrasonography, urodynamic testing, and cystoscopy are not recommended for the initial evaluation of uncomplicated urinary incontinence. Treatments include weight loss, pelvic floor training, lifestyle modifications including appropriate fluid intake, smoking cessation, and pharmacologic therapy. Patients with refractory urinary incontinence can be referred for more invasive treatments, such as mechanical devices, intravesical onabotulinumtoxinA injection, and surgical procedures.

References

Hu JS, Pierre EF. Urinary incontinence in women: evaluation and management. *Am Fam Physician*. 2019;100(6):339-348.

Item 9

ANSWER: A

Immune thrombocytopenia (ITP) is caused by autoimmune destruction of platelets in the spleen and causes chronic thrombocytopenia of varying severity. Although hematologists typically oversee care for patients with ITP, family physicians may need to make acute treatment decisions in collaboration with these patients. The American Society of Hematology recommends not to treat ITP unless there is active bleeding or very low platelet counts, defined as $<30,000/\text{mm}^3$. This patient does not require treatment at this time because she does not have any current bleeding and her platelet count is above the threshold for treatment. Corticosteroids and rituximab are both used to treat ITP, along with intravenous immunoglobulin. Hydroxychloroquine sulfate is not typically recommended in the treatment of ITP.

References

Liu XG, Hou Y, Hou M. How we treat primary immune thrombocytopenia in adults. *J Hematol Oncol*. 2023;16(4).

Gauer RL, Whitaker DJ. Thrombocytopenia: evaluation and management. *Am Fam Physician*. 2022;106(3):288-298.

Item 10

ANSWER: E

The Infectious Diseases Society of America recommends performing urinalysis at least once yearly and obtaining a glomerular filtration rate at least twice yearly in stable patients on highly active antiretroviral therapy (HAART) to detect kidney function abnormalities and drug nephrotoxicity. While cardiovascular risk assessment is important in persons living with HIV, the Framingham Risk Score is not well validated in this population. The American College of Cardiology/American Heart Association recommend using traditional risk factors and clinical judgment for cardiovascular risk assessment in this population.

Neurocognitive screening is typically performed based on clinical indications rather than as an annual routine screening for all patients on HAART. Liver fibrosis screening is appropriate in patients with coinfection with hepatitis B or C, but it is not a routine annual screening. Although bone mineral density screening is important in persons living with HIV, especially those at risk for osteoporosis, it is not recommended annually for all patients on HAART.

References

Lucas GM, Ross MJ, Stock PG, et al. Clinical practice guideline for the management of chronic kidney disease in patients infected with HIV: 2014 update by the HIV Medicine Association of the Infectious Diseases Society of America. *Clin Infect Dis*. 2014;59(9):e96-e138.

Grundy SM, Stone NJ, Bailey AL, et al. 2018 AHA/ACC/AACVPR/AAPA/ABC/ACPM/ADA/AGS/APhA/ASPC/NLA/PCNA guideline on the management of blood cholesterol: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2019;73(24):e285-e350.

Chu C, Pollock LC, Selwyn PA. HIV-associated complications: a systems-based approach. *Am Fam Physician*. 2017;96(3):161-169.

Saag MS. HIV infection — screening, diagnosis, and treatment. *N Engl J Med*. 2021;384(22):2131-2143.

Item 11

ANSWER: B

For this intermediate-risk patient with no history of coronary artery disease (CAD) who presents with angina and a negative workup for acute coronary syndrome plus negative stress testing within 1 year, coronary CT angiography would be the next step to determine if he has obstructive CAD or atherosclerotic plaques. A repeat stress test would not be indicated. Fractional flow reserve with CT is useful to help guide revascularization after an abnormal coronary CT angiogram reveals vessel stenosis. Cardiac MRI would be useful to determine structural abnormalities.

References

Gulati M, Levy PD, Mukherjee D, et al. 2021 AHA/ACC/ASE/CHEST/SAEM/SCCT/SCMR guideline for the evaluation and diagnosis of chest pain: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2021;144(22):e368-e454.

Item 12

ANSWER: A

This patient meets criteria for moderate alcohol use disorder (AUD). Although AUD is a leading cause of preventable death, less than 2% of patients with AUD are prescribed medications. Acamprosate is recommended for patients who have achieved abstinence and desire to maintain it (SOR A). Gabapentin is useful for the prevention of withdrawal and is safe for those who are still drinking (SOR A). Topiramate is useful in decreasing heavy drinking (SOR B). There is insufficient evidence to recommend disulfiram.

References

Poorman E, McQuade BM, Messmer S. Medications for alcohol use disorder. *Am Fam Physician*. 2024;109(1):71-78.

Item 13

ANSWER: C

Poison ivy is perhaps the most common cause of allergic contact dermatitis in the United States, causing up to 10 million persons annually to seek medical care. Caused by the oleoresin urushiol that is present in *Toxicodendron* species (poison ivy, sumac, oak), allergic contact dermatitis is a type IV delayed hypersensitivity reaction. Symptoms develop 12–72 hours after exposure. Prior sensitization is necessary. The rash has varying presentations, including irregular erythematous patches, linear streaks of vesicles, and larger fluid-filled bullae. The blister fluid does not cause the rash to spread. This illusion of spreading is due to the time variability in when the rash develops (hours to days) based on how much resin was in contact with the skin.

Systemic corticosteroids are indicated for facial, genital, or significant vesiculobullous involvement. In type IV hypersensitivity reactions, rebound inflammation occurs if treatment with corticosteroids is insufficient in dose or duration. The most effective treatment in this patient would be a 15-day prednisone taper, starting with 1 mg/kg (60 mg at maximum). A single injection of triamcinolone has variable efficacy and often does not take effect as quickly as oral prednisone; however, if used, it should be administered intramuscularly, never subcutaneously, due to the likelihood of lipatrophy.

Topical corticosteroids, even the most potent formulations, do not work on bullous lesions and are contraindicated on the face and genitals. A common prescription for poison ivy dermatitis is a 5-day dose pack of methylprednisolone. This provides temporary improvement but is frequently followed by a rebound in rash, which indicates an insufficient dose and duration of the corticosteroid. There is no indication for draining the bullae or vesicles in this patient.

References

Lawley LP, Cheeley JT, Swerlick RA. Eczema, psoriasis, cutaneous infections, acne, and other common skin disorders. In: Loscalzo J, Kasper DL, Longo DL, Fauci AS, Hauser SL, Jameson JL, eds. *Harrison's Principles of Internal Medicine*. Volume I. 21st ed. McGraw Hill; 2022:375-376.

Dinulos JGH. Contact dermatitis and patch testing. In: Dinulos JGH, ed. *Habif's Clinical Dermatology: A Color Guide to Diagnosis and Therapy*. 7th ed. Elsevier; 2021:128-135.

Weinberg L, Malani PN. Poison ivy, poison oak, and poison sumac. *JAMA*. 2024;331(21):1872.

Item 14

ANSWER: B

Evaluation of a pulmonary nodule balances the benefits of minimal testing for benign nodules with the risks of malignant nodules. It requires knowledge of the factors associated with the probability that the nodule is malignant, the performance characteristics of additional testing, and the advantages and disadvantages of more or less intensive follow-up. The probability of malignancy and the effectiveness of diagnostic tests depend on the size and characteristics of the nodule, which can be classified into 3 categories: small solid nodules, larger solid nodules, and subsolid nodules. A solid nodule between 6 mm and 8 mm should be monitored with CT of the chest in 6–12 months. Flexibility in these time intervals (eg, 3–6 months, 6–12 months) is included in the guidelines to accommodate nodule features other than size, as well as clinician and patient preferences. A solid nodule that has remained unchanged in size on chest CT over a period of 2 years or longer is considered benign. Primary care physicians should consider referral to a pulmonologist for patients with a larger solid nodule (8–30 mm) or a subsolid nodule due to their higher risk of malignancy.

References

Mazzone PJ, Lam L. Evaluating the patient with a pulmonary nodule: a review. *JAMA*. 2022;327(3):264-273.

MacMahon H, Naidich DP, Goo JM, et al. Guidelines for management of incidental pulmonary nodules detected on CT images: from the Fleischner Society 2017. *Radiology*. 2017;284(1):228-243.

Item 15

ANSWER: A

This patient likely has an upper gastrointestinal (GI) bleed from erosive gastritis or peptic ulcer disease. This is suggested by a history of abdominal pain and nausea and the consistent use of ibuprofen with alcohol. The patient's BUN is elevated, and his hemoglobin level shows anemia. Given that the patient is hemodynamically stable and his hemoglobin level is above the transfusion threshold, he does not need excessive workup prior to treatment, but guidelines recommend initiating a proton pump inhibitor on presentation with no delay regardless of status for endoscopy.

Urgent endoscopy is only indicated in patients with hemodynamic instability and signs of upper GI bleeding. There are no randomized, controlled trials or guidelines to support the use of nasogastric lavage. Current guidelines recommend blood transfusion for patients with upper gastrointestinal bleeding when the hemoglobin level is <7 g/dL in most cases or <8 g/dL (80 g/L) in patients with coronary artery disease, recent cardiac surgery, or hematologic malignancies. Arterial embolization is only performed if endoscopy has failed to control bleeding.

References

Wilkins T, Wheeler B, Carpenter M. Upper gastrointestinal bleeding in adults: evaluation and management. *Am Fam Physician*. 2020;101(5):294-300.

Item 16

ANSWER: A

Fluoride is widely recognized for its role in preventing dental caries, and community water fluoridation remains a safe, effective, and evidence-based strategy for improving oral health. Dietary supplements are recommended for infants and children when the community water supply is nonfluoridated. It is also recommended for parents to brush the teeth of infants and toddlers once or twice daily with a rice-sized amount of fluoridated toothpaste, increasing to a pea-sized amount of toothpaste at age 3. At age 6, children are less likely to swallow toothpaste, which can lead to fluorosis (permanent brownish staining of teeth). Children may be more independent in toothbrushing starting at age 6, although some may still need assistance for adequate brushing.

Mouthwash with or without fluoride should not be used in children younger than 6 years. It can be used after age 6 if the child can reliably spit out the mouthwash rather than swallow it. Starting at the emergence of the first tooth, fluoride varnish should be applied every 3 months for children at high risk of caries and every 6 months for all other children.

References

Clark MB, Keels MA, Slayton RL; Section on Oral Health. Fluoride use in caries prevention in the primary care setting. *Pediatrics*. 2020;146(6):e2020034637.

Item 17

ANSWER: B

This patient likely presents with Parkinson disease based on the hallmark symptoms and physical examination findings. First-line medications for Parkinson disease are carbidopa/levodopa, monoamine oxidase-B inhibitors, and nonergot dopamine agonists.

Amitriptyline is a tricyclic antidepressant that is commonly used for treatment of depression and neuropathic pain. It is on the Beers Criteria list and must be used with great caution in older patients. Fluoxetine is typically prescribed for major depressive disorder, which is likely not the primary diagnosis in this patient. Furthermore, the Beers Criteria list cautions against prescribing fluoxetine in older patients. Gabapentin is typically used for partial seizures and postherpetic neuralgia. Propranolol hydrochloride is effective for management of essential tremor, but this patient's clinical picture is more indicative of Parkinson disease.

References

Reich SG, Savitt JM. Parkinson's disease. *Med Clin North Am*. 2019;103(2):337-350.

Halli-Tierney AD, Luker J, Carroll DG. Parkinson disease. *Am Fam Physician*. 2020;102(11):679-691.

Armstrong MJ, Okun MS. Diagnosis and treatment of Parkinson disease: a review. *JAMA*. 2020;323(6):548-560.

Item 18**ANSWER: D**

Pituitary adenomas are classified according to their primary cell origin and the specific hormones they produce. Somatotroph adenomas secrete growth hormone and cause symptoms consistent with acromegaly, including an increase in hand and foot size, enlargement of the mandible, carpal tunnel symptoms, fatigue, proximal muscle weakness, and menstrual irregularities. Somatotroph adenomas are diagnosed by a high serum insulin-like growth factor 1 level. Corticotroph adenomas secrete corticotropin (ACTH) and cause symptoms consistent with Cushing disease such as mood changes, weight gain, proximal muscle weakness, easy bruising, and menstrual irregularity. Corticotroph adenomas can be diagnosed by an abnormal result on a dexamethasone suppression test. Lactotroph adenomas secrete prolactin and cause symptoms consistent with hyperprolactinemia such as galactorrhea, decreased libido, and oligomenorrhea or amenorrhea. Lactotroph adenomas can be diagnosed with an elevated serum prolactin level. Thyrotroph adenomas secrete TSH and cause symptoms consistent with hyperthyroidism such as weight loss, palpitations, heat intolerance, and anxiety. Thyrotroph adenomas can be diagnosed with elevated TSH, free T₄, and total T₄ levels.

References

Tritos NA, Miller KK. Diagnosis and management of pituitary adenomas: a review. *JAMA*. 2023;329(16):1386.

Item 19

ANSWER: A

Oral or transdermal estrogen is the most effective treatment for vasomotor symptoms in menopausal women (SOR A). Antidepressants, including SSRIs and SNRIs, are effective alternatives to hormone therapy for the treatment of vasomotor symptoms in menopause (SOR A). The SSRI paroxetine is FDA approved for this indication. Testosterone therapy is not effective for vasomotor symptoms. Vaginal estrogen cream is recommended for women whose primary symptoms are vaginal rather than vasomotor. Phytoestrogen supplements such as soy have not been shown to consistently improve vasomotor symptoms associated with menopause (SOR B).

References

American College of Obstetricians and Gynecologists' Committee on Practice Bulletins—Gynecology. ACOG Practice Bulletin No. 141: management of menopausal symptoms. *Obstet Gynecol* 2014;123(1):202-216.

Pinkerton JV, Sánchez Aguirre F, Blake J, et al; North American Menopause Society 2017 Hormone Therapy Position Statement Advisory Panel. The 2017 hormone therapy position statement of the North American Menopause Society. *Menopause*. 2017;24(7):728-753.

Item 20

ANSWER: B

When a patient has either an advance directive or demonstrates decision-making capacity, their autonomous refusal of treatment must be respected, even if it leads to adverse medical outcomes or conflicts with the physician's values or ethical beliefs. Blood refusal directives are considered legally binding and physicians have been found liable for violating religious freedom when violating these directives. Transfusion of blood products may be permissible for patients who are Jehovah's Witnesses if they are incapacitated and require transfusion but do not have a blood refusal directive or surrogate decision-maker.

References

Wiser Z, Gauer R. Approaches to a patient with anemia who is refusing a blood transfusion. *Am Fam Physician*. 2024;109(1):online.

Item 21**ANSWER: A**

The initial treatment of a hemodynamically stable patient with epistaxis is to pinch the lower third of the nose for 15–20 minutes. If digital compression is not successful in stopping the bleeding and an anterior bleeding site is visualized on anterior rhinoscopy, then the next step would be to either apply topical vasoconstrictors or use chemical/electrical cautery. If bleeding continues after local measures fail, then anterior nasal packing would be indicated. Acute reversal of anticoagulation should be considered in a patient with ongoing bleeding who is hemodynamically unstable.

References

Seikaly H. Epistaxis. *N Engl J Med*. 2021;384(10):944-951.

Item 22

ANSWER: D

β -Blockers, calcium channel blockers, and long-acting nitrates are first-line medications to decrease symptoms in patients with angina. If symptoms persist despite the use of a maximally tolerated dose of a single first-line medication, an additional drug from a different class should be introduced. Adding isosorbide mononitrate is the most appropriate option for this patient; however, it may decrease his blood pressure and will require close attention on follow-up. Ranolazine is another potential recommendation but is often more costly than alternatives. It is recommended in patients who are already taking medications from 2 other classes. Increasing the patient's carvedilol is not recommended due to his relative bradycardia. Neither increasing empagliflozin nor adding clopidogrel helps to decrease anginal symptoms. Ivabradine is contraindicated in patients with chronic coronary disease and normal ejection fraction and is potentially harmful.

References

Virani SS, Newby LK, Arnold SV, et al. 2023 AHA/ACC/ACCP/ASPC/NLA/PCNA guideline for the management of patients with chronic coronary disease: a report of the American Heart Association/American College of Cardiology Joint Committee on clinical practice guidelines. *J Am Coll Cardiol*. 2023;82(9):833-955.

Item 23

ANSWER: C

Acute insomnia is a sleep continuity disturbance, defined as difficulty initiating and/or maintaining sleep, that occurs ≥ 3 days per week with a duration of 1 week to 3 months. It is often associated with a precipitating factor such as life events or psychological distress. The first step in the management of acute insomnia is identifying and treating the underlying cause. Cognitive behavioral therapy for insomnia (CBT-I) is an effective treatment for insomnia and may be offered as first-line therapy, especially in patients who wish to avoid pharmacotherapy. CBT-I has 3 categories: sleep hygiene education, sleep restriction, and stimulus control. Waking up at the same time every day and not napping during the day are among the 8 components of sleep hygiene education. Stimulus control strategies include going to bed only when sleepy and reserving the bedroom for sleep and sex. Sleep restriction techniques involve reducing time spent in bed followed by gradually increasing it by 15 minutes each week once sleep efficiency reaches 90%.

References

Vargas I, Nguyen AM, Muench A, Bastien CH, Ellis JG, Perlis ML. Acute and chronic insomnia: what has time and/or hyperarousal got to do with it? *Brain Sci.* 2020;10(2):71.

Item 24

ANSWER: D

Hypothyroidism affects approximately 1 in 300 individuals in the United States, with a higher prevalence among women and older adults. Symptoms can range from mild to life-threatening. Because the signs and symptoms of hypothyroidism can be nonspecific, diagnosis is based on blood levels of TSH and free T₄. Subclinical hypothyroidism is defined by a TSH level that is abnormal but <10 µU/mL while the free T₄ level remains within the normal range. Nonpregnant patients with subclinical hypothyroidism should not be treated with thyroid hormone replacement therapy unless the thyroid peroxidase antibody level is elevated or the TSH level is >10 µU/mL (SOR A).

Evaluating the total or free T₃ level is not routinely indicated. Treatment-resistant depression is an indication for treating subclinical hypothyroidism; however, this is not the case for treatment-resistant constipation.

References

Wilson SA, Stem LA, Bruehlman RD. Hypothyroidism: diagnosis and treatment. *Am Fam Physician*. 2021;103(10):605-613.

Item 25**ANSWER: C**

Abnormal uterine bleeding is a common clinical entity with a broad differential diagnosis and can be grouped into structural and nonstructural causes. Clinicians should determine if the bleeding is severe or life-threatening and appropriately ensure stability with resuscitative measures. While this patient's bleeding is severe and she has symptoms of acute blood loss anemia, her vital signs indicate hemodynamic stability. The most appropriate next step is to administer intravenous tranexamic acid. Given this patient's recent cancer history, it is advisable to avoid therapies with estrogen or progesterone such as a high-dose combined oral contraceptive or depot medroxyprogesterone acetate if clinically appropriate. Intravenous estrogen should be reserved for patients who are hemodynamically unstable given the risk for clotting. While the levonorgestrel IUD is an effective long-term management for abnormal uterine bleeding, insertion in the emergency department is impractical, and it would not be anticipated to improve bleeding for 3–6 months.

References

Wouk N, Helton M. Abnormal uterine bleeding in premenopausal women. *Am Fam Physician*. 2019;99(7):435-443.

Item 26

ANSWER: C

Topical metronidazole is a first-line treatment for rosacea. The cost and side-effect profile of topical metronidazole make it most appropriate. Topical azelaic acid is also a first-line treatment for rosacea.

Oral doxycycline is a reasonable next step if topical metronidazole is not successful in managing symptoms. The side-effect profile of doxycycline makes it less desirable than topical metronidazole. Additionally, increased sun sensitivity is a common side effect of doxycycline that is problematic for management because rosacea is generally clinically worse in the warmer months.

Laser and light-based therapies are second-line treatments for rosacea. They are reasonable options for patients who have not had success with topical metronidazole or azelaic acid. Brimonidine is an α -2 agonist that is another reasonable option in a patient who has not had success with topical metronidazole or topical azelaic acid. However, due to its cost and tendency to exacerbate skin irritation, it is considered a second-line treatment option. Topical cyclosporine is another second-line treatment that has been recalcitrant to first-line options.

References

Frazier W, Zemtsov RK, Ge Y. Rosacea: common questions and answers. *Am Fam Physician*. 2024;109(6):533-542.

Searle T, Al-Niaimi F, Ali FR. Rosacea. *Br J Hosp Med*. 2021;82(2):1-8.

Item 27**ANSWER: E**

The U.S. Preventive Services Task Force recommends against cervical cancer screening in women older than 65 years only if they have had adequate prior screening. Adequate screening is defined as 3 consecutive negative cytology results or 2 consecutive negative co-testing results within 10 years before stopping screening, with the most recent test occurring within 5 years. Because this patient has not had adequate prior screening, she requires either 2 consecutive negative co-testing results that are 5 years apart or 3 consecutive negative cytology results that are 3 years apart before stopping screening.

References

Final recommendation statement: cervical cancer: screening. US Preventive Services Task Force. Updated August 21, 2018.

Item 28

ANSWER: A

Suicidal ideation is much more common than suicide attempts or suicide completions and should be carefully addressed; up to half of patients characterized as low risk ultimately complete suicide. Current recommendations for evaluating suicidal ideation suggest focusing on individual assessment and understanding certain risk factors that may exacerbate suicidal ideation for a given patient. This includes asking questions, such as whether the patient has a plan and means to complete the plan, as well as assessing for underlying mental illness, substance use, and life stressors.

SSRIs and cognitive behavioral therapy are first-line treatments for the majority of people with depression and suicidality, but this patient requires further evaluation for suicide risk before initiating pharmacologic therapy. Lithium has been shown to decrease suicide attempts and all-cause mortality among patients with major depression and bipolar disorder, and SSRIs are often used in conjunction with lithium; however, an underlying diagnosis has not yet been established for this patient, so this treatment is not yet warranted. The use of ketamine for rapid improvement in depression has shown promise, with patients experiencing significant improvement in depression symptoms and suicidality within 24 hours, but the evidence quality is considered low, and effects dissipated after 2 weeks. Ketamine treatment is recommended only if depression is resistant to other medications. If the patient endorses suicide plans or any other factors that are consistent with high risk for suicide completion, transfer to an inpatient treatment facility may be indicated. Although, stabilization in a standard emergency department is often necessary while awaiting an inpatient bed due to facility limitations.

References

Norris DR, Clark MS. The suicidal patient: evaluation and management. *Am Fam Physician*. 2021;103(7):417-421.

Buelt A. Management of major depression: guidelines from the VA/DoD. *Am Fam Physician*. 2023;107(3):323-325.

Marzani G, Price Neff A. Bipolar disorders: evaluation and treatment. . 2021;103(4);227-239.

Gaddey HL, Mason B, Naik A. Depression: managing resistance and partial response to treatment. *Am Fam Physician*. 2024;109(5):410-416.

Item 29

ANSWER: E

For average-risk patients with 1–2 tubular adenomas <10 mm in size found on screening colonoscopy, repeat colonoscopy in 7–10 years is recommended. The risk of small adenomas progressing to malignancy is quite low and the rate of progression is slow, likely >10 years. Using GRADE (Grading of Recommendations Assessment, Development, and Evaluation) criteria in its consensus guidelines from 2020, the U.S. Multi-Society Task Force on Colorectal Cancer recommends follow-up in 7–10 years for 1–2 tubular adenomas <10 mm in size (moderate-quality evidence), 3–5 years for 3–4 tubular adenomas <10 mm in size (very low-quality evidence), 3 years for 5–10 tubular adenomas <10 mm in size (moderate-quality evidence), and yearly for >10 adenomas (very low-quality evidence).

In the United States, only colonoscopy is recommended for follow-up of findings of colonic polyps. International guidelines allow for stool-based testing.

This patient is healthy, and colon cancer screening should continue until at least age 75 (SOR A).

References

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Dornblaser D, Young S, Shaukat A. Colon polyps: updates in classification and management. *Curr Opin Gastroenterol*. 2024;40(1):14-20.

Gupta S, Lieberman D, Anderson JC, et al. Recommendations for follow-up after colonoscopy and polypectomy: a consensus update by the US Multi-Society Task Force on Colorectal Cancer. *Gastroenterology*. 2020;158(4):1131-1153.e5.

Ford B. Colonoscopy follow-up: U.S. Multi-Society Task Force on Colorectal Cancer Updates recommendations. *Am Fam Physician*. 2021;103(5):314-316.

Item 30

ANSWER: D

The 2018 ARRIVE (A Randomized Trial of Induction Versus Expectant Management) trial was a randomized, controlled trial that included more than 6000 low-risk nulliparous patients at 41 sites across the United States who were assigned to either induction of labor (IOL) between 39 weeks 0/7 days and 39 weeks 4/7 days or expectant management. The results showed no significant differences in a neonatal composite outcome, including perinatal mortality and severe perinatal morbidity between the intervention and control groups (4.3% compared to 5.4%, relative risk [RR] 0.80, 95% CI 0.64 to 1.00). A significantly lower rate of cesarean delivery was found in the intervention group than the control group (18.6% compared to 22.2%, RR 0.84, 95% CI 0.76 to 0.93).

Other findings included: lower rates of gestational hypertension and preeclampsia (9.1% compared to 14.1%, RR 0.64, 95% CI 0.56 to 0.74), increased length of stay in the labor and delivery unit (a median duration of 20 hours with an interquartile range [IQR] of 13 to 28 compared to a median duration of 14 hours with an IQR of 9 to 20), and reduced need for neonatal respiratory support within the first 72 hours of life (3.0% compared to 4.2%, RR 0.71, 95% CI 0.55 to 0.93). The intervention group had significantly lower median birth weights than the control group (3300 g compared to 3380 g; P value <0.001); it did not reduce the risk of low infant birth weight. There were no significant differences in neonatal ICU admission and maternal risk of postpartum hemorrhage.

This information should be shared with patients as part of a shared decision-making conversation regarding timing of delivery. Not all labor and delivery units can accommodate elective IOL at 39 weeks' gestation for all patients. It is important that clinicians and hospital systems assess their policies and procedures on a regular basis to ensure equitable access to IOL for those who choose this approach.

References

Wheeler V, Hoffman A, Bybel M. Cervical ripening and induction of labor. *Am Fam Physician*. 2022;105(2):177-186.

Grobman WA, Rice MM, Reddy UM, et al. Labor induction versus expectant management in low-risk nulliparous women. *N Engl J Med*. 2018;379(6):513-523.

American College of Obstetricians and Gynecologists; Torbenson VE, Caughey AB, Hoang M, Saposnik SR. Management of full-term nulliparous individuals without a medical indication for delivery: ACOG clinical practice update. *Obstet Gynecol*. 2025;145(1):e45-e50.

Item 31

ANSWER: D

This patient presents with hip osteoarthritis. Intra-articular corticosteroid injections have been shown to relieve hip and osteoarthritis pain for at least 4 months. In addition to pain relief, these injections may improve activity levels and quality of life. In contrast to intra-articular corticosteroid injections, intra-articular hyaluronic acid injections have not been shown to be substantially beneficial in relieving symptoms of hip osteoarthritis. Topical lidocaine patches have not been shown to be effective for management of pain due to hip osteoarthritis and are not a sustainable solution for this patient. Opioids such as hydrocodone are advised against for treatment of chronic osteoarthritis. They are not effective over the long term, have significant side effects, and carry a high risk of addiction and misuse. Tramadol is not effective for pain relief or functional improvement in hip osteoarthritis and carries similar risks as opioids.

References

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Nandi S, Hannon CP, Fillingham Y. 2023 American Academy of Orthopaedic Surgeons management of osteoarthritis of the hip evidence-based clinical practice guideline: case studies. *J Am Acad Orthop Surg*. 2025;33(4):e220-e223.

Alkaramany E, Murshid A, Alhaneedi GA. Appropriateness of hip osteoarthritis management in clinical practice compared with the American Academy of Orthopaedic Surgeons (AAOS) criteria. *Int Orthop*. 2021;45(11):2805-2810.

Item 32

ANSWER: A

This patient has transient global amnesia (TGA), which is characterized by the abrupt onset of anterograde amnesia without other neurologic deficits or cognitive impairment. TGA can be triggered by intense physical or psychological stress such as sexual intercourse or acute illness. Symptoms usually resolve within approximately 6 hours. No intervention is indicated for TGA other than reassurance. The etiology of TGA is unrelated to seizure, TIA, or stroke. Anticoagulants, antiplatelet medications, antiseizure medications, and thrombolytic therapy are not indicated in the treatment of TGA.

References

Sealy D, Tiller RJ, Johnson K. Transient global amnesia. *Am Fam Physician*. 2022;105(1):50-54.

Ropper AH. Transient global amnesia. *N Engl J Med*. 2023;388(7):635-640.

Item 33

ANSWER: A

In the United States, multiple diseases are carried by ticks including Lyme disease, Rocky Mountain spotted fever, ehrlichiosis, anaplasmosis, and Colorado tick fever. Tick-borne illnesses should be considered in patients who present with influenza-like symptoms during the spring and summer months and who have spent time outdoors.

Lyme disease can be diagnosed clinically in patients with erythema migrans (SOR C). In patients such as this who present without a rash, a 2-tiered testing regimen is recommended. Doxycycline is the preferred first-line treatment, but for patients who are pregnant or do not tolerate it, an appropriate alternative is amoxicillin, 500 mg 3 times daily for 14–21 days. It is important to prescribe a medication that the patient can tolerate for its full course, as partial treatment of the infection could lead to more serious complications. Cefuroxime axetil is another treatment option.

Intravenous ceftriaxone is indicated for patients with Lyme disease who present with late neurologic manifestations, but that is not the case for this patient. Chloramphenicol is the preferred treatment for Rocky Mountain spotted fever in patients with contraindications to doxycycline therapy.

References

Pace EJ, O'Reilly M. Tickborne diseases: diagnosis and management. *Am Fam Physician*. 2020;101(9):530-540.

Item 34**ANSWER: A**

The recombinant zoster vaccine (RZV) is recommended for healthy adults ≥ 50 years of age, and this patient can receive it today. Because most studies of RZV effectiveness have excluded patients with a prior episode of shingles, direct evidence in this subgroup is limited; however, available data and expert reviews support vaccination after a prior episode because the risk for recurrence remains and RZV is safe and immunogenic. The CDC recommends that patients wait until the zoster rash has resolved prior to receiving RZV, but it is not necessary to wait until postherpetic neuralgia has resolved. Patients who have previously received live zoster vaccine should receive RZV as well.

References

Shingles vaccine recommendations. Centers for Disease Control and Prevention. Updated October 22, 2024.

Item 35

ANSWER: E

This patient's unilateral orbital swelling, erythema, pain with extraocular movements, and decreased visual acuity are concerning for orbital cellulitis. Prompt recognition and early treatment are vital to avoid worsening outcomes associated with this ocular emergency. A patient who presents in the outpatient setting with clinical findings that are suspicious for orbital cellulitis should be promptly directed to the emergency department (ED) for expedited evaluation and anticipated hospitalization for urgent ophthalmology consultation and broad-spectrum intravenous antibiotics, if the diagnosis is confirmed. CT with intravenous contrast is the imaging modality of choice, although MRI is an alternative. Ordering noncontrast CT would potentially yield a less accurate diagnosis and could delay definitive treatment. Same-day outpatient evaluation by an ophthalmologist would cause a diagnostic delay, compared to transfer to the ED. Symptomatic treatment with warm compresses and close outpatient follow-up would cause a treatment delay with a risk of further clinical deterioration. Empiric treatment with an oral antibiotic would not provide adequate initial treatment of the infection compared to hospitalization for intravenous antibiotic treatment and closer clinical monitoring.

References

Pflipsen M, Massaquoi M, Wolf S. Evaluation of the painful eye. *Am Fam Physician*. 2016;93(12):991-998.

Pelletier J, Koyfman A, Long B. High risk and low prevalence diseases: orbital cellulitis. *Am J Emerg Med*. 2023;68:1-9.

Item 36

ANSWER: D

This patient has an acute deep vein thrombosis (DVT) with no evidence of pulmonary embolism. Outpatient treatment would be appropriate and a low-molecular-weight heparin such as enoxaparin is a reasonable option in this patient. Renal dosing is the reason for daily, rather than twice-daily, dosing for enoxaparin. Dabigatran is not preferred due to a higher risk for bleeding. Direct oral anticoagulants such as rivaroxaban are not appropriate in this patient due to his glomerular filtration rate $<30 \text{ mL/min/1.73 m}^2$. Furthermore, the initial dose of rivaroxaban, 15 mg twice daily, should be 21 days, not 14 days. Warfarin is not recommended for the acute phase treatment of DVT in this patient. Admission for intravenous heparin is generally not indicated for uncomplicated DVT.

References

Stevens SM, Woller SC, Baumann Kreuziger L, et al. Antithrombotic therapy for VTE disease: compendium and review of CHEST guidelines 2012-2021. *Chest*. 2024;166(2):388-404.

Item 37

ANSWER: C

This patient has chronic bacterial prostatitis (CBP) and should be treated with a fluoroquinolone. Fluoroquinolones are the first-line treatment for CBP because they favorably cover the common uropathogens that cause CBP, penetrate prostatic tissue well, and have high oral bioavailability. Amoxicillin and ceftriaxone are not preferred for the treatment of prostatitis. CT is not necessary to make the diagnosis. Referral to a urologist is unnecessary in this clinical scenario because the diagnosis and management of CBP is within the scope of family medicine.

References

Lam JC, Stokes W. Acute and chronic prostatitis. *Am Fam Physician*. 2024;110(1):45-51.

Dickson G. Prostatitis--diagnosis and treatment. *Aust Fam Physician*. 2013;42(4):216-219.

Holt JD, Garrett WA, McCurry TK, Teichman JM. Common questions about chronic prostatitis. *Am Fam Physician*. 2016;93(4):290-296.

Item 38

ANSWER: B

For patients who are stable with a small (<15% of the hemithoracic cavity) primary pneumothorax, conservative management with initiation of oxygen via nasal cannula and observation for at least 6 hours is recommended, followed by a repeat chest x-ray to assess for stability in size of the pneumothorax. If stable at 6 hours, patients can return home with a repeat follow-up in 24 hours. CT imaging can aid in diagnosis if the initial chest x-ray is nondiagnostic or equivocal; however, it is not needed to confirm definitive findings of pneumothorax on chest x-ray. Patients who are clinically unstable with a large pneumothorax or severe symptoms should receive emergent needle decompression, followed by an ipsilateral tube thoracostomy.

References

Kosowsky JM, Harbison Kimberly H. Pleural disease. In: Walls RM, Hockberger RS, Gausche-Hill M, et al, eds. *Rosen's Emergency Medicine: Concepts and Clinical Practice*. Vol 1. 9th ed. Elsevier; 2018:881-885.

Brown SGA, Ball EL, Perrin K, et al. Conservative versus interventional treatment for spontaneous pneumothorax. *N Engl J Med*. 2020;382(5):405-415.

Item 39

ANSWER: B

A sports preparticipation evaluation is important for assessing health risks and ensuring an athlete is not unnecessarily restricted from play. An athlete with an acutely enlarged spleen or liver faces an increased risk of organ rupture during contact sports. This patient should be evaluated for a cause of the organomegaly, if present, prior to participation.

Obesity in an athlete increases the risk for heat-related illness, but it is not an indication to restrict play. Exercise is beneficial for adolescents with diabetes mellitus and may require modification of insulin and eating regimens. Asymptomatic mild mitral valve prolapse is not a reason to deny participation. Tibial tubercle apophysitis is not a contraindication to play, and most experts encourage activity and conservative management.

References

Rice SG; American Academy of Pediatrics Council on Sports Medicine and Fitness. Medical conditions affecting sports participation. *Pediatrics*. 2008;121(4):841-848.

Bernhardt DT, Roberts WO, eds; American Academy of Family Physicians. *PPE: Preparticipation Physical Evaluation*. 5th ed. American Academy of Pediatrics; 2019.

Item 40

ANSWER: C

Meniere disease is characterized by spontaneous recurring episodes of vertigo with associated unilateral hearing loss, tinnitus, and nystagmus. Even though there are no specific diagnostic tests, audiometry should be performed in patients suspected of having this condition. Sensorineural hearing loss may or may not be present on audiometric evaluation.

Acute attacks should be treated with medication such as promethazine, diazepam, or prochlorperazine to suppress vestibular function. Lifestyle interventions, including salt restriction and reduction in alcohol and caffeine intake, have traditionally been recommended. However, there is no sufficient evidence to support these recommendations. The Epley maneuver is a repositioning treatment used for benign paroxysmal positional vertigo and not Meniere disease. There is no good-quality evidence that diuretics, antivirals, or corticosteroids treat or prevent Meniere disease.

References

Basura GJ, Adams ME, Monfared A, et al. Clinical practice guideline: Meniere's Disease Executive Summary. *Otolaryngol Head Neck Surg*. 2020;162(4):415-434.

Webster KE, Galbraith K, Harrington-Benton NA, et al. Systemic pharmacological interventions for Meniere's disease. *Cochrane Database Syst Rev*. 2023;2(2):CD015171.

Rogers TS, Noel MA, Garcia B. Dizziness: evaluation and management. *Am Fam Physician*. 2023;107(5):514-523.

Webster KE, George B, Lee A, et al. Lifestyle and dietary interventions for Meniere's disease. *Cochrane Database Syst Rev*. 2023;(2):CD015244.

Item 41

ANSWER: B

The number needed to treat (NNT) is the number of patients who need to receive a particular treatment in order for 1 patient to benefit. The NNT is the inverse of the absolute risk reduction (ARR), the amount by which a particular treatment reduces the risk of a negative outcome. The ARR must be calculated first in order to determine the NNT.

To calculate the ARR, the experimental event rate (EER) is subtracted from the control event rate (CER). In this example, the low-intensity treatment group had a risk of acute myocardial infarction (bad outcome) of 0.085 (CER) whereas the high-intensity treatment group had a risk of 0.045 (EER). This would make the equation: $0.085 - 0.045 =$ an ARR of 0.04. The NNT is then calculated by dividing 1 by the ARR: $1 / 0.04 = 25$. This means that 25 patients would need to receive the high-intensity treatment to prevent 1 additional myocardial infarction.

References

Centre for Evidence-Based Medicine. Number needed to treat (NNT). University of Oxford Nuffield Department of Primary Care Health Sciences.

Item 42

ANSWER: D

Midodrine has been found to be the safest and best-tolerated medication for the management of neurogenic orthostatic hypotension (SOR B). Atomoxetine is an adjunctive treatment for neurogenic orthostatic hypotension that is generally successful when added to medication such as midodrine, but it is not effective as monotherapy. Although fludrocortisone is sometimes used off-label for this indication, it should be used with caution because of the risk of left ventricular hypertrophy, renal failure, and hypokalemia. Methylphenidate has not been shown to have a role in the successful treatment of neurogenic orthostatic hypotension. Treatment with sodium tablets is usually inadequate, does not correct the neurogenic issue, and substantially increases the risk of hyponatremia and resultant renal injury.

References

Kim MJ, Farrell J. Orthostatic hypotension: a practical approach. *Am Fam Physician*. 2022;105(1):39-49.

Palma JA, Kaufmann H. Management of orthostatic hypotension. *Continuum*. 2020;26(1):154-177.

Item 43

ANSWER: C

The mainstay of treatment for diabetic ketoacidosis is intravenous fluid replacement and insulin therapy along with careful monitoring and frequent reassessment. Once plasma glucose levels reach 201–250 mg/dL, glucose should be added to the saline infusion to maintain plasma glucose levels in the 150–200 mg/dL range. Because acidosis and ketosis resolve more slowly than hyperglycemia, glucose needs to be added while insulin and saline therapy are continued.

References

Powers AC, Fowler MJ, Rickels MR. Diabetes mellitus: management and therapies. In: Loscalzo J, Fauci AS, Kasper DL, Hauser SL, Longo DL, Jameson JL. *Harrison's Principles of Internal Medicine*. 21st ed. Vol 2. McGraw-Hill; 2022:3104-3120.

Veauthier B, Levy-Grau B. Diabetic ketoacidosis: evaluation and treatment. *Am Fam Physician*. 2024;110(5):476-486.

Item 44**ANSWER: C**

Prazosin is an α_2 -agonist that decreases sympathetic nervous tone during sleep and has been found to be helpful for midsleep awakening due to hyperarousal and nightmares in patients with posttraumatic stress disorder. Amitriptyline, mirtazapine, and risperidone have significant side-effect profiles and would not be appropriate initial treatments for this patient's sleep symptoms. Temazepam is a benzodiazepine and should be avoided if possible due to its potential for misuse and chronic long-term effects.

References

Sartor Z, Kelley L, Laschober R. Posttraumatic stress disorder: evaluation and treatment. *Am Fam Physician*. 2023;107(3):273-281.

Item 45

ANSWER: E

This patient presents with a symmetrical peripheral neuropathy with sensory and motor components that are characteristic of beriberi caused by vitamin B₁ deficiency. Risk factors include alcohol use, bariatric surgery, older age, diabetes mellitus, and HIV infection. Severe thiamine deficiency can lead to heart failure. Blood and urine tests for thiamine levels are recommended to confirm the diagnosis.

Diabetic peripheral neuropathy usually presents as a symmetric polyneuropathy, but it does not usually involve motor components, except in severe cases. Neurologic manifestations of Lyme disease include meningitis, facial nerve palsy, radiculoneuritis with radicular pain in a dermatomal distribution with accompanied weakness, sensory deficits, abnormal reflexes, other cranial neuropathies, and Lyme encephalomyelitis. Peripheral neuropathy is rare with Lyme disease and typically presents as a mononeuropathy, not a polyneuropathy.

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) typically occurs after an acute infection, presenting with symptoms such as severe fatigue, sleep changes, and cognitive changes that are exacerbated by physical activity. ME/CFS is not associated with peripheral neuropathy. Secondary syphilis typically presents with a rash and constitutional symptoms. Neurologic symptoms associated with secondary syphilis include ocular symptoms and otosyphilis with vertigo, tinnitus, and hearing loss. Secondary syphilis is not associated with peripheral neuropathy.

References

Smith TJ, Johnson CR, Koshy R, et al. Thiamine deficiency disorders: a clinical perspective. *Ann N Y Acad Sci.* 2021;1498(1):9-28.

Wilson RB. Pathophysiology, prevention, and treatment of beriberi after gastric surgery. *Nutr Rev.* 2020;78(12):1015-1029.

Williams AL, Bevan J, Arnold MJ. Lyme disease: updated recommendations from the IDSA, AAN, and ACR. *Am Fam Physician.* 2021;104(6):652-654.

Latimer KM, Gunther A, Kopec M. Fatigue in adults: evaluation and management. *Am Fam Physician.* 2023;108(1):58-69.

Bragg S, Marrison ST, Haley S. Diabetic peripheral neuropathy: prevention and treatment. *Am Fam Physician*. 2024;109(3):226-232.

Ricco J, Westby A. Syphilis: far from ancient history. *Am Fam Physician*. 2020;102(2):91-98.

Item 46**ANSWER: C**

According to the Ottawa knee rules, a well-validated tool used to limit unnecessary radiography, a knee x-ray is indicated in this patient to rule out fracture because she is ≥ 55 years of age. Radiography is currently the preferred modality over ultrasonography for diagnosing a fracture in an adult. Because ligament and meniscal testing is negative, MRI is not indicated. Arthrocentesis with culture and Gram stain would be indicated if the history and physical examination suggest an infection of the joint space.

References

Taljanovic MS, Chang EY, Ha AS, et al. ACR appropriateness criteria acute trauma to the knee. *Journal of the American College of Radiology*. 2020;17(5):S12-S25.

Sims JI, Chau MT, Davies JR. Diagnostic accuracy of the Ottawa Knee Rule in adult acute knee injuries: a systematic review and meta-analysis. *Eur Radiol*. 2020;30(8):4438-4446.

Item 47**ANSWER: D**

Doxycycline is the recommended treatment for all individuals with presumed Rocky Mountain spotted fever (RMSF). RMSF can be fatal, so testing for the condition is not recommended because it would delay treatment. Short courses (5–10 days) of doxycycline have been shown to be safe in children and pregnant persons. Inpatient treatment with rapid desensitization is recommended for individuals with a true allergy to doxycycline. Chloramphenicol is another treatment option for RMSF, but it is associated with a higher risk for death than doxycycline. There is no role for amoxicillin or erythromycin in the treatment of RMSF.

References

Clinical care of Rocky Mountain spotted fever. Centers for Disease Control and Prevention. Reviewed March 6, 2025.

Item 48

ANSWER: D

Primary biliary cholangitis is characterized by chronic cholestasis, positive antimitochondrial antibodies (AMA), and elevated alkaline phosphatase (ALP). The American Association for the Study of Liver Diseases (AASLD) guidelines state that the diagnosis of primary biliary cholangitis can be established with the presence of at least 2 of the following: biochemical evidence of cholestasis (elevated ALP), presence of AMA, and histopathologic evidence of nonsuppurative destructive cholangitis. Elevated AMA is particularly helpful as it is found in >95% of cases of primary biliary cholangitis.

Autoimmune hepatitis usually presents with elevated aminotransferases and positive autoantibodies such as antinuclear antibodies or anti-smooth muscle antibodies, rather than AMA. It is not primarily a cholestatic disease and elevated ALP is typically not the predominant finding. Depending on the instigating agent, drug-induced liver injury can present with a variety of liver enzyme abnormalities; however, the presence of positive AMA and the chronicity of symptoms make it less likely in this patient scenario. Metabolic dysfunction–associated steatotic liver disease (MASLD) (formerly known as nonalcoholic fatty liver disease) typically presents with elevated aminotransferases and is associated with metabolic syndrome. It is not characterized by cholestasis or positive AMA. Primary sclerosing cholangitis typically presents with cholestatic liver enzyme abnormalities and is often associated with inflammatory bowel disease. It is diagnosed based on characteristic imaging findings of bile duct strictures using MRI cholangiography or endoscopic retrograde cholangiopancreatography. Positive AMA is rarely found in primary sclerosing cholangitis.

References

Lindor KD, Bowlus CL, Boyer J, Levy C, Mayo M. Primary biliary cholangitis: 2021 practice guidance update from the American Association for the Study of Liver Diseases. *Hepatology*. 2022;75(4):1012-1013.

Bowlus CL, Arrivé L, Bergquist A, et al. AASLD practice guidance on primary sclerosing cholangitis and cholangiocarcinoma. *Hepatology*. 2023;77(2):659-702.

Item 49

ANSWER: C

This patient has features of bipolar disorder, a mental health condition affecting >1% of the world's population. The *DSM-5* criteria can be used to suggest a prognosis and determine episode severity. Patients with bipolar disorder have a high suicide rate, so proper diagnosis and prompt treatment are important. The risk of relapse increases in patients who are untreated. Treatment improves prognosis and medications are the mainstay of therapy. FDA-approved medications for the treatment of bipolar disorder include lithium, anticonvulsants, and atypical antipsychotics such as quetiapine.

Although this patient has symptoms of depression, monotherapy with an antidepressant is contraindicated in patients with mixed features, manic episodes, or bipolar I disorder. Augmenting therapies such as omega-3 fatty acids, light therapy, ketamine, and probiotics may be helpful but current evidence is insufficient. Cognitive behavioral therapy, exercise, and nutrition have been shown to be beneficial as well. Electroconvulsive therapy can be an effective treatment for psychotic depression, mania, refractory illness, catatonia, acute psychosis, and suicidal behavior. However, pharmacotherapy is the first-line treatment for bipolar disorder.

References

Marzani G, Price Neff A. Bipolar disorders: evaluation and treatment. *Am Fam Physician*. 2021;103(4):227-239.

Item 50

ANSWER: C

The U.S. Preventive Services Task Force and the American College of Obstetricians and Gynecologists recommend aspirin prophylaxis for pregnant patients who are at increased risk for preeclampsia. Indications include 1 or more high-risk factors or 2 or more moderate-risk factors.

High-risk factors include a history of preeclampsia, multifetal gestation, kidney disease, autoimmune disease, type 1 or type 2 diabetes, and chronic hypertension. Moderate-risk factors include first pregnancy, maternal age ≥ 35 years, BMI >30 kg/m², family history of preeclampsia, sociodemographic characteristics, and personal history factors.

This patient is nulliparous and has a pre-pregnancy BMI >30 kg/m², giving her 2 moderate-risk factors. The recommended aspirin prophylaxis for preeclampsia is 81 mg per day and should be started at 12 weeks' gestation.

Low-dose aspirin has been shown to decrease the frequency of preeclampsia as well as other adverse pregnancy outcomes such as intrauterine growth restriction and preterm birth. It also has favorable fetal and maternal safety profiles.

References

Henderson JT, Vesco KK, Senger CA, Thomas RG, Redmond N. *Aspirin Use to Prevent Preeclampsia and Related Morbidity and Mortality. Agency for Healthcare Research and Quality (US); 2021.*

American College of Obstetricians and Gynecologists' Committee on Obstetric Practice; Society for Maternal-Fetal Medicine. ACOG Committee Opinion No. 743: low-dose aspirin use during pregnancy. *Obstet Gynecol.* 2018;132(1):254-256.

Item 51

ANSWER: A

This patient presents with a recent history of cardioembolic stroke. In patients who have a TIA or small cardioembolic stroke and nonvalvular atrial fibrillation, guidelines recommend starting anticoagulation without antiplatelet agents 2–14 days after the event. Nonvalvular atrial fibrillation is defined by the lack of moderate to severe mitral stenosis or a mechanical heart valve; direct-acting oral anticoagulants are preferred to warfarin in these patients. The 2024 Optimal Timing of Anticoagulation After Acute Ischaemic Stroke (OPTIMAS) trial confirmed the safety of starting anticoagulation within 4 days of a stroke. In patients with larger strokes that may be at high risk for hemorrhagic conversion, initiating anticoagulation may be delayed to 14 days.

References

Werring DJ, Dehbi HM, Ahmed N, et al. Optimal timing of anticoagulation after acute ischaemic stroke with atrial fibrillation (OPTIMAS): a multicentre, blinded-endpoint, phase 4, randomised controlled trial. *Lancet*. 2024:S0140-6736(24)02197-4.

Kleindorfer DO, Towfighi A, Chaturvedi S, et al. 2021 guideline for the prevention of stroke in patients with stroke and transient ischemic attack: a guideline from the American Heart Association/American Stroke Association. *Stroke*. 2021;52(7):e364-e467.

Item 52

ANSWER: B

Advances in angiography, pharmacology, and systems of care have improved morbidity and mortality for acute myocardial infarction (AMI) over the last several decades. This has reduced the incidence of mechanical complications of AMI; however, the hospital mortality for these complications, such as acute papillary muscle rupture, remains high. Acute papillary muscle rupture occurs 3–5 days after a transmural infarction involving the right coronary artery or circumflex artery, depending on coronary anatomy. This leads to acute mitral regurgitation, resulting in flash pulmonary edema and shock, as seen in this patient.

Cardiac tamponade can occur after an AMI if the ventricular free wall ruptures, which is rare. Clinical clues include chest pain, shortness of breath, muffled heart sounds, and distended neck veins. Chest pain is also a core feature of pericarditis, making it an unlikely diagnosis in this patient. Pulmonary embolism (PE) is a common complication seen in immobile hospitalized patients and can be reduced with appropriate mechanical prophylaxis and chemoprophylaxis. While astute clinicians may rule out PE in this patient, her admission has likely included anticoagulation, and lack of edema makes a venous thromboembolism etiology for PE less likely. Arrhythmias can also occur after an AMI. Supraventricular tachyarrhythmias such as atrial fibrillation may present similarly to this patient, but a ventricular rate of 110 beats/min is unlikely to result in hemodynamic compromise as described in this case.

References

Damluji AA, van Diepen S, Katz JN, et al; American Heart Association Council on Clinical Cardiology; Council on Arteriosclerosis, Thrombosis and Vascular Biology; Council on Cardiovascular Surgery and Anesthesia; Council on Cardiovascular and Stroke Nursing. Mechanical complications of acute myocardial infarction: a scientific statement from the American Heart Association. *Circulation*. 2021;144(2):e16-e35.

Frampton J, Ortengren AR, Zeitler EP. Arrhythmias after acute myocardial infarction. *Yale J Biol Med*. 2023;96(1):83-94.

Item 53

ANSWER: C

Based on the Global Initiative for Asthma (GINA) guideline, patients who have symptoms <3–5 days per week with normal or mildly reduced lung function are classified as having mild asthma. For patients 12 years and older with mild asthma, the guidelines recommend treatment with as-needed low-dose inhaled corticosteroid plus formoterol (ICS/formoterol). As-needed low-dose ICS/formoterol reduces the risk of severe asthma exacerbations by 60%–64% compared to as-needed short-acting β -agonists (SABAs) such as albuterol. ICS/formoterol also reduces the risk of severe exacerbations compared to other regimens that include a SABA reliever. Formoterol should not be used without an ICS.

References

Global Initiative for Asthma. *Global Strategy for Asthma Management and Prevention (2024 Update)*. Updated May 2024.

Item 54

ANSWER: A

The most likely diagnosis is cutaneous larva migrans. The typical rash that occurs with this infection is an intensely pruritic, erythematous, serpiginous rash. It is typically seen on the feet and gluteal areas. There are no diagnostic tests for this infection and the diagnosis is made clinically. Infections are generally self-limited but can be treated with albendazole. Cutaneous larva migrans infections are mainly found in Asia and Africa where macaque monkeys are the reservoir. It has an incubation period of 1–5 days but can take up to 1 month.

Cutaneous leishmaniasis generally presents with small papules that develop into nonhealing open sores with a raised border and central ulceration. The lesions are generally painless. Herpesvirus B presents with a vesicular rash near the site of inoculation and influenza-like symptoms. The clinical manifestations of *Neisseria meningitidis* range from asymptomatic dermatologic findings such as petechiae to severe systemic disease. Scabies presents with a very pruritic rash that is red with small bumps, typically appears along skinfolds, and often forms a line.

References

Rathjen NA, Shahbodaghi SD. The ill returning traveler. *Am Fam Physician*. 2023;108(4):396-403.

Item 55**ANSWER: E**

Infection with *Cryptosporidium* is caused by ingesting oocysts, which can survive for up to 10 days in treated water. Diarrhea is the predominant symptom, and a stool sample is used for diagnosis. Treatment for infection with *Cryptosporidium* is supportive with fluid and electrolyte replacement. Nitazoxanide can be used in cases of severe or prolonged disease. Because shedding of oocysts continues even after symptoms resolve, it is recommended that patients refrain from swimming for 2 weeks after resolution of symptoms.

References

Perkins A, Trimmier M. Recreational waterborne illnesses: recognition, treatment, and prevention. *Am Fam Physician*. 2017;95(9):554-560.

Item 56

ANSWER: D

Hemolysis, elevated liver enzymes, and low platelet count (HELLP) syndrome should be suspected in pregnant women presenting with malaise, headache, hypertension, and proteinuria. Women with HELLP syndrome have 10%–15% maternal morbidity and mortality rates compared to those without. The elevated risks are due to the potential of severe complications such as cerebral hemorrhage, cardiopulmonary arrest, disseminated intravascular coagulation, acute respiratory distress syndrome, renal failure, sepsis, and hepatic hemorrhage.

Early diagnosis and management are critical. Quick and targeted evaluation with urgent delivery in patients over 34 weeks' gestation is key to avoiding maternal deaths. Corticosteroids can be used to treat low platelet count, but will not address the symptoms of liver failure. Acute liver failure due to HELLP is secondary to pregnancy. Although this patient has been taking acetaminophen, the examination and laboratory results are consistent with HELLP syndrome; therefore, treatment for acetaminophen overdose with *N*-acetylcysteine is not warranted.

References

Alexander V, Benjamin SJ, Subramani K, Sathyendra S, Goel A. Acute liver failure in pregnancy. *Indian J Gastroenterol*. 2024;43(2):325-337.

Terrault NA, Williamson C. Pregnancy-associated liver diseases. *Gastroenterology*. 2022;163(1):97-117.e1.

Item 57

ANSWER: E

Before a diagnosis of chronic kidney disease can be made, evidence of renal impairment or damage must be present for 3 months in order to differentiate the renal abnormalities from an acute kidney injury (AKI). This is most often an estimated glomerular filtration rate <60 mL/min/1.73 m²; however, the Kidney Disease: Improving Global Outcomes (KDIGO) guidelines do allow for other evidence of persistent renal damage or decreased function, such as albuminuria or proteinuria, that persists for at least 3 months. Previous documentation of kidney damage or impairment can also be sought. Renal imaging abnormalities such as decreased kidney volume, cortical thickening, or multiple cysts are also evidence of chronicity.

References

Levey AS, Eckardt KU, Dorman NM, et al. Nomenclature for kidney function and disease: report of a Kidney Disease: Improving Global Outcomes (KDIGO) Consensus Conference. *Kidney Int.* 2020;97(6):1117-1129.

Goodbred AJ, Langan RC. Chronic kidney disease: prevention, diagnosis, and treatment. *Am Fam Physician.* 2023;108(6):554-561.

Item 58**ANSWER: A**

This patient presents with Kawasaki disease, which commonly occurs in children <5 years of age. Classic Kawasaki disease presents with 4 of the following features: oral mucous membrane changes, bilateral conjunctivitis, diffuse rash, hand and foot edema, and unilateral cervical adenopathy ≥ 1.5 cm. Inflammation occurs and can affect multiple organ systems. The most serious complication is cardiac. If untreated, 15%–25% of patients will develop cardiovascular abnormalities, including coronary artery dilation, aortic root dilation, cardiac aneurysms, tachycardia, hypotension, and shock. Echocardiography must be performed as soon as possible. Less serious complications of Kawasaki disease include anemia, elevated liver function test results, pyuria, and pulmonary edema.

References

Darby JB, Jackson JM. Kawasaki disease and multisystem inflammatory syndrome in children: an overview and comparison. *Am Fam Physician*. 2021;104(3):244-252.

Item 59**ANSWER: A**

For this patient, reassurance only is indicated. The U.S. Preventive Services Task Force recommends against routine screening, risk assessment, counseling, and testing in all women for *BRCA1/2* mutations (D recommendation). Only women with a personal or family history of breast, ovarian, tubal, or peritoneum cancer or specific susceptibility to *BRCA1* and *BRCA2* mutations should be formally evaluated with a risk assessment tool. Furthermore, genetic testing for *BRCA1* and *BRCA2* is indicated only after a positive risk assessment score is calculated and subsequent genetic counseling is received.

References

Final recommendation statement: BRCA-related cancer: risk assessment, genetic counseling, and genetic testing. US Preventive Services Task Force. Updated August 20, 2019.

Item 60

ANSWER: C

Supplementation with folic acid leading up to and during pregnancy can significantly reduce the risk of neural tube defects (NTDs). Ideally, supplementation should be started 1–3 months prior to conception. Those with pregnancies at higher risk of NTDs are candidates for high-dose folic acid supplementation, which is more effective at prevention. This patient is considered high risk as she has a family history of NTDs. The recommended dose of folic acid is 4 mg per day until 12 weeks after conception, and then 0.4 mg per day after 12 weeks. The rationale for decreasing the dose after 12 weeks is that organogenesis is largely complete by this stage and there is theoretical concern for fetal toxicity. Folic acid supplementation is recommended in addition to the usual dietary sources of folic acid.

References

American College of Obstetricians and Gynecologists' Committee on Practice Bulletins—Obstetrics. ACOG Practice Bulletin No. 187: neural tube defects. *Obstet Gynecol.* 2017;130(6):e279-e290.

Folic acid: facts for clinicians. Centers for Disease Control and Prevention. Reviewed May 20, 2025.

Final recommendation statement: folic acid supplementation to prevent neural tube defects: preventive medication. US Preventive Services Task Force. Updated August 1, 2023.

Item 61**ANSWER: E**

Gout is a common crystalline arthropathy. Long-term treatment with urate-lowering medications is effective in preventing recurrent attacks and is generally well tolerated. Allopurinol is typically used as first-line treatment for gout in all patients, even in those with kidney disease. The American College of Rheumatology (ACR) strongly recommends that an anti-inflammatory medication, such as colchicine, be given when allopurinol is initiated to prevent an acute flare. The allopurinol dosage should be adjusted to target a uric acid level <6 mg/dL. Colchicine alone is useful in the setting of an acute flare, as are NSAIDs or corticosteroids. Febuxostat is a urate-lowering medication that works by inhibiting xanthine oxidase, which interferes with the conversion of hypoxanthine and xanthine to uric acid. While this medication is indicated for chronic gout, it is not a first-line choice. The ACR strongly recommends against pegloticase as a first-line therapy due to cost, lack of evidence regarding safety, and other options with more favorable benefit to harm ratios.

References

FitzGerald JD, Dalbeth N, Mikuls T, et al. 2020 American College of Rheumatology guideline for the management of gout. *Arthritis Care Res.* 2020;72(6):744-760.

Item 62**ANSWER: E**

Once symptoms occur in patients with aortic stenosis, the clinical outcome is extremely poor, with 2-year survival rates below 50%. It is well established that this prognosis can be reversed by valve replacement, which has acceptable levels of operative morbidity and mortality, and postoperative survival rates similar to those of age-matched normal adults. Angina could be an indication of the seriousness of aortic valve stenosis. Patients should also be evaluated for coronary artery disease.

References

Carabello BA, Kodali S. Valvular heart disease. In: Goldman L, Cooney KA, eds. *Goldman-Cecil Medicine*. 27th ed. Vol. 1. Elsevier; 2024:400-406.

Grimard BH, Safford RE, Burns EL. Aortic stenosis: diagnosis and treatment. *Am Fam Physician*. 2016;93(5):371-378.

Item 63**ANSWER: B**

This patient has mild allergic rhinitis. Intranasal corticosteroids are considered first-line therapy. Intranasal or oral antihistamines may be considered given the patient's mild symptoms, but they are less effective. Intranasal cromolyn is less effective and requires multiple dosing. Since there is no apparent infection, oral antibiotics are not indicated. Oral corticosteroids are also not indicated at this time. Some experts recommend nasal saline irrigation to remove allergens from the nares, but this approach has limited evidence and would augment, not replace, pharmacotherapy.

References

Weaver-Agostoni J, Kosak Z, Bartlett S. Allergic rhinitis: rapid evidence review. *Am Fam Physician*. 2023;107(5):466-473.

Item 64

ANSWER: E

While there is no single test to diagnose polycystic ovary syndrome (PCOS), the diagnosis can be made based on the presence of at least 2 of the 3 Rotterdam criteria: oligoanovulation, hyperandrogenism, and polycystic ovaries on ultrasonography. Other conditions that may mimic PCOS must be excluded, including an androgen-secreting tumor, exogenous androgen exposure, Cushing syndrome, nonclassical congenital adrenal hyperplasia, primary hypothalamic amenorrhea, primary ovarian insufficiency, acromegaly, thyroid disease, prolactin disorders, and genetic disorders that impede insulin action. Hyperandrogenism may be diagnosed clinically with evidence of hirsutism, androgenic alopecia, acne, or seborrhea. Additionally, it may be diagnosed biochemically with elevated androgen levels, such as an elevated free testosterone level.

An elevated 17-hydroxyprogesterone level would not suggest PCOS, but rather a possible diagnosis of congenital adrenal hyperplasia, which may share overlapping clinical features with PCOS. An elevated cortisol level would similarly suggest an etiology other than PCOS, such as Cushing syndrome or adrenal tumor, which may also share overlapping clinical features with PCOS. Although insulin resistance is often part of the complex pathophysiology of PCOS, neither elevated insulin nor glucose levels are part of the Rotterdam criteria for PCOS.

References

Williams T, Moore JB, Regehr J. Polycystic ovary syndrome: common questions and answers. *Am Fam Physician*. 2023;107(3):264-272.

Nelson M, LaRouche V. Polycystic ovary syndrome: assessment and management guidelines. *Am Fam Physician*. 2024;110(5):547-548.

Item 65

ANSWER: B

This patient's vertigo is caused by vestibular neuritis, as evidenced by its spontaneity, rotatory vertigo, nausea, and listing to the affected side with walking. In this scenario, head-impulse testing shows saccade, and the head-impulse, nystagmus, test of skew (HINTS) examination is significant for horizontal nystagmus to the affected side. Vestibular neuritis is most often found in patients 30–50 years of age and is thought to have a viral etiology. It involves prolonged, continuous vertigo, while benign paroxysmal positional vertigo (BPPV) is characterized by brief, episodic vertigo associated with head movements. Vestibular neuritis usually resolves on its own within a few days, although some patients have recurrent symptoms for months. The treatment for vestibular neuritis is reassurance. Antiemetics can be used for up to 3 days. Patients with symptoms that last longer should be referred for vestibular rehabilitation. Home Epley maneuvers are appropriate for vertigo caused by BPPV, but are neither indicated nor effective for vestibular neuritis (SOR A). Corticosteroids and antiviral medications have not been shown to be effective in treating vestibular neuritis. MRI of the brain is indicated for patients who exhibit signs of central vertigo or those with hearing loss (SOR C).

References

Muncie HL, Sirmans SM, James E. Dizziness: approach to evaluation and management. *Am Fam Physician*. 2017;95(3):154-162.

Rogers TS, Noel MA, Garcia B. Dizziness: evaluation and management. *Am Fam Physician*. 2023;107(5):514-523.

Item 66**ANSWER: E**

Melasma is a common condition in pregnancy that occurs in up to 75% of patients. It is caused by increased levels of progesterone, estrogen, and melanocyte-stimulating hormone. While it resolves in 90% of cases during the postpartum period, it can persist and may recur in subsequent pregnancies or with the use of oral contraception. Topical tretinoin, hydroquinone, and a corticosteroid such as fluocinolone can all be used in the treatment of melasma, and a combination of the 3 is the most effective. These may be prescribed as a single formulation, which may be cost prohibitive, or as individual components. Azelaic acid and benzoyl peroxide are used to treat acne and are not effective for treating melasma. Imiquimod is an immunomodulator that is helpful in the treatment of condyloma acuminata but has no role in the treatment of melasma.

References

Erlandson M, Wertz MC, Rosenfeld E. Common skin conditions during pregnancy. *Am Fam Physician*. 2023;107(2):152-158.

Item 67**ANSWER: A**

The lowest risk of developing future allergies to many foods, including peanuts and eggs, occurs with early exposure at 4–6 months of age. Peanut butter can be thinned with water, breast milk, cow’s milk, or formula. Peanuts should not be given in their original hard form because it is a choking hazard. Eggs should be cooked well prior to eating.

References

Halken S, Muraro A, de Silva D, et al; European Academy of Allergy and Clinical Immunology Food Allergy and Anaphylaxis Guidelines Group. EAACI guideline: preventing the development of food allergy in infants and young children (2020 update). *Pediatr Allergy Immunol*. 2021;32(5):843-858.

Bright DM, Stegall HL, Slawson DC. Food allergies: diagnosis, treatment, and prevention. *Am Fam Physician*. 2023;108(2):159-165.

Item 68

ANSWER: D

In addition to trauma-informed evaluation and counseling, women who experience sexual assault should be offered forensic examination, emergency contraception, and screening and prevention for sexually transmitted infections. Empiric treatment for *Neisseria gonorrhoeae*, *Chlamydia trachomatis*, and *Trichomonas vaginalis* is appropriate and should include ceftriaxone, azithromycin, and either metronidazole or tinidazole.

Postexposure prophylaxis for HIV should be made available with triple therapy for 28 days, rather than the double therapy regimens used in pre-exposure prophylaxis.

Sofosbuvir/velpatasvir is used to treat hepatitis C infection, but routine postexposure prophylaxis is not recommended. Long-term contraception may be offered, but patients should first be provided with effective emergency contraception, such as ulipristal, levonorgestrel, or the copper IUD.

References

Farahi N, McEachern M. Sexual assault of women. *Am Fam Physician*. 2021;103(3):168-176.

Item 69

ANSWER: B

Symptoms of premenstrual syndrome (PMS) with mood instability can be reduced with 1000–1200 mg of calcium carbonate daily (SOR B). Patients with PMS symptoms have been found to have cyclic changes in calcium levels that may account for the symptoms, but the mechanism of action is still unknown.

All SSRIs are first-line treatments for symptoms of PMS and premenstrual dysphoric disorder (PMDD), but bupropion has not been shown to be effective. Continuous combined oral contraceptives are also first-line treatment (SOR A). High-dose dutasteride (2.5 mg daily) has been shown to alleviate PMDD symptoms, but it can have a deleterious effect on mood and is teratogenic. Low-dose dutasteride has not demonstrated beneficial effect on PMDD symptoms. Patients with PMS have not been found to have low levels of vitamin D, and supplementing with vitamin D₃ (cholecalciferol) has not been clearly demonstrated to alleviate PMS symptoms. There is insufficient evidence to prove the benefit of acupuncture in reducing PMS symptoms.

References

Hofmeister S, Bodden S. Premenstrual syndrome and premenstrual dysphoric disorder. *Am Fam Physician*. 2016;94(3):236-240.

Cary E, Simpson P. Premenstrual disorders and PMDD - a review. *Best Pract Res Clin Endocrinol Metab*. 2024;38(1):101858.

Item 70**ANSWER: E**

Women ≥ 65 years of age should be tested for osteoporosis. A bone mineral density score ≤ -2.5 indicates the need for treatment, as seen in this patient. Oral bisphosphonates, such as alendronate, are considered first-line treatment.

To minimize gastrointestinal irritation, oral bisphosphonates should be taken on an empty stomach with 8 oz of water, and the patient should remain upright and avoid food and beverages other than plain water for at least 30 minutes afterward. If oral bisphosphonates are not tolerated despite correct administration, intravenous bisphosphonates should be considered. In this case, the patient should discontinue alendronate and transition to intravenous bisphosphonate therapy. Raloxifene is not the preferred initial treatment.

References

Harris K, Zagar CA, Lawrence KV. Osteoporosis: common questions and answers. *Am Fam Physician*. 2023;107(3):238-246.

Item 71

ANSWER: B

Despite optimal therapy, the mortality and loss of quality-adjusted life-years from bacterial meningitis is significant. Empiric therapy for adults includes vancomycin and ceftriaxone. Intravenous (IV) dexamethasone, when given shortly before or at the time of administration of IV antibiotics, has been shown to improve mortality (odds ratio 0.90) and neurologic sequelae. Bactericidal antibiotics help to reduce inflammation.

Acyclovir would be indicated if there is clinical concern for herpes simplex virus or symptoms of encephalitis. While optimal fluid management for meningitis should be tailored based on clinical evaluation, this patient has also been diagnosed with sepsis and should receive appropriate fluid resuscitation; therefore, IV furosemide would be inappropriate. Glycerol is a hyperosmolar agent intended to reduce intracranial pressure in bacterial meningitis, but it cannot be recommended for treatment due to an increased risk of death according to available data in adults with a high prevalence of HIV. IV immunoglobulin is not a recognized treatment for bacterial meningitis.

References

Hasbun R. Progress and challenges in bacterial meningitis: a review. *JAMA*. 2022;328(21):2147-2154.

Mount HR, Boyle SD. Aseptic and bacterial meningitis: evaluation, treatment, and prevention. *Am Fam Physician*. 2017;96(5):314-322.

Ajdukiewicz KM, Cartwright KE, Scarborough M, et al. Glycerol adjuvant therapy in adults with bacterial meningitis in a high HIV seroprevalence setting in Malawi: a double-blind, randomised controlled trial. *Lancet Infect Dis*. 2011;11(4):293-300.

Brouwer MC, McIntyre P, Prasad K, van de Beek D. Corticosteroids for acute bacterial meningitis. *Cochrane Database Syst Rev*. 2015;2015(9):CD004405.

Item 72

ANSWER: B

Eosinophilic esophagitis is an inflammatory condition triggered by food antigens in sensitive individuals. Endoscopic findings of linear grooves, concentric rings, and whitish exudates may suggest its presence, but the diagnosis is confirmed histologically based on the presence of >15 eosinophils/hpf and the exclusion of other causes of eosinophilia. The diagnosis should be suspected in patients with heartburn who have atopic illnesses such as allergic rhinitis, asthma, or eczema and who do not respond to proton pump inhibitor therapy. Eosinophilic esophagitis is more common in men in their 20s and 30s, and dysphagia may be present for 4 or more years before diagnosis. More than half of patients with this condition may have a history of food impaction. First-line pharmacologic treatment is oral budesonide. Alternatively, inhalant corticosteroids may be sprayed into the oral cavity during a breath hold and then swallowed to minimize pulmonary deposition. Treatment with proton pump inhibitors may help, but they do not provide complete relief in two-thirds of cases. Dietary changes can be very helpful, but consultation with an allergist may be needed.

Immunosuppressed individuals such as patients with HIV or diabetes mellitus may have *Candida* esophagitis, which is treated with fluconazole rather than clotrimazole. Adding an H₂-blocker such as famotidine would help in cases of erosive esophagitis. Metoclopramide was previously used to treat severe GERD, but it is no longer encouraged due to side effects. While montelukast is used to treat asthma and allergic rhinitis, it is not as effective for eosinophilic esophagitis.

References

Kroch DA, Madanick RD. Medical treatment of gastroesophageal reflux disease. *World J Surg.* 2017;41(7):1678-1684.

Clayton S, Emerson JF. Eosinophilic esophagitis: a mimic of gastroesophageal reflux disease. *Am Fam Physician.* 2018;97(10):628-629.

Mohamed AA, Lu XL, Mounmin FA. Diagnosis and treatment of esophageal candidiasis: current updates. *Can J Gastroenterol Hepatol.* 2019;2019:3585136.

Gómez-Aldana A, Jaramillo-Santos M, Delgado A, Jaramillo C, Lúquez-Mindiola A. Eosinophilic esophagitis: current concepts in diagnosis and treatment. *World J Gastroenterol.* 2019;25(32):4598-4613.

Arnold MJ. Eosinophilic esophagitis: management guidelines from the AGA and JTF. *Am Fam Physician.* 2021;103(9):573-574.

Item 73

ANSWER: A

Adults with inflammatory bowel disease (IBD) should receive a pneumococcal vaccine. Meningococcal vaccine is currently recommended for adolescents with IBD. The MMR vaccine is contraindicated in patients already on immunosuppressive treatment. Because this patient's Tdap vaccine was administered <10 years ago, he is not due for another dose. Similarly, his next dose of zoster vaccine will be due when he is 50 years of age, and there is no recommendation to administer the next vaccine before the age of 50 in patients with IBD.

References

Farraye FA, Melmed GY, Lichtenstein GR, Kane SV. ACG clinical guideline: preventive care in inflammatory bowel disease. *Am J Gastroenterol*. 2017;112(2):241-258.

Kobayashi M, Pilishvili T, Farrar JL, et al. Pneumococcal vaccine for adults aged ≥ 19 years: recommendations of the Advisory Committee on Immunization Practices, United States, 2023. *MMWR Recomm Rep*. 2023;72(3):1-39.v

Item 74

ANSWER: C

This patient's chronic corticosteroid use, symptoms, and electrolyte abnormalities suggest she is experiencing an adrenal crisis; the correct initial treatment of this is intravenous hydrocortisone. Acute illness, dehydration, severe physical stress, or nonadherence to treatment are all possible causes of acute adrenal crisis in patients with chronic adrenal insufficiency. These patients may present with weakness, severe fatigue, abdominal pain, tachycardia, and hypotension. Laboratory abnormalities can include hyponatremia, hyperkalemia, and, in some cases, hypoglycemia. If left untreated, adrenal crisis can be fatal.

Glucagon is primarily used to treat hypoglycemia in patients with insulin-dependent diabetes, but it will not correct the cardiovascular abnormalities of adrenal crisis. Dexamethasone is a corticosteroid that can be used to treat adrenal crisis, but it is not the preferred therapy according to Endocrine Society practice guidelines. Additionally, a 1-mg dose is subtherapeutic for adrenal crisis. Dexamethasone is sometimes used at this dose to test for primary hyperadrenalism. Levofloxacin is not appropriate for treatment of this patient's adrenal crisis. Although ondansetron may help reduce vomiting, it will not correct the underlying cause of this patient's illness.

References

Michels A, Michels N. Addison disease: early detection and treatment principles. *Am Fam Physician*. 2014;89(7):563-568.

Bornstein SR, Allolio B, Arlt W, et al. Diagnosis and treatment of primary adrenal insufficiency: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab*. 2016;101(2):364-389.

Item 75**ANSWER: B**

Electrocardiography (EKG) is a primary evaluation tool, along with a comprehensive history and physical examination, to investigate the etiology of a syncopal episode (SOR C). Electroencephalography (EEG) is not recommended as a routine test. Carotid ultrasonography is not recommended for evaluation of syncope in patients with a normal physical examination (SOR B). Similarly, head CT and MRI are not recommended as neurologic imaging studies in patients with a normal neurologic examination after a syncopal episode (SOR B). If the patient were noted to experience syncope with exertion, an exercise stress test would be appropriate (SOR C). However, this patient was observed to be at rest when the "fainting" occurred.

References

Runser LA, Gauer RL, Houser A. Syncope: evaluation and differential diagnosis. *Am Fam Physician*. 2017;95(5):303-312B.

Item 76

ANSWER: B

Men with lower urinary tract symptoms may have overactive bladder, bladder outlet obstruction, or both. Differentiating overactive bladder and bladder outlet obstruction can be difficult due to significant symptom overlap. Overactive bladder is most often due to detrusor overactivity, and bladder outlet obstruction is most often secondary to benign prostatic hyperplasia. The most appropriate initial treatment for these symptoms is an α -blocker, such as tamsulosin, alfuzosin, or terazosin (Evidence Level A), regardless of the predominance of storage symptoms (eg, daytime frequency, nocturia, urgency, incontinence) or voiding symptoms (eg, weak stream, hesitancy, straining, dribbling).

The American Urological Association recommends 5- α -reductase inhibitors if prostate volume is ≥ 30 mL on imaging, the prostate is enlarged on examination, or PSA levels exceed 1.5 ng/mL. 5- α -Reductase inhibitors can take up to a year to take full effect. Anticholinergics (oxybutynin) and β_3 -agonists (mirabegron) may be used alone or in combination with α -blockers for patients with predominantly storage-related symptoms if α -blockers are not adequately effective. Some guidelines recommend assessing postvoid residual volume before initiating these agents. A daily phosphodiesterase-5 inhibitor (typically tadalafil) can be used as initial monotherapy or as combination therapy in patients with erectile dysfunction, although it is not generally recommended as first-line treatment. The mechanism of action is not fully understood.

References

Sandhu JS, Bixler BR, Dahm P, et al. Management of lower urinary tract symptoms attributed to benign prostatic hyperplasia (BPH): AUA guideline amendment 2023. *J Urol*. 2024;211(1):11-19.

Lerner LB, McVary KT, Barry MJ, et al. Management of lower urinary tract symptoms attributed to benign prostatic hyperplasia: AUA guideline part I-initial work-up and medical management. *J Urol*. 2021;206(4):806-817.

Item 77**ANSWER: B**

Recognition of adequate newborn feeding and parental lactation counseling is important in the care of mothers and infants. An infant who is feeding adequately should produce 6 or more wet diapers and 3 or more stools after 4 days of life. Infant weight loss is normal in the first 3 days of life, but a weight loss of more than 7% from birth weight or ongoing weight loss after 3 days of life should raise concern. Maternal breasts that appear smaller or less full do not necessarily indicate poor feeding; however, lack of any breast changes or leaking of milk by day 5 post partum may indicate a problem with milk supply. Maternal nipple pain is normal with breastfeeding, but pain that persists beyond the first 30–60 seconds should prompt evaluation and counseling. If supplementation with formula is deemed medically necessary, protocol from the Academy of Breastfeeding Medicine should be followed.

References

Westerfield KL, Koenig K, Oh R. Breastfeeding: common questions and answers. *Am Fam Physician*. 2018;98(6):368-373.

Item 78**ANSWER: A**

According to the CDC, antiviral treatment for influenza is not recommended for non–high-risk individuals who present more than 48 hours after symptom onset. Thus, supportive therapy is most important for this patient who is at low risk for hospitalization and death. Methylprednisolone is not recommended to treat influenza. The risk for hospitalization and death is not reduced in adults and children treated with a neuraminidase inhibitor in the outpatient setting, as supported by systematic and unpublished randomized clinical trials. While oseltamivir is recommended as first-line therapy for high-risk patients, it is not indicated for this patient, who has no underlying medical conditions or known exposure to high-risk individuals and presents more than 48 hours after symptom onset. Rimantadine is not active against influenza B and is also not considered first-line medical therapy for influenza.

References

Gaitonde DY, Moore FC, Morgan MK. Influenza: diagnosis and treatment. *Am Fam Physician*. 2019;100(12):751-758.

Item 79**ANSWER: E**

The most effective medication therapy options for obsessive-compulsive disorder (OCD) are SSRIs. Fluoxetine, fluvoxamine, sertraline, and paroxetine all have approval from the FDA for the treatment of OCD. Dosages that are higher than usual are often required to achieve symptom control. Clomipramine, a tricyclic antidepressant, has an indication from the FDA for treatment of OCD, but a higher toxicity risk makes it less favorable for first-line treatment. Bupropion (a norepinephrine-dopamine reuptake inhibitor), buspirone (an atypical anxiolytic), and quetiapine (an atypical antipsychotic) are not first-line treatment for OCD.

References

Semenya AM, Bhatnagar P. Diagnosis and management of obsessive-compulsive disorder in the primary care setting. *Am Fam Physician*. 2024;110(4):385-392.

Item 80

ANSWER: C

Falls are a common cause of serious injury and death in older adults in the United States. A 2018 study found that 27.5% of adults ≥ 65 years of age had fallen at least once in the past year (714 falls per 1000 community-dwelling older adults). More than 300,000 older adults are hospitalized each year for hip fracture, which is one of the most serious potential complications from falls in older adults. Falls are also the leading cause of injury-related death in older adults. A history of a fall is a key risk factor for future falls. Other risks include sensory and cognitive deficits, acute and chronic health conditions, alcohol and drug use, and certain other medication use.

In community-dwelling older adults who are identified as being at higher risk of falls, the U.S. Preventive Services Task Force found that exercise interventions lower the risk of falls and should be recommended. Both individual and group exercise programs are beneficial and there is no data to recommend any single type of exercise over others. Multifactorial interventions to reduce falls typically include an assessment of modifiable risk factors followed by a customized approach to reducing risk. Evaluations may include balance, gait, vision, cognition, postural blood pressure, medication use, and the home environment. This assessment differs from a focused diagnostic assessment of a patient with a specific concern such as vision change, dizziness, or numbness in the feet. Interventions to address risk factors identified in a multifactorial screening program may offer benefit to high-risk individuals, although studies have found a small net benefit on average. A 2023 Cochrane review comprising 22 studies on environmental interventions found evidence of benefit for reducing fall hazards at home by using a validated tool to identify and address hazards, usually via a home visit. Patient education, assistive devices, vision impairment correction, and footwear selection were not found to be beneficial.

References

Final recommendation statement: falls prevention in community-dwelling older adults: interventions. US Preventive Services Task Force. Updated June 4, 2024.

Herness J, Sanders J, James J. Environmental interventions for preventing falls in older people living in the community. *Am Fam Physician*. 2023;108(5):450-451.

Older adult falls data. Centers for Disease Control and Prevention. Reviewed September 6, 2023.

Item 81

ANSWER: D

Infection with *Helicobacter pylori* should be treated when it causes symptoms. Several regimens exist for the treatment of *H pylori* infection. Guidelines currently recommend optimized quadruple therapy with bismuth, metronidazole, omeprazole, and tetracycline. This combination is approximately 91% effective at eradicating *H pylori* infection. Dual therapy with vonoprazan and amoxicillin is an option for patients who are unable or unwilling to take quadruple therapy, but its eradication rates only range from 70% to 80%. Regimens containing clarithromycin or levofloxacin should be limited to infections with documented susceptibility to macrolides or fluoroquinolones, respectively.

References

Malfertheiner P, Megraud F, Rokkas T, et al. Management of *Helicobacter pylori* infection: the Maastricht VI/Florence consensus report. *Gut*. Published online August 8, 2022.

Chey WD, Howden CW, Moss SF, et al. ACG clinical guideline: treatment of *Helicobacter pylori* infection. *Am J Gastroenterol*. 2024;119(9):1730-1753.

Item 82

ANSWER: A

Burns are a common cause of injury in the United States, but most are minor and can be treated in the outpatient setting. The initial treatment goals for burn injuries include limiting the extent of the injury, cleaning the area, and managing pain. Removing clothing limits the injury related to scalding. The most effective way to limit the extent of injury is to cool the burn surface by running cool tap water over the burn surface for at least 20 minutes. This intervention has been shown to reduce burn depth, improve healing time, and decrease skin graft requirements. Ideally, burns should be cooled within the first 30 minutes of injury, but some evidence suggests benefit up to 3 hours after the initial injury. Applying ice to the burn surfaces is not recommended as a cooling strategy due to the risk of vasoconstriction, which can contribute to tissue damage. Wet dressings are not effective as a cooling strategy and are not recommended. For superficial burns such as those described in this patient, aloe vera and NSAIDs can be used to address pain after the initial cooling strategy, but these are not the recommended first step.

Burns are classified by their depth: superficial (first degree), partial thickness (second degree), and full thickness (third degree). Superficial burns are limited to the epidermal layer. They are painful and appear red (blanchable), with intact skin and no blistering. Partial-thickness burns are deeper and involve injury to the dermis. Superficial partial-thickness burns may blister and be exudative, whereas deep partial-thickness burns are drier and do not blanch. Full-thickness burns involve the entire dermis and extend into the fat and connective tissue layers. They appear white rather than red, and are dry with a waxy appearance. Patients often experience less pain due to the destruction of nerve endings. Deep partial-thickness and full-thickness burns should be evaluated and treated at a burn center.

References

Lanham JS, Nelson NK, Hendren B, Jordan TS. Outpatient burn care: prevention and treatment. *Am Fam Physician*. 2020;101(8):463-470.

Item 83**ANSWER: A**

Direct-acting oral anticoagulants (DOACs) are now recommended as monotherapy for outpatient management of deep vein thrombosis. DOACs reach therapeutic levels quickly; therefore, concomitant initiation of unfractionated or low-molecular-weight heparin (LMWH) is not necessary. DOACs have less major bleeding and require less monitoring, making them preferred over warfarin. Previously, unfractionated heparin was administered during hospitalization while warfarin was simultaneously initiated; however, subcutaneous LMWH can now be started with warfarin as an outpatient and continued until the goal INR is reached if a DOAC is unaffordable.

References

Stevens SM, Woller SC, Kreuziger LB, et al. Antithrombotic therapy for VTE disease: compendium and review of CHEST guidelines 2012-2021. *Chest*. 2024;166(2):388-404.

Item 84**ANSWER: E**

Women who received chest radiotherapy as children are at increased risk for breast cancer as adults, with those who received whole lung radiation at especially high risk. Other risk factors for breast cancer include the onset of menopause at >55 years of age, a first pregnancy at >35 years of age, and a first-degree relative with breast cancer. Patients with dense breasts have a higher amount of glandular and stromal tissue compared with adipose tissue on mammography and are at higher risk for breast cancer.

References

Khan M, Chollet A. Breast cancer screening: common questions and answers. *Am Fam Physician*. 2021;103(1):33-41.

Item 85

ANSWER: B

Poststroke depression (PSD) is a common complication that occurs in 20%–60% of patients with stroke, depending on the definition used in diagnosis and the timing of symptom onset. The exact pathophysiology of PSD is unknown but appears complex and multifactorial. The incidence of PSD varies with the location of the stroke. Abnormal neurotransmitter and inflammatory marker levels have been associated with PSD. SSRIs such as escitalopram are considered first-line therapy for PSD. Other options include SNRIs, tricyclic antidepressants (TCAs), and newer multimodal agents such as vortioxetine.

Bupropion may be considered if SSRIs are ineffective or cannot be tolerated. As a class, TCAs have a higher anticholinergic effect than SSRIs and are therefore not preferred, although nortriptyline has been evaluated in randomized, controlled trials and found to be effective in treating PSD. There is a growing body of evidence to support the use of repetitive transcranial magnetic stimulation, but it is not recommended as first-line treatment at this time.

References

Lekoubou A, Nguyen C, Kwon M, Nyalundja AD, Agrawal A. Post-stroke everything. *Curr Neurol Neurosci Rep*. 2023;23(11):785-800.

Frank D, Gruenbaum BF, Zlotnik A, Semyonov M, Frenkel A, Boyko M. Pathophysiology and current drug treatments for post-stroke depression: a review. *Int J Mol Sci*. 2022;23(23):15114.

Masuccio FG, Grange E, Di Giovanni R, Rolla M, Solaro CM. Post-stroke depression in older adults: an overview. *Drugs Aging*. 2024;41(4):303-318.

Item 86**ANSWER: A**

The 2022 CDC clinical practice guideline for prescribing opioids recommends that clinicians offer naloxone to patients at increased risk of overdose, including patients with a history of substance use, overdose, and sleep-disordered breathing, as well as patients who take higher dosages of opioids (≥ 50 MME/day), take benzodiazepines with opioids, or are at risk of returning to higher opioid dosages with loss of tolerance, such as when tapering opioid dosages.

References

Dowell D, Ragan KR, Jones CM, Baldwin GT, Chou R. CDC clinical practice guideline for prescribing opioids for pain—United States, 2022. *MMWR Recomm Rep*. 2022;71(3):1-95.

Item 87

ANSWER: D

The most likely cause of this patient's symptoms is late dumping syndrome. Dumping syndrome is a common postbariatric surgery complication occurring in about 40% of patients. There are 2 types of dumping syndrome: early dumping syndrome and late dumping syndrome. Early dumping syndrome is caused by the rapid passage of nutrients into the small bowel, resulting in intestinal hormonal release and fluid shift. Symptoms occur within 30–60 minutes of eating and include abdominal pain, diarrhea, bloating, nausea, flushing, palpitations, sweating, tachycardia, hypotension, and syncope without chemical evidence of hypoglycemia. Late dumping syndrome is caused by the rapid absorption of carbohydrates, leading to exaggerated insulin secretion, resulting in chemical evidence of hypoglycemia occurring 1–3 hours after eating.

Gastroparesis, unlike dumping syndrome, results from significantly delayed gastric emptying without hypoglycemia, often due to complications of diabetes mellitus. Idiopathic postprandial syndrome describes hypoglycemic-like symptoms that occur after eating without chemical evidence of hypoglycemia. Insulinoma is a rare insulin-producing tumor of the pancreas that causes hypoglycemia regardless of food intake timing.

References

Vavricka SR, Greuter T. Gastroparesis and dumping syndrome: current concepts and management. *J Clin Med*. 2019;8(8):1127.

Banerjee ES, Schroeder R, Harrison TD. Metabolic surgery for adult obesity: common questions and answers. *Am Fam Physician*. 2022;105(6):593-601.

Lawler HM. Idiopathic postprandial syndrome. In: McDermott MT, ed. *Management of Patients with Pseudo-Endocrine Disorders*. Springer; 2019:91-97.

Item 88

ANSWER: C

The American Academy of Pediatrics recommends antibiotic therapy for children ≥ 6 months of age with severe signs and symptoms of acute otitis media (AOM), including moderate or severe otalgia, otalgia lasting longer than 48 hours, or a temperature $\geq 39^\circ\text{C}$ (102°F), whether the AOM is unilateral or bilateral (SOR B). Children younger than 24 months without severe symptoms should receive antibiotic therapy for bilateral AOM, whereas older children or those with unilateral AOM can be offered the option of observation and follow-up.

The usual treatment for AOM is amoxicillin, but an antibiotic with additional β -lactamase coverage, such as amoxicillin/clavulanate, should be given if the child has received amoxicillin within the past 30 days, has concurrent purulent conjunctivitis, or has a history of AOM unresponsive to amoxicillin (SOR C). Penicillin-allergic patients should be treated with an alternative antibiotic such as cefdinir, cefuroxime, cefpodoxime, or ceftriaxone. Ciprofloxacin should not be used in children. Sulfamethoxazole/trimethoprim is no longer effective for the treatment of acute otitis media due to evidence of *Streptococcus pneumoniae* resistance.

References

Paul CR, Frohna JG. Acute otitis media. *Pediatrics in Review*. 2025;46(3):139-147.

Item 89

ANSWER: A

This patient likely has a mild traumatic brain injury (mTBI), or concussion, based on the mechanism of injury and the presence of vomiting. This patient's TBI would be classified as mild due to the lack of worsening symptoms or additional concerning findings on examination, such as the child being fussier or holding his head. For these reasons, reassurance and outpatient observation with follow-up is the most appropriate next step.

Routine imaging is not recommended for mTBI in children. There is no role for radiography of the skull in the management of mTBI or concussion. CT of the head or neck would only be indicated if the patient were experiencing acutely worsening symptoms, reporting or showing signs of headache, or had a suspected head or neck injury. There is no need to contact child protective services because there has been no delay in seeking care or the presence of other unusual physical findings or behaviors between the parents and child.

References

Shaughnessy AF. Imaging guidelines for children with mild traumatic brain injury. *Am Fam Physician*. 2024;110(2):202-203.

Lumba-Brown A, Yeates KO, Sarmiento K, et al. Centers for Disease Control and Prevention guideline on the diagnosis and management of mild traumatic brain injury among children. *JAMA Pediatr*. 2018;172(11):e182853.

Item 90

ANSWER: A

Behavioral interventions are effective treatments for smoking cessation. Advising a patient who wishes to stop smoking to create a smoke-free home and car environment is a highly effective behavioral intervention. This advice is an important aspect of a brief 3-step tobacco cessation intervention model (Ask, Advise, Assist/Connect). Caffeine intake does not play a role in smoking cessation. Switching to menthol cigarettes does not have a role in supporting smoking cessation. There is no high- or moderate-quality evidence to support hypnotherapy or acupuncture for smoking cessation.

In addition to behavioral interventions, the U.S. Preventive Services Task Force also recommends providing FDA-approved pharmacotherapy such as nicotine replacement therapy, sustained-release bupropion hydrochloride, or varenicline. Combined interventions result in increased tobacco-cessation rates.

References

Rigotti NA, Kruse GR, Livingstone-Banks J. Treatment of tobacco smoking: a review. *JAMA*. 2022;327(6):566-577.

Final recommendation statement: tobacco smoking cessation in adults, including pregnant persons: interventions. US Preventive Services Task Force. Updated January 19, 2021.

Levett JY, Filion KB, Reynier P, Prell C, Eisenberg MJ. Efficacy and safety of e-cigarette use for smoking cessation: a systematic review and meta-analysis of randomized controlled trials. *Am J Med*. 2023;136(8):804-813.e4.

Item 91**ANSWER: A**

Long-acting muscarinic antagonists (LAMAs) such as umeclidinium are recommended as the preferred initial management of COPD that is classified as GOLD group A. Long-acting β -agonists (LABAs) such as formoterol may be used as an alternative for these patients. LAMA/LABA combinations such as aclidinium bromide/formoterol are recommended as initial management of COPD that is classified as GOLD group B. LAMA/LABA combinations are also recommended for management of COPD that is classified as GOLD group E. Inhaled corticosteroid (ICS)/LAMA/LABA combinations such as fluticasone furoate/umeclidinium/vilanterol can also be considered for initial management of COPD that is classified as GOLD group E, especially if blood eosinophil levels are ≥ 300 cells/ μ L.

References

Cagle SD Jr, Landrum LS, Kennedy AM. Chronic obstructive pulmonary disease: diagnosis and management. *Am Fam Physician*. 2023;107(6):604-612.

Item 92

ANSWER: C

This patient presents with acute confusion, dehydration, hyperglycemia, and hyperosmolality. The appropriate diagnosis is hyperglycemic hyperosmolar state (HHS), rather than diabetic ketoacidosis (DKA), given her lack of significant ketonuria and acidosis. She is hypovolemic and dehydrated as evidenced by her change in vital signs. Initial treatment consists of intravenous fluids with isotonic solutions, such as 0.9% normal saline or lactated Ringer solution. She will also need an intravenous insulin infusion of short-acting insulin at a dose of 0.05 units/kg/hr. After the initial fluid bolus, potassium chloride can be added to intravenous fluids to correct her total body potassium depletion. Sodium bicarbonate should only be considered if the arterial pH is <7.0 (see Figure 4 of the American Diabetes Association guideline for treatment pathways of DKA and HHS).

References

Umpierrez GE, Davis GM, ElSayed NA, et al. Hyperglycemic crises in adults with diabetes: a consensus report. *Diabetes Care*. 2024;47(8):1257-1275.

Stoner GD. Hyperosmolar hyperglycemic state. *Am Fam Physician*. 2017;96(11):729-736.

Item 93

ANSWER: C

Treatment with an SGLT2 inhibitor is recommended to decrease all-cause mortality in patients with heart failure with reduced ejection fraction (HFrEF), regardless of the presence or absence of diabetes mellitus. According to guidelines from the American College of Cardiology/American Heart Association, first-line medical management for symptomatic HFrEF is angiotensin receptor/neprilysin inhibitor (ARNi) therapy. If ARNi therapy is not feasible, an ACE inhibitor or angiotensin receptor blocker is second-line therapy. These patients should also receive therapy with a cardioselective β -blocker, a loop diuretic (for pulmonary congestion or edema), and a mineralocorticoid receptor antagonist.

Neither calcium channel blockers (CCBs) nor thiazide diuretics provide mortality benefits in HFrEF; although thiazide diuretics may be used for the adjunctive management of symptomatic congestion and edema, CCBs may lead to worse cardiac outcomes. The potential mortality benefits of GLP-1 receptor agonists in patients with HFrEF are still being investigated. Metformin has been shown to improve mortality in patients with heart failure with preserved ejection fraction (left ventricular ejection fraction >50%).

References

Shchendrygina A, Rakisheva A, Giverts I, Rustamova Y, Soloveva A. Effects of glucagon-like peptide-1 receptor agonists on cardiac function, exercise capacity and quality of life. *Card Fail Rev.* 2024;10:e10.

Halabi A, Sen J, Huynh Q, Marwick TH. Metformin treatment in heart failure with preserved ejection fraction: a systematic review and meta-regression analysis. *Cardiovasc Diabetol.* 2020;19(1):124.

Heidenreich PA, Bozkurt B, Aguilar D, et al. 2022 AHA/ACC/HFSA guideline for the management of heart failure: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. 2022;145(18):e895-e1032.

Item 94

ANSWER: D

According to guidelines from the American Diabetes Association, diabetes mellitus can be diagnosed with a random blood glucose level ≥ 200 mg/dL in patients with classic symptoms of hyperglycemia, such as polyuria and polydipsia, as seen in this patient. An 8-hour fasting plasma glucose level ≥ 126 mg/dL, a hemoglobin A_{1c} $\geq 6.5\%$, or a 2-hour plasma glucose level ≥ 200 mg/dL on 75-g oral glucose tolerance testing can also indicate diabetes but would require confirmatory testing with a second abnormal laboratory value to confirm the diagnosis.

A fasting plasma glucose level 100–125 mg/dL or a hemoglobin A_{1c} 5.7%–6.4% suggests a diagnosis of prediabetes. A 1-hour plasma glucose level ≥ 140 mg/dL on 50-g oral glucose testing is an abnormal first step in a 2-step screening approach for gestational diabetes mellitus in pregnant individuals.

References

Final recommendation statement: prediabetes and type 2 diabetes: screening. US Preventive Services Task Force. August 24, 2021.

Final recommendation statement: gestational diabetes: screening. US Preventive Services Task Force. August 10, 2021.

American Diabetes Association Professional Practice Committee. 2. Diagnosis and classification of diabetes: standards of care in diabetes—2025. *Diabetes Care*. 2025;48(suppl 1):S27-S49.

Item 95**ANSWER: B**

This patient meets the Budapest criteria for complex regional pain syndrome (CRPS) including hyperalgesia, temperature asymmetry, evidence of edema, and motor weakness following trauma to the limb. CRPS is a clinical diagnosis. The absence of fever or other signs of infection makes cellulitis unlikely. The examination findings make factitious disorder and functional neurologic symptom disorder (conversion disorder) unlikely. The absence of the characteristic stocking-glove altered sensation makes peripheral neuropathy unlikely.

References

Harden RN, McCabe CS, Goebel A, et al. Complex regional pain syndrome: practical diagnostic and treatment guidelines, 5th edition. *Pain Med.* 2022;23(suppl 1):S1-S53.

Lloyd ECO, Dempsey B, Romero L. Complex regional pain syndrome. *Am Fam Physician.* 2021;104(1):49-55.

Item 96**ANSWER: A**

This patient's likely diagnosis is pelvic inflammatory disease (PID), and the diagnosis can be made clinically in most cases. In women at risk of PID, treatment should be initiated if any of the following are present: cervical motion tenderness, uterine tenderness, or adnexal tenderness. These criteria are considered sensitive for the diagnosis, but the clinician should be mindful to evaluate for other causes, such as appendicitis or tubo-ovarian abscess (TOA) as indicated by the history and physical examination. Appendicitis would likely cause unilateral pain. TOA often presents acutely with unilateral pain on examination and an adnexal mass. These findings may prompt the clinician to pursue additional testing such as pelvic ultrasonography. A pelvic ultrasound may show fluid-filled or thickened tubes in the setting of PID, but it is not routinely indicated. MRI may be necessary if an ultrasound were to demonstrate TOA or findings other than simple PID. CT of the abdomen may be completed to evaluate complications above the pelvis. Laparoscopy would not be the initial evaluation. This patient is acutely ill and repeating a pregnancy test a week later would not be appropriate.

References

Sexually transmitted infections treatment guidelines, 2021: pelvic inflammatory disease (PID). Centers for Disease Control and Prevention. Reviewed September 21, 2022.

Curry A, Williams T, Penny ML. Pelvic inflammatory disease: diagnosis, management, and prevention. *Am Fam Physician*. 2019;100(6):357-364.

Item 97**ANSWER: D**

In patients presenting with a moderate- to high-risk TIA or minor stroke within 24 hours of symptom onset, dual antiplatelet therapy (DAPT) for 21 days prior to transitioning to a single antiplatelet agent decreases the 90-day stroke risk from 7.8% to 5.2% compared to aspirin alone. This patient has symptoms consistent with a high-risk TIA based on an ABCD² score (age, blood pressure, clinical features of TIA, duration, and diabetes mellitus history) of 4. Recommended management includes loading doses of clopidogrel (300–600 mg) and aspirin (50–325 mg) within 24 hours of symptom onset followed by clopidogrel (75 mg) and aspirin (50–325 mg daily) for 21 days. The patient can then transition to a single antiplatelet agent. Initial treatment with a single antiplatelet agent or with DAPT for less than 21 days does not reduce stroke risk compared to DAPT for 21 days.

References

Mendelson SJ, Prabhakaran S. Diagnosis and management of transient ischemic attack and acute ischemic stroke: a review. *JAMA*. 2021;325(11):1088-1098.

Item 98

ANSWER: A

Treatment of hepatitis C in patients with metabolic dysfunction–associated steatohepatitis (formerly known as nonalcoholic steatohepatitis) can improve fibrotic changes, as can bariatric surgery. Prophylactic antibiotics should be initiated to prevent spontaneous bacterial peritonitis (SBP) in patients with a history of SBP; low ascitic fluid protein levels (<1.5 g/dL); advanced hepatic disease as indicated by a Child-Pugh score ≥ 9 or a bilirubin level ≥ 3 mg/dL; or renal disease as indicated by a serum creatinine level ≥ 1.2 mg/dL or a serum sodium level ≤ 130 mEq/L. Nonselective β -blockers should be started in patients with significant varices seen on upper endoscopy. Coinfection with hepatitis C virus and HIV increases a patient's risk of fibrotic progression to cirrhosis. A conversation regarding risks and benefits of initiating pre-exposure prophylaxis (PrEP) would be beneficial; however, PrEP has not been shown to improve fibrotic changes. Offering treatment for viral hepatitis along with PrEP increases access for patients who are at risk of coinfection.

In individuals with metabolic dysfunction–associated steatotic liver disease (MASLD) (formerly known as nonalcoholic fatty liver disease) and cirrhosis, antibiotic prophylaxis is recommended in specific scenarios: in patients with a history of SBP, during episodes of acute upper gastrointestinal bleeding, and in select high-risk patients with low ascitic fluid protein levels (<1.5 g/dL) along with either advanced liver disease (Child-Pugh score ≥ 9 and a bilirubin level ≥ 3 mg/dL) or signs of renal impairment (a creatinine level ≥ 1.2 mg/dL, a BUN level ≥ 25 mg/dL, or a sodium level ≤ 130 mEq/L).

References

Smith A, Baumgartner K, Bositis C. Cirrhosis: diagnosis and management. *Am Fam Physician*. 2019;100(12):759-770.

Differentiated and simplified pre-exposure prophylaxis for HIV prevention: update to WHO implementation guidance. World Health Organization; July 27, 2022.

Biggins SW, Angeli P, Garcia-Tsao G, et al. Diagnosis, evaluation, and management of ascites, spontaneous bacterial peritonitis and hepatorenal syndrome: 2021 practice guidance by the American Association for the Study of Liver Diseases. *Hepatology*. 2021;74(2):1014-1048.

Item 99

ANSWER: B

Pelvic organ prolapse is a common problem, and can be described as apical, of the anterior wall, or of the posterior wall. The Pelvic Organ Prolapse Quantification system can be used to evaluate the prolapse and determine grading. A grade 2 prolapse indicates a prolapse that extends within 1 cm above or below the hymenal ring. Pelvic floor muscle training can reduce prolapse by 1–2 cm and provide subjective improvement in symptoms. There is no clear evidence of the benefit of estrogen alone for pelvic organ prolapse. Surgical management is often the ultimate treatment for pelvic organ prolapse, but it is important to note that urinary incontinence can worsen after repair. This patient is minimally bothered by her symptoms; therefore, surgical treatment would not be recommended at this time but may be indicated in the future. Pelvic organ prolapse is generally progressive until menopause, after which it can alternate between regression and progression.

References

Iglesia CB, Smithling KR. Pelvic organ prolapse. *Am Fam Physician*. 2017;96(3):179-185.

Taithongchai A, Johnson EE, Ismail SI, Barron-Millar E, Kernohan A, Thakar R. Oestrogen therapy for treating pelvic organ prolapse in postmenopausal women. *Cochrane Database Syst Rev*. 2023;7(7):CD014592.

Qiu J, Jiang D. Pessaries for managing pelvic organ prolapse in women. *Am Fam Physician*. 2021;103(11):660-661.

Item 100

ANSWER: E

This patient's presentation is concerning for acute limb ischemia and he requires immediate transfer to an emergency facility with vascular surgeon accessibility (SOR C). Anticoagulation with unfractionated heparin should be started soon after arrival. Acute limb ischemia is a vascular emergency as skeletal muscle can only survive 4–6 hours without blood flow. Acute limb ischemia is characterized by the 6 *P*'s: pain, pallor, poikilothermia, pulselessness, paresthesia, and paralysis.

Direct-acting oral anticoagulants can be used to manage symptomatic peripheral artery disease (PAD), but not for acute management of limb ischemia. While duplex ultrasonography or CT angiography can be used to determine the presence and location of the obstructed vessel, transfer to an emergency facility should not be delayed for imaging. Referral to a vascular surgeon would delay treatment and would be more appropriate for chronic PAD symptoms.

References

Rosenberg H, Rosenberg E, Kubelik D. Acute limb ischemia. *CMAJ*. 2023;195(40):E1383.

Olinic DM, Stanek A, Tătaru DA, Homorodean C, Olinic M. Acute limb ischemia: an update on diagnosis and management. *J Clin Med*. 2019;8(8):1215.

Firnhaber JM, Powell CS. Lower extremity peripheral artery disease: diagnosis and treatment. *Am Fam Physician*. 2019;99(6):362-369.

Item 101**ANSWER: D**

Bone stress injuries are typically overuse injuries that are caused by an accumulation of bone microdamage. They often occur after an increase in the intensity, frequency, or interval of training. They can also occur when there is structurally compromised bone (osteoporosis, tumors, infections) as normal physiologic loads can damage such compromised bone. Radiography is the initial imaging modality because of its high specificity, despite its low sensitivity. If the radiograph is negative and the patient remains symptomatic, then MRI is the preferred follow-up study for definitive diagnosis. It is preferred over CT and nuclear bone scintigraphy because it has higher sensitivity and specificity and avoids the use of radiation. Repeat radiography is not the preferred definitive imaging modality because radiographs are often negative when taken within the first 2–3 weeks of symptoms. Point-of-care ultrasonography can be used by trained clinicians as an adjunct modality, but its sensitivity is not as high as the sensitivity of MRI.

References

Schroeder JD, Trigg SD, Capo Dosal GE. Bone stress injuries: diagnosis and management. *Am Fam Physician*. 2024;110(6):592-600.

Item 102**ANSWER: B**

The first-line treatment of chronic urticaria is an oral second-generation H₁-antihistamine. If that is not effective, a reasonable next step is to increase the H₁-antihistamine, up to 4 times the usual FDA-recommended dosing. Current evidence does not strongly support the effectiveness of a pseudoallergen-free diet in managing chronic urticaria. Nasal corticosteroids are effective for allergic rhinitis but have not been found to be useful for chronic urticaria. A short course of oral corticosteroids, such as prednisone, can be effective as a short-term solution for a flare-up but is not recommended for chronic control. Cyclosporine is only recommended for patients with severe refractory symptoms that have not responded to increased dosages of second-generation antihistamines and a trial of H₂-blockers. Omalizumab is effective for chronic urticaria and would be indicated if aggressive antihistamine therapy is unsuccessful.

References

Schaefer P. Acute and chronic urticaria: evaluation and treatment. *Am Fam Physician*. 2017;95(11):717-724.

Zuberbier T, Aberer W, Asero R, et al. The EAACI/GA²LEN/EDF/WAO guideline for the definition, classification, diagnosis and management of urticaria. *Allergy*. 2018;73(7):1393-1414.

Item 103**ANSWER: C**

It is difficult to adequately distinguish between viral sinusitis and acute bacterial rhinosinusitis. However, most instances of acute bacterial rhinosinusitis improve without antibiotics. Major symptoms of sinusitis include purulent nasal discharge, nasal congestion, facial pain or pressure, decreased smell, and fever. Minor symptoms include headaches, cough, fatigue, ear pain or pressure, halitosis, and tooth pain.

If adequate follow-up can be ensured, watchful waiting and symptomatic management is recommended for up to 10 days. Patients with severe symptoms (fever $\geq 39^{\circ}\text{C}$ with purulent nasal discharge and/or facial pain) or those with double sickening (new symptoms following improvement of an upper respiratory infection) should be treated immediately with empiric antibiotics.

References

Chow AW, Benninger MS, Brook I, et al. IDSA clinical practice guideline for acute bacterial rhinosinusitis in children and adults. *Clin Infect Dis*. 2012;54(8):e72-e112.

Item 104**ANSWER: C**

Lung cancer is the leading cause of cancer death in the United States, with approximately 90% of cases linked to smoking. In 2021, the U.S. Preventive Services Task Force recommended annual screening with low-dose CT in adults 50–80 years of age who have a 20-pack-year smoking history and either currently smoke or have quit within the past 15 years.

References

Final recommendation statement: lung cancer: screening. US Preventive Services Task Force. Updated March 9, 2021.

Item 105

ANSWER: A

This patient has nongonococcal urethritis, which should be treated with azithromycin. Symptoms of urethritis include dysuria, purulent or mucopurulent urethral discharge, and urethral discomfort and/or erythema. Diagnostic criteria include typical signs and symptoms or a history of exposure along with ≥ 10 WBCs/hpf on a first-void urinalysis, ≥ 2 WBCs/oil immersion field found on Gram stain of the discharge, or a positive leukocyte esterase with first-void urine.

At least half of nongonococcal urethritis cases have no known etiologic agent. Of the cases with known etiologies, *Mycoplasma genitalium* is responsible for 15%–20%. Treatment options for *M genitalium* include azithromycin, 1 g orally; doxycycline, 100 mg twice daily for 1 week; or moxifloxacin, 400 mg orally for 7–14 days.

If the results of nucleic acid amplification testing for gonorrhea and chlamydia are unavailable, then intramuscular ceftriaxone plus a 7-day course of oral doxycycline would be appropriate and can be used for empiric therapy while awaiting results. Intramuscular gentamicin plus oral azithromycin is used as an alternative treatment regimen for *Neisseria gonorrhoeae*. Metronidazole is used to treat *Trichomonas*, which represents 1%–8% of cases in men and should be diagnosed using a molecular-based test. Levofloxacin is a treatment option for chlamydia and is not first line.

References

Sell J, Nasir M, Courchesne C. Urethritis: rapid evidence review. *Am Fam Physician*. 2021;103(9):553-558.

Diseases characterized by urethritis and cervicitis. Sexually Transmitted Infections Treatment Guidelines, 2021. Centers for Disease Control and Prevention. Reviewed September 21, 2022.

Mycoplasma genitalium. Sexually Transmitted Infections Treatment Guidelines, 2021. Centers for Disease Control and Prevention. Reviewed July 22, 2021.

Item 106

ANSWER: C

This patient's clinical picture is most consistent with a subarachnoid hemorrhage. Patients typically describe this as a sudden onset of the worst headache they have experienced. The most immediate and meaningful next step is to order a CT scan of the head to verify the diagnosis of subarachnoid hemorrhage. The diagnosis and management of subarachnoid hemorrhage is time sensitive. Neurosurgery should be consulted immediately if subarachnoid hemorrhage is verified. Brain MRI would likely take longer to obtain than head CT and is not diagnostically advantageous in subarachnoid hemorrhage compared to a head CT. Pain and nausea control are reasonable, but a CT scan of the head should take precedence because of the time sensitivity of diagnosis and management. A lumbar puncture should be considered if the head CT scan is negative but clinical suspicion for subarachnoid hemorrhage remains high, especially if the presentation occurs >6 hours after onset.

References

Edlow J. Diagnosing subarachnoid hemorrhage. *J Emerg Med*. 2021;61(5):623-624.

Claassen J, Park S. Spontaneous subarachnoid haemorrhage. *Lancet*. 2022;400(10355):846-862.

Arnold MJ. Management of patients with aneurysmal subarachnoid hemorrhage: guidelines from the AHA and ASA. *Am Fam Physician*. 2024;110(2):204-206.

Hoh BL, Ko NU, Amin-Hanjani S, et al. 2023 guideline for the management of patients with aneurysmal subarachnoid hemorrhage: a guideline from the American Heart Association/American Stroke Association. *Stroke*. 2023;54(7):e314-e370.

Item 107**ANSWER: B**

This patient has cannabinoid hyperemesis syndrome, characterized by heavy chronic cannabis use and nausea that improves with hot baths or showers. Current evidence suggests that haloperidol is more effective than more traditionally used antiemetics, such as metoclopramide or prochlorperazine, or corticosteroids in managing nausea associated with this syndrome. Topical capsaicin has also been shown to be effective. Topical capsaicin, when applied to the upper abdomen, may help alleviate symptoms by activating transient receptor potential vanilloid 1 (TRPV1) receptors. This mechanism can lead to desensitization of sensory nerve fibers, potentially reducing the need for antiemetic medications. Benzodiazepines may help acutely, and tricyclic antidepressants may have efficacy for longer-term management. Cessation of cannabis use is necessary for definitive treatment. Diclofenac gel is not appropriate pharmacotherapy for cannabinoid hyperemesis syndrome.

References

Rubio-Tapia A, McCallum R, Camilleri M. AGA clinical practice update on diagnosis and management of cannabinoid hyperemesis syndrome: commentary. *Gastroenterology*. 2024;166(5):930-934.e1.

Perisetti A, Gajendran M, Dasari CS, et al. Cannabis hyperemesis syndrome: an update on the pathophysiology and management. *Ann Gastroenterol*. 2020;33(6):571-578.

Rubio-Tapia A, McCallum R, Camilleri M. AGA clinical practice update on diagnosis and management of cannabinoid hyperemesis syndrome: commentary. *Gastroenterology*. 2024;166(5):930-934.e1.

Item 108**ANSWER: B**

Duloxetine is an SNRI that is effective for treating fibromyalgia pain (SOR A). Other medications with level A evidence of effectiveness in reducing fibromyalgia-related pain include amitriptyline, cyclobenzaprine, and pregabalin. The SNRI milnacipran also has level A evidence for reducing fibromyalgia-related pain and has evidence for reducing fatigue and improving cognition in patients with fibromyalgia. Other SNRIs, including venlafaxine, have not been studied as extensively. Current evidence is insufficient to support acetaminophen or gabapentin use to treat fibromyalgia pain. Similarly, there is insufficient evidence to support the routine use of opioids such as hydrocodone, which are associated with a significant risk of adverse effects, dependence, misuse, and worsening hyperalgesia. NSAIDs such as ibuprofen are no better than placebo and introduce significant risk of adverse effects.

References

Winslow BT, Vandal C, Dang L. Fibromyalgia: diagnosis and management. *Am Fam Physician*. 2023;107(2):137-144.

Item 109**ANSWER: A**

Behavioral methods and enuresis alarms are first-line treatments for monosymptomatic nocturnal enuresis without an identifiable cause. If pharmacotherapy is initiated, intranasal or oral desmopressin is acceptable. Imipramine and oxybutynin are additional options if there is no response to oral desmopressin. Tamsulosin is not recommended for this indication.

References

Lauters RA, Garcia KW, Arnold JJ. Enuresis in children: common questions and answers. *Am Fam Physician*. 2022;106(5):549-556.

Item 110**ANSWER: A**

Stimulants are first-line pharmacotherapy for adults with attention-deficit/hyperactivity disorder (ADHD) given the extensive data supporting their effectiveness. However, there are potential adverse effects that can prevent patients from taking them, such as headaches, insomnia, poor appetite, weight loss, anxiety, depression, elevated heart rate, and elevated blood pressure. When stimulants are not tolerated or do not provide an adequate response, the second-line therapy would be an SNRI such as atomoxetine or viloxazine. Clonidine and guanfacine are only approved for the treatment of ADHD in children and adolescents. Although they are sometimes used off-label in adults, there is limited evidence to support such use. Tricyclic antidepressants such as nortriptyline have limited evidence and are not approved for ADHD treatment. Escitalopram is not effective for treating ADHD.

References

Olagunju AE, Ghoddusi F. Attention-deficit/hyperactivity disorder in adults. *Am Fam Physician*. 2024;110(2):157-166.

Item 111**ANSWER: D**

Although anterior shoulder dislocations have a high rate of recurrence (up to 90%), evidence from cohort studies has consistently found a lower rate of recurrence in active patients <25 years of age who have received surgical treatment for first-time dislocations. Because of this patient's age and activity level, referral for surgical repair would be appropriate. Other treatment options for anterior shoulder dislocation include the use of a sling for 2–4 weeks with graduated rehabilitation of passive and active range of motion. If no surgery is warranted, athletes can return to play when they are pain free with symmetrical shoulder range of motion and able to perform sport-specific motions, which can be as early as 2–3 weeks after injury. Intra-articular corticosteroids are not advised after a shoulder dislocation.

References

Simon LM, Nguyen V, Ezinwa NM. Acute shoulder injuries in adults. *Am Fam Physician*. 2023;107(5):503-512.

Item 112

ANSWER: A

Mastitis is a spectrum of inflammatory breast conditions, which includes lactational or inflammatory mastitis, bacterial mastitis, breast abscess, postpartum breast engorgement, and more. Patients with inflammatory mastitis exhibit unilateral breast erythema, swelling, and pain and may experience systemic inflammatory responses such as fever, malaise, or headache. The initial steps in managing inflammatory mastitis are supportive care with NSAIDs, cold compresses, and watchful waiting. Inflammatory mastitis is diagnosed clinically and does not require laboratory testing or imaging.

If a patient's symptoms worsen or do not improve within 24–48 hours with conservative measures, narrow-spectrum antibiotics targeting the most common organisms (*Staphylococcus* species and *Streptococcus* species) should be empirically initiated for bacterial mastitis. Warm compresses and frequent breast pumping for complete breast emptying to avoid milk stasis are no longer recommended. CBC and C-reactive protein testing are not helpful as these levels can be elevated in both inflammatory and bacterial mastitis. Breast milk cultures are seldom necessary because they may produce false-positive results caused by normal skin bacteria colonization.

References

Mitchell KB, Johnson HM, Rodríguez JM, et al. Academy of Breastfeeding Medicine Clinical Protocol #36: the mastitis spectrum, revised 2022. *Breastfeed Med*. 2022;17(5):360-376.

Lawrence KV. Supporting the paradigm shift: watchful waiting for lactational mastitis. *Am Fam Physician*. 2024;110(2):118-119.

Morcomb EF, Dargel CM, Anderson SA. Mastitis: rapid evidence review. *Am Fam Physician*. 2024;110(2):174-182.

Item 113**ANSWER: B**

Cystic fibrosis is a genetic disorder with an autosomal recessive pattern of inheritance. People who are affected by the disease are homozygous for the condition, with 2 copies of the mutated gene. Carriers are heterozygous for the condition with 1 mutated gene and 1 normal gene. When both partners in a couple are carriers, any potential offspring has a 25% chance of being affected, a 50% chance of being an unaffected carrier, and a 25% chance of being an unaffected noncarrier.

References

Dictionary of genetics terms: autosomal recessive inheritance. National Cancer Institute at the National Institutes of Health. Accessed February 16, 2025.

Collins MS, Mansilla-Rivera K. Diagnostic and therapeutic advances in cystic fibrosis: how family physicians can partner in care. *Am Fam Physician*. 2024;109(5):388-390.

Cystic fibrosis: causes. National Heart, Lung, and Blood Institute. Updated November 15, 2024.

Item 114**ANSWER: C**

Based on the National Academy of Medicine criteria for myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS), this patient has demonstrated the core symptoms of ME/CFS (unrefreshing sleep, new-onset fatigue, and postexertional malaise) for at least 6 months, in addition to brain fog and lightheadedness. Notably, their symptoms developed after a previous COVID-19 infection, an illness that is commonly associated with ME/CFS.

A Patient Health Questionnaire–9 (PHQ-9) score of 3 indicates no or minimal depression, which would not account for severe fatigue or energy crashes after minimal exertion. Exercise may actually improve depression. Secondary causes of fatigue, such as fibromyalgia and systemic lupus erythematosus, typically improve with exercise. Although obstructive sleep apnea may cause unrefreshing sleep, it is not associated with profound fatigue and postexertional malaise.

References

Latimer KM, Gunther A, Kopec M. Fatigue in adults: evaluation and management. *Am Fam Physician*. 2023;108(1):58-69.

Item 115

ANSWER: B

Heart failure with reduced ejection fraction (HFrEF) is a major public health concern affecting more than 6.5 million U.S. adults. Family physicians should be familiar with its typical clinical presentations and diagnosis, which is best established using echocardiography. Symptoms vary and may include dyspnea, ankle edema, and fatigue. Orthopnea, as evident by modification of sleeping from supine to reclined or semi-reclined, is also common. Symptoms of right-sided heart failure, such as early satiety, abdominal bloating, ill-fitting clothes, and shortness of breath when bent forward, are also frequently encountered, but are less specific. Physical examination may reveal signs of congestion such as elevated jugular venous pressure, S₃ gallop, or a laterally displaced apical impulse. Some predictors of HFrEF include increased heart rate, coronary artery disease (hazard ratio [HR] 1.73), and previous myocardial infarction (MI) (HR 3.49).

The diagnosis is made by documenting left ventricular systolic dysfunction with a left ventricular ejection fraction $\leq 40\%$ and an elevated BNP level. An EKG is a key component of the workup and evaluation of contributing factors, such as arrhythmia or acute MI, but not diagnosis. A chest x-ray, while quick and readily available, may identify cardiomegaly or vascular congestion but is normal in approximately 20% of those with acute HFrEF, limiting diagnostic utility. Given this patient's history of MI, CT angiography would be inappropriate and is typically reserved for evaluating coronary artery disease in low-risk patients. CT angiography is used to evaluate for pulmonary embolism, which is less likely, given this patient's lack of immobility, prolonged travel, or other risk factors, as well as a history and physical examination that favor heart failure (displaced apical impulse, orthopnea, abdominal distention/weight gain).

References

Heidenreich PA, Bozkurt B, Aguilar D, et al. 2022 AHA/ACC/HFSA guideline for the management of heart failure: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2022;145(18):e895-e1032.

Murphy SP, Ibrahim NE, Januzzi JL. Heart failure with reduced ejection fraction: a review. *JAMA*. 2020;324(5):488–504.

Item 116

ANSWER: A

The U.S. Preventive Services Task Force (USPSTF) recommends vision screening at least once in all children 3–5 years of age to detect amblyopia or its risk factors (B recommendation). Screening may be performed using visual acuity testing with picture identification, Snellen or "tumbling E," or by an automated device that assesses visual acuity. Early identification of vision disorders can help prevent permanent vision loss due to amblyopia.

Anemia screening is recommended for all children at 1 year of age by the American Academy of Pediatrics (AAP), but this approach was found to have insufficient evidence of benefit by the USPSTF (I recommendation). Screening for autism spectrum disorder is recommended by the AAP at 18 and 24 months of age, but the USPSTF found insufficient evidence that universal screening results in improved care in families without concerns about their child's development (I recommendation). The USPSTF found insufficient evidence of benefit of screening for dental caries in children <5 years of age (I recommendation). Instead, it recommends that fluoride varnish be applied every 3–6 months in all children 6 months to 5 years of age to prevent dental caries (B recommendation). This should be performed in the family physician's office if it has not been applied by a dentist. Additionally, an oral fluoride supplement is recommended to prevent dental caries in children without adequate fluoride in their water supply (B recommendation). Universal hearing screening is recommended for newborns, but screening at this age is not recommended.

References

Final recommendation statement: autism spectrum disorder in young children: screening. US Preventive Services Task Force. Updated February 16, 2016.

Hyman SL, Levy SE, Myers SM, et al; Council on Children with Disabilities, Section on Developmental and Behavioral Pediatrics. Identification, evaluation, and management of children with autism spectrum disorder. *Pediatrics*. 2020;145(1):e20193447.

Baker RD, Greer FR; Committee on Nutrition. Diagnosis and prevention of iron deficiency and iron-deficiency anemia in infants and young children (0–3 years of age). *Pediatrics*. 2010;126(5):1040-1050.

Final recommendation statement: iron deficiency anemia in young children: screening. US Preventive Services Task Force. Updated September 7, 2015.

Final recommendation statement: prevention of dental caries in children younger than 5 years: screening and interventions. US Preventive Services Task Force. Updated December 7, 2021.

Final recommendation statement: vision in children ages 6 months to 5 years: screening. US Preventive Services Task Force. Updated September 5, 2017.

Item 117**ANSWER: D**

While counseling on diet and exercise and screening for diabetes mellitus and hyperlipidemia are important, malignancy screening is the most important component of the wellness visit in this patient with dermatomyositis diagnosed after the age of 40. Cancer is a leading cause of death in this patient population and is diagnosed in up to 25% of these patients within 3 years of their dermatomyositis diagnosis. The most likely cancers in this population include adenocarcinomas of the ovary, lung, pancreas, stomach, and colon as well as non-Hodgkin lymphoma. Population-level cancer screening recommendations are generally insufficient for this population, and advanced cancer screening modalities, including CT scans and PET scans, may be recommended based on the patient's individual risk factors to target these types of cancers.

Chronic kidney disease and obesity have not been observed to have an increased risk in patients with dermatomyositis.

References

Vaughan H, Rugo HS, Haemel A. Risk-based screening for cancer in patients with dermatomyositis: toward a more individualized approach. *JAMA Dermatol*. 2022;158(3):244-247.

Oldroyd AGS, Callen JP, Chinoy H, et al; International Myositis Assessment and Clinical Studies Group Cancer Screening Expert Group. International guideline for idiopathic inflammatory myopathy-associated cancer screening: an International Myositis Assessment and Clinical Studies Group (IMACS) initiative. *Nat Rev Rheumatol*. 2023;19(12):805-817.

Item 118

ANSWER: A

Anchoring bias describes the practice of physicians prioritizing information and data that supports their initial impressions, even when these impressions are incorrect. In other words, it is when physicians are biased toward their first impression. *Availability bias* describes the likelihood of a diagnosis being erroneously linked to how easy it is to think of examples. *Implicit bias* is defined as attitudes and beliefs about race, ethnicity, age, ability, gender, or other characteristics. Implicit bias operates on an unconscious level and can only be measured indirectly. *Outcomes bias* refers to the attribution of all results to prior decisions, regardless of whether there is a reason to do so.

References

Doherty TS, Carroll AE. Believing in overcoming cognitive biases. *AMA J Ethics*. 2020;22(9):E773-E778.

Sabin JA. Tackling implicit bias in health care. *N Engl J Med*. 2022;387(2):105-107.

Catalogue of Bias Collaboration; Banerjee A, Nunan D. Availability bias. Catalogue of Bias. 2019.

Item 119**ANSWER: A**

Heavy menstrual bleeding (HMB) is very common in adolescents and may be a sign of an underlying bleeding disorder. Although bleeding disorders affect 1%–2% of the general population, the prevalence increases to 20% among adolescent girls with heavy menstrual bleeding. Platelet dysfunction and von Willebrand disease are most commonly identified. Evaluation for bleeding disorders is indicated when there is frequent pad or tampon use (>1 every 1–2 hours); menstrual bleeding lasting longer than 7 days; passage of clots larger than 1 inch in diameter; a personal history of anemia treatment, excessive bleeding with tooth extraction or surgery, or excessive bruising; or a family history of a bleeding disorder.

The initial workup for HMB in an adolescent includes a CBC; a prothrombin time, partial thromboplastin time, and INR; a von Willebrand panel; and hCG, ferritin, fibrinogen, and TSH levels. Testing for chlamydia and gonorrhea should be considered in sexually active patients. Testosterone, prolactin, and DHEA levels should be obtained if there are signs of polycystic ovary syndrome or other endocrine disorders.

Levels of FSH, LH, estrogen, and progesterone are not helpful in the evaluation of excessive uterine bleeding. Pelvic imaging is not routinely necessary but should be considered for HMB that does not respond to initial management. Hormonal contraceptives and tranexamic acid may be used for the treatment of HMB, but workup for a bleeding disorder is the most important first step.

References

Oleka C, Dietrich JE. HMB in the adolescent: a review of the modern approach to diagnosis and management. *Clin Obstet Gynecol*. 2020;63(3):553-560.

Item 120

ANSWER: B

This patient presents with symptoms of dyspepsia. Dyspepsia is a diagnosis of exclusion and should be considered if at least 1 of the following symptoms is present: early satiety, epigastric pain, epigastric burning, and postprandial fullness. The majority of patients diagnosed with dyspepsia have functional dyspepsia with no evidence of organic disease on endoscopy. Testing for and treating *Helicobacter pylori* infection is recommended in patients with dyspepsia, with evidence of improvement 1 year after treatment. The stool antigen test and the urea breath test are the preferred noninvasive options for detecting *H pylori*. Serum testing for *H pylori* is not recommended because it has lower sensitivity and specificity than other noninvasive tests.

Sucralfate has not been shown to be effective for dyspepsia. Abdominal ultrasonography, particularly of the upper abdomen, is indicated when evaluating for liver pathology or gallbladder disease but is not useful for investigating dyspepsia. Endoscopy is recommended for patients >60 years of age with new dyspepsia, and for patients <60 years of age with multiple or severe alarm symptoms, such as marked weight loss, rapidly progressive dysphagia, an epigastric mass, or supraclavicular lymphadenopathy.

References

McConaghy JR, Decker A, Nair S. Peptic ulcer disease and H. pylori infection: common questions and answers. *Am Fam Physician*. 2023;107(2):165-172.

Moayyedi PM, Lacy BE, Andrews CN, Enns RA, Howden CW, Vakil N. Corrigendum: ACG and CAG clinical guideline: management of dyspepsia. *Am J Gastroenterol*. 2017;112(9):1484. doi:10.1038/ajg.2017.238

Du LJ, Chen BR, Kim JJ, Kim S, Shen JH, Dai N. Helicobacter pylori eradication therapy for functional dyspepsia: systematic review and meta-analysis. *World J Gastroenterol*. 2016;22(12):3486-3495.

Mounsey A, Barzin A, Rietz A. Functional dyspepsia: evaluation and management. *Am Fam Physician*. 2020;101(2):84-88.

Item 121**ANSWER: B**

Hypercalcemia is commonly identified through a metabolic panel and should be confirmed with a corrected or ionized calcium level. The most common cause of hypercalcemia is primary hyperparathyroidism, but any underlying malignancy should be ruled out. A thorough history should assess for symptoms such as arrhythmias, hypertension, abdominal pain, arthralgias, fractures, mental status changes, anxiety, depression, and nephrolithiasis. Dietary intake of calcium and phosphorus, use of medications such as lithium, and a family history of multiple endocrine neoplasia are also important to evaluate.

After confirming hypercalcemia, the initial test should be a parathyroid hormone (PTH) level. If the PTH level is elevated, then 25-hydroxyvitamin D, creatinine, and 24-hour urinary calcium levels should be measured in addition to ordering a bone density test. Imaging of the parathyroid glands is not necessary for diagnosis, but is useful if surgical management is indicated. A low PTH level could suggest malignancy and may indicate the need for a CT scan.

References

Sell J, Ramirez S, Partin M. Parathyroid disorders. *Am Fam Physician*. 2022;105(3):289-298.

Item 122**ANSWER: C**

This patient presents with a relatively acute onset of facial weakness affecting only the right side of the face, but she is able to close her right eye and raise her eyebrows. Sparing of the forehead muscles is consistent with a supranuclear lesion because the nucleus of cranial nerve VII receives fibers from both sides of the cortex for the upper face, meaning the forehead would be unaffected. Cranial nerve VII includes visceral efferent fibers (facial muscles and stapedius), visceral motor fibers (salivary and lacrimal glands), and special sensory fibers (taste to the anterior two-thirds of the tongue). In Bell palsy, which involves edema of this cranial nerve, the entire side of the face is affected, including the forehead. Therefore, this presentation is not consistent with Bell palsy and warrants further evaluation, including brain MRI, to assess for a central cause.

Prednisone and valacyclovir are not appropriate for this patient, as they are treatments used for Bell palsy. Plasmapheresis is not indicated for Bell palsy, stroke, or tumor. Surgical decompression is not indicated.

References

Dalrymple SN, Row JH, Gazewood J. Bell palsy: rapid evidence review. *Am Fam Physician*. 2023;107(4):415-420.

Item 123

ANSWER: D

In an older adult, head trauma with a loss of consciousness, vomiting, anterograde amnesia, or a previous history of head trauma requires CT of the head. Being older than 65 years is considered an independent risk factor for head trauma. Older adults with a traumatic brain injury experience higher morbidity and mortality due to age-related changes in vasculature that leave vessels more vulnerable to rupture and changes in white matter that render white matter tracts more susceptible to shear injury. As a result of these physiological changes, older patients are more susceptible to injury, and even those with minor mechanisms of injury and no abnormalities on a neurologic examination may still have subdural or epidural bleeding. NSAIDs carry a risk of bleeding and should be avoided following a traumatic brain injury.

References

Gardner RC, Dams-O'Connor K, Morrissey MR, Manley GT. Geriatric traumatic brain injury: epidemiology, outcomes, knowledge gaps, and future directions. *J Neurotrauma*. 2018;35(7):889-906.

Pahor M. Falls in older adults: prevention, mortality, and costs. *JAMA*. 2019;321(21):2080-2081.

Mori K, Abe T, Matsumoto J, Takahashi K, Takeuchi I. Indications for computed tomography in older adult patients with minor head injury in the emergency department. *Acad Emerg Med*. 2021;28(4):435-443.

Item 124**ANSWER: A**

Capillary refill time is the most reliable clinical sign for assessing dehydration in infants. An abnormal capillary refill time (>2 seconds) is a strong indicator of dehydration and has been shown to have a high likelihood ratio for predicting dehydration in children. While decreased urine output is a sensitive indicator of dehydration, it is not specific and can be influenced by other factors such as feeding patterns and ambient temperature. Dry mucous membranes are a useful sign but can be subjective and less reliable compared to capillary refill time. Abnormal skin turgor is more commonly used in older children and is less reliable in neonates due to their higher skin elasticity. A sunken fontanelle can indicate dehydration, but it is less specific and can be influenced by other factors, such as the patient's position during the examination.

References

Steiner MJ, DeWalt DA, Byerley JS. Is this child dehydrated?. *JAMA*. 2004;291(22):2746-2754.

Item 125**ANSWER: C**

Two SSRIs, fluoxetine and escitalopram, have been approved by the FDA for the treatment of depression in adolescents. Other SSRIs such as sertraline, NDRI such as bupropion, and SNRIs such as venlafaxine are often used off-label for this purpose but have not been approved by the FDA for use in adolescents. Esketamine is a ketamine nasal spray used in conjunction with an oral antidepressant for treatment-resistant depression in adults.

References

FDA approves new nasal spray medication for treatment-resistant depression; available only at a certified doctor's office or clinic. PR Newswire, 2019.

Miller L, Campo JV. Depression in adolescents. *N Engl J Med*. 2021;385(5):445-449.

Item 126**ANSWER: E**

Acute bronchitis is a lower respiratory tract infection that is most commonly caused by a virus. Treatment is generally supportive as it is a self-limited illness. Guaifenesin, honey, and dextromethorphan have been shown to decrease the frequency of cough (SOR B). In otherwise healthy patients without evidence of pneumonia, acute bronchitis can last 2–3 weeks. Patients may exhibit purulent sputum, rhonchi, and wheezing with acute bronchitis but are typically afebrile with normal oxygen saturation.

β_2 -Agonists such as albuterol are not recommended for the treatment of acute bronchitis unless wheezing is present (SOR B). While 90% of acute bronchitis cases are caused by viruses, a significant percentage are treated inappropriately with antibiotics. Antibiotics can decrease the duration of cough by 0.46 days, but the overall net benefit of antibiotics is negligible and often outweighed by potential side effects (SOR A). Studies show codeine has no benefit in the treatment of acute bronchitis and has rather significant side effects.

References

Smith SM, Fahey T, Smucny J, Becker LA. Antibiotics for acute bronchitis. *Cochrane Database Syst Rev*. 2017;6(6):CD000245.

Kinkade S, Long NA. Acute bronchitis. *Am Fam Physician*. 2016;94(7):560-565.

Killeen BM, Wolfson AB. Antibiotics for acute bronchitis. *Am Fam Physician*. 2020;102(9):online.

Item 127**ANSWER: D**

When a foot or ankle fracture is suspected, a weight-bearing foot and ankle x-ray should be obtained. The Ottawa foot and ankle rules are a useful tool for determining the need for an x-ray for an acute ankle inversion injury. According to the Ottawa foot and ankle rules, a foot x-ray should be obtained if any of the following requirements are met: bone tenderness at the posterior edge or the tip of the lateral or medial malleolus, bone tenderness at the base of the fifth metatarsal or navicular bone, or inability to bear weight on the affected leg or foot for at least 4 steps after the injury.

References

Silver S, Williams E, Plunkett ML. Common foot fractures. *Am Fam Physician*. 2024;109(2):119-129.

Item 128**ANSWER: A**

This patient meets 3 of the Rome IV diagnostic criteria for functional constipation (≤ 2 bowel movements per week, painful hard stools, and stools that clog the toilet). There is good evidence that polyethylene glycol is effective for treating functional constipation in children and adolescents. There is no evidence that probiotics are effective. This patient does not have any red flag symptoms that would prompt a need for either magnetic resonance enterography to evaluate for inflammatory bowel disease or a colonoscopy to evaluate for a structural lesion.

References

Mulhem E, Khondoker F, Kandiah S. Constipation in children and adolescents: evaluation and treatment. *Am Fam Physician*. 2022;105(5):469-478.

Item 129**ANSWER: A**

Patients 40–75 years of age with diabetes mellitus and an LDL-cholesterol level ≥ 70 mg/dL should be treated with a moderate-intensity statin without calculating a 10-year risk of atherosclerotic cardiovascular disease. However, high-intensity statin therapy should be considered to lower the LDL-cholesterol level by at least 50% for patients such as this one who are at elevated risk, especially those 50–75 years of age or with multiple risk factors.

Of the options listed, atorvastatin, 40–80 mg daily, is the only therapy that is considered high-intensity statin therapy. Pravastatin, 20 mg daily, is low-intensity statin therapy. Simvastatin, 20–40 mg daily, is considered moderate-intensity therapy and is not recommended at doses >40 mg daily. Ezetimibe and fenofibrate are not recommended first-line medications for the treatment of hyperlipidemia and the prevention of cardiovascular disease.

References

Grundy SM, Stone NJ, Bailey AL, et al. 2018
AHA/ACC/AACVPR/AAPA/ABC/ACPM/ADA/AGS/APhA/ASPC/NLA/PCNA guideline on the management of blood cholesterol: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Circulation*. 2019;139(25):e1082-e1143.

Item 130

ANSWER: E

Tarsal tunnel syndrome is an entrapment neuropathy of the foot caused by compression of the tibial nerve or one of its branches in the fibro-osseous tunnel under the flexor retinaculum and within the abductor hallucis muscle of the foot. It may occur following an injury, such as a fracture or sprain of the ankle. Patients can present with paresthesias, local pain behind the medial malleolus, and loss of sensation in the sole. Paresis of the small muscle of the foot can lead to impaired abduction.

There is no criterion standard for the diagnosis of tarsal tunnel syndrome. The most helpful clinical diagnostic sign is a positive Tinel sign at the ankle, and electrophysiologic studies can help in making the diagnosis. MRI can be useful in cases of compression due to a mass lesion. Patients can begin a trial of conservative management, but advanced cases are better treated surgically.

Achilles tendinopathy causes posterior heel pain that is localized at the site of insertion of the Achilles tendon and worsens with increased activity or pressure on the area. Haglund deformity presents with pain in the upper part of the posterior calcaneus and is more common in middle-aged women who repeatedly wear ill-fitting shoes. X-rays demonstrate the deformity on the superior aspect of the posterior calcaneus. Patients with heel pad syndrome report a deep-seated pain that feels like a bruise in the middle of the heel. The pain can be reproduced by palpating the area. It is typically an inflammatory response to damage of the heel pad secondary to walking barefoot for prolonged periods of time. Plantar fasciitis causes tenderness along the medial calcaneal tuberosity and along the plantar fascia. Patients typically report pain with their first steps after getting out of bed in the morning.

References

Lin CY, Lin CC, Chou YC, Chen PY, Wang CL. Heel pad stiffness in plantar heel pain by shear wave elastography. *Ultrasound Med Biol*. 2015;41(11):2890-2898.

Tu P. Heel pain: diagnosis and management. *Am Fam Physician*. 2018;97(2):86-93.

Rodriguez-Merchan EC, Moracia-Ochagavia I. Tarsal tunnel syndrome: current rationale, indications and results. *EFORT Open Rev*. 2021;6(12):1140-1147.

Nelson SC. Tarsal tunnel syndrome. *Clin Podiatr Med Surg*. 2021;38(2):131-141.

Item 131

ANSWER: A

Hearing protection is essential while shooting firearms because noise exposure often exceeds 140 dB. Even a single gunshot can cause permanent hearing loss. Earmuffs and earplugs with a noise reduction rate of 28–31 dB are common and inexpensive protective devices. Electronic hearing protection can amplify quiet sounds such as a range safety officer's instructions, but also safely protect against loud sounds. Many ranges require their use. Noise-canceling headphones do not provide enough noise reduction for gunshots. Wearing loose-fitting clothing can trap the ejected hot brass casings, leading to unsafe gun handling. Safety eyeglasses that meet the necessary certifications can protect against shrapnel from the bullet striking the target or a malfunction. Regular glasses do not provide sufficient eye protection.

References

Carter PM, Cunningham RM. Clinical approaches to the prevention of firearm-related injury. *N Eng J Med*. 2024;391(10):926-940.

Meinke DK, Finan DS, Flamme GA, et al. Prevention of noise-induced hearing loss from recreational firearms. *Semin Hear*. 2017;38(4):267-281.

Olson DE, Weber J, Allen BF. Eye injuries. In: Madden CC, Putukian M, McCarty EC, et al, eds. *Netter's Sports Medicine*. 3rd ed. Elsevier; 2023:371-378.

Item 132**ANSWER: C**

Approximately 1% of primary care office visits are for evaluation of chest pain, in which 2%–4% of these patients have unstable angina or myocardial infarction. Twelve-lead EKG is recommended to look for ST-segment changes, new-onset left bundle branch block, presence of Q waves, and new T-wave inversions. The presence of new-onset left bundle branch block increases the likelihood of acute coronary syndrome (ACS) and acute myocardial infarction. Patients with suspected ACS or the aforementioned changes on EKG should be transported immediately to the emergency department. The presence of first-degree atrioventricular block, frequent PVCs, sinus tachycardia, or voltage criteria for left ventricular hypertrophy does not increase the likelihood of ACS.

References

McConaghy JR, Sharma M, Patel H. Acute chest pain in adults: outpatient evaluation. *Am Fam Physician*. 2020;102(12):721-727.

Item 133

ANSWER: B

Although this patient's T-score indicates osteopenia, the presence of fragility fractures indicates osteoporosis. The T-score is obtained by measuring bone mineral density using DXA at the spine, total hip, or femoral neck. A T-score between -1.0 and -2.49 indicates osteopenia, whereas a T-score of ≤ -2.5 denotes osteoporosis. However, postmenopausal osteoporosis may be diagnosed by bone mineral density or by the presence of a fragility fracture, which is a fracture that occurs with no associated trauma or after a fall from standing height or lower. Hence, osteoporosis can be diagnosed and the patient further evaluated and treated even though her T-score is > -2.5 .

Spondylolisthesis is the acquired anterior displacement of one vertebral body over another and is typically associated with degenerative changes. Spondylolysis is a cause of acute back pain resulting from a stress fracture involving the pars interarticularis of the lumbar vertebral body. It is typically found in young athletes.

References

NASS Evidence-Based Clinical Guidelines Committee; Matz P, Meagher RJ, Lamer T, et al. *Evidence-Based Clinical Guidelines for Multidisciplinary Spine Care: Diagnosis and Treatment of Degenerative Lumbar Spondylolisthesis*. 2nd ed. North American Spine Society; 2014.

Walker MD, Shane E. Postmenopausal osteoporosis. *N Engl J Med*. 2023;389(21):1979-1991.

Item 134**ANSWER: C**

Blood type/Rh factor testing should be ordered when evaluating second-trimester bleeding. This patient would require a dose of Rho(D) immune globulin (RhoGAM) if found to be Rh negative. Therefore, of all the tests listed, an Rh factor test is the most time sensitive and affects the plan of care. Patients may have had unattended home births with prior pregnancies or refused laboratory tests altogether. With the potential exposure to mixing of maternal and fetal blood during the patient's first delivery, it is crucial during this pregnancy to ensure maternal and fetal blood are compatible because there is a high risk of contact between maternal and fetal blood.

An antiphospholipid antibody profile would be appropriate for evaluation of an underlying hypercoagulable disorder, such as in cases of late pregnancy loss (>10 weeks), recurrent early pregnancy loss (<10 weeks), certain preterm deliveries, or thrombosis. Although HIV testing is an important component of prenatal laboratory studies, it would not be the most important test to order at this time. A toxoplasmosis antibody test may be considered if there are signs of infection or exposure to toxoplasmosis. While cocaine use can cause placental abruption, a urine toxicology screen for cocaine is not the most pressing laboratory study to order at this time.

References

American College of Obstetricians and Gynecologists' Committee on Practice Bulletins—Obstetrics. ACOG Practice Bulletin No. 181: prevention of Rh D alloimmunization. *Obstet Gynecol.* 2017;130(2):e57-e70.

Item 135**ANSWER: A**

Postexposure prophylaxis with hepatitis B immune globulin or hepatitis B vaccine is not recommended in those who have completed the hepatitis B vaccine series and have proven immunity to hepatitis B virus. The presence of hepatitis B surface antibody titers of >10 mIU/mL are indicative of immunity either from previous vaccination or from previous infection with hepatitis B (both hepatitis B core and surface antibodies are reactive). Hepatitis B immune globulin would be recommended in a patient with negative hepatitis B serology testing.

References

Schillie S, Vellozzi C, Reingold A, et al. Prevention of hepatitis B virus infection in the United States: recommendations of the Advisory Committee on Immunization Practices. *MMWR Recomm Rep*. 2018;67(1):1-31.

Item 136**ANSWER: D**

This patient is a healthy young adult female who presents with an elevated BMI but no other identified risk factors. The U.S. Preventive Services Task Force recommends HIV screening for all persons 15–65 years of age (A recommendation). Cervical cancer screening is not recommended in women before age 21 (D recommendation), thus precluding the patient from screening. Chlamydia screening is recommended for all sexually active females under 25 years of age (B recommendation), but this patient has no history of sexual activity. Diabetes mellitus screening is recommended for persons 35–70 years of age with overweight or obesity (B recommendation). While the patient's age precludes her from this screening, she should be offered intensive multicomponent behavioral intervention for weight loss. Preventive counseling for skin cancer is recommended for persons with fair skin types who are 6 months to 24 years of age (B recommendation), but screening examinations are not recommended (I recommendation). This patient should also be offered screening for hepatitis C.

References

Final recommendation statement: human immunodeficiency virus (HIV) infection: screening. US Preventive Services Task Force. Updated June 11, 2019.

Final recommendation statement: cervical cancer: screening. US Preventive Services Task Force. August 21, 2018.

Final recommendation statement: chlamydia and gonorrhea: screening. US Preventive Services Task Force. September 14, 2021.

Final recommendation statement: prediabetes and type 2 diabetes: screening. US Preventive Services Task Force. August 24, 2021.

Final recommendation statement: skin cancer: screening. US Preventive Services Task Force. April 18, 2023.

Item 137**ANSWER: C**

Family physicians frequently evaluate ophthalmologic symptoms and should be familiar with conditions requiring emergent evaluation. Peripheral flashing lights and floaters are common symptoms of retinal detachment. This patient should be referred for urgent ophthalmologic evaluation including a dilated fundoscopic examination.

While positive and negative visual features are seen with migraine with aura, these features should be transient and followed by a headache typical of migraine. New-onset migraine in a 65-year-old is also uncommon and would require further evaluation. Central retinal artery occlusion, often ischemic or thromboembolic, typically occurs in those with risk factors such as diabetes mellitus, vascular disease, and tobacco use. Rapid, painless vision loss is often accompanied by afferent pupillary defect and the lesion can be visualized on a dilated fundoscopic examination as a cherry-red spot. Globe rupture is unlikely to occur without a history of trauma. Acute ischemic stroke may cause vision loss but is unlikely to cause ocular symptoms of flashing lights and floaters.

References

Gelston CD, Deitz GA. Eye emergencies. *Am Fam Physician*. 2020;102(9):539-545.

Raharja A, Whitefield L. Clinical approach to vision loss: a review for general physicians. *Clin Med (Lond)*. 2022;22(2):95-99.

Bagheri N, Mehta S. Acute vision loss. *Prim Care*. 2015;42(3):347-361.

Item 138**ANSWER: E**

Early advanced imaging for low back pain increases the probability of surgery and is associated with a higher use of opioids, a higher cost of care, higher pain scores, and more work absenteeism. While there is no consensus on what constitutes red flags for low back pain, new urinary retention is concerning for cauda equina syndrome. Other symptoms associated with cauda equina syndrome are bladder fullness and saddle anesthesia. The severity of pain does not correlate clearly with pathologic back pain. While cigarette use increases the chance of an underlying cancer, without a known history of cancer or other signs, it is not an indication for advanced imaging. While intravenous drug use increases the risk of epidural abscess, with no other signs of infection or history of infection, it is not solitarily an indication for advanced imaging. Nocturnal pain is not uncommon in benign low back pain and may be associated with a more specific underlying cause if the pain occurs in the second part of the night (early morning).

References

The Diagnosis and Treatment of Low Back Pain Work Group; Office of Quality and Patient Safety, Veterans Health Administration; Clinical Quality Improvement Program, Defense Health Agency. *VA/DoD Clinical Practice Guideline for the Diagnosis and Treatment of Low Back Pain*. Version 3.0. US Government Printing Office; 2022.

Buelt A, McCall S, Coster J. Management of low back pain: guidelines from the VA/DoD. *Am Fam Physician*. 2023;107(4):435-437.

Item 139**ANSWER: D**

After intranasal administration of naloxone, peak plasma concentrations are attained within 15–30 minutes, although its effect on the patient may be seen sooner. Absorption may be affected by substances or conditions that impact the nasal mucosa function or its blood flow. The half-life of naloxone is 2 hours, which is shorter than many long-acting opioids. Ongoing monitoring is important even after reversal of opioid respiratory depression is achieved.

References

Ortega R, Nozari A, Baker W, Surani S, Edwards M. Intranasal naloxone administration. *N Engl J Med*. 2021;384(12):e44.

Item 140**ANSWER: E**

This patient presents with an inguinal hernia that is apparent on examination. When the hernia is apparent, no further imaging is needed. Recent studies have found that nearly 50% of patients sent for inguinal imaging have been unnecessarily referred. When the hernia is symptomatic or limits function, surgical referral is appropriate. If the hernia is asymptomatic, then watchful waiting would be a viable option. Physical therapy would not be appropriate for this patient with a symptomatic inguinal hernia.

References

Fitzgibbons RJ Jr, Forse RA. Groin hernias in adults. *N Engl J Med*. 2015;372(8):756-763.

Fitzgibbons RJ Jr, Giobbie-Hurder A, Gibbs JO, et al. Watchful waiting vs repair of inguinal hernia in minimally symptomatic men: a randomized clinical trial. *JAMA*. 2006;295(3):285-292.

Shakil A, Aparicio K, Barta E, Munez K. Inguinal hernias: diagnosis and management. *Am Fam Physician*. 2020;102(8):487-492.

Liu N, Prout TM, Xu Y, et al. Unnecessary use of radiology studies in the diagnosis of inguinal hernias: a retrospective cohort study. *Surg Endosc*. 2021;35(8):4444-4451.

Item 141**ANSWER: D**

This patient presents with the classic triad symptoms associated with normal-pressure hydrocephalus: gait change, urinary incontinence, and dementia. MRI will show evidence of hydrocephalus, and gait changes will often improve with high-volume removal of cerebrospinal fluid. Treatment involves placing a ventriculoperitoneal shunt to relieve hydrocephalus. Carbidopa/levodopa, donepezil, levothyroxine, and surgery with chemotherapy are all treatments for other causes of dementia including Parkinson disease with Lewy body dementia, Alzheimer dementia, hypothyroidism, and brain tumor or neoplasm, respectively.

References

Falk N, Cole A, Meredith TJ. Evaluation of suspected dementia. *Am Fam Physician*. 2018;97(6):398-405.

Item 142

ANSWER: D

The American College of Physicians recommends that all adults with gross hematuria be referred for further urologic evaluation, including cystoscopy and imaging, due to the high association with malignancy. The sensitivity and specificity of cystoscopy for detecting bladder cancer are high, with a sensitivity ranging from 87% to 100% and a specificity of 64% to 100%. The positive predictive value of gross hematuria for urologic malignancy in older men is high.

Gross hematuria warrants immediate investigation to rule out serious underlying conditions; thus a repeat urinalysis in 1 month would not be appropriate. Initiating empiric antibiotic therapy and re-evaluating in 1 week would not be appropriate without evidence of infection. Urine cytology has limited sensitivity and specificity and should not replace cystoscopy and imaging for the initial evaluation of hematuria.

References

Nielsen M, Qaseem A; High Value Care Task Force of the American College of Physicians. Hematuria as a marker of occult urinary tract cancer: advice for high-value care from the American College of Physicians. *Ann Intern Med*. 2016;164(7):488-497.

Wolfman DJ, Marko J, Nikolaidis P, et al; Expert Panel on Urological Imaging. ACR Appropriateness Criteria hematuria. *J Am Coll Radiol*. 2020;17(suppl 5):S138-S147.

Devlies W, de Jong JJ, Hofmann F, et al. The diagnostic accuracy of cystoscopy for detecting bladder cancer in adults presenting with haematuria: a systematic review from the European Association of Urology Guidelines Office. *Eur Urol Focus*. 2024;10(1):115-122.

Item 143

ANSWER: C

Motivational interviewing (MI) is the counseling strategy that best fits the described approach. MI is an evidence-proven strategy that is particularly helpful in situations of ambivalence toward change. In primary care settings, MI has been shown to help people achieve weight loss, improve blood pressure, and reduce alcohol use (SOR A). The key elements of MI include establishing rapport, eliciting the patient's values, asking open-ended questions, offering affirmations, practicing reflective listening, and recapping discussions with summarizing statements. The acronym OARS (open-ended questions, affirmations, reflective listening, summaries) serves as a reminder of these 4 MI components.

The BATHE (background, affect, troubles, handling, empathy) model provides a structure for addressing various psychosocial problems. BATHE uses more targeted lines of questioning, in contrast to the MI approach of asking open-ended questions that give way to affirming, reflective, and summarizing statements. The five A's (ask, advise, assess, assist, arrange) approach provides a stepwise guide for assessing and counseling on health-related behaviors. In this strategy, the lines of questioning are targeted, rather than open-ended, leading to more directive recommendations and focused follow-up questions. The transtheoretical (stages of change) model can be helpful when patients are ambivalent, but this approach focuses more on identifying the patient's current level of readiness to modify a particular health behavior. The stages are precontemplation, contemplation, preparation, action, and maintenance. Identifying a patient's current stage can facilitate movement to the next stage of change and can provide a helpful framework to select another counseling strategy, such as MI.

References

Schwenker R, Dietrich CE, Hirpa S, et al. Motivational interviewing for substance use reduction. *Cochrane Database Syst Rev*. 2023;12(12):CD008063.

Stewart EE, Fox CH. Encouraging patients to change unhealthy behaviors with motivational interviewing. *Fam Pract Manag*. 2011;18(3):21-25.

Searight HR. Counseling patients in primary care: evidence-based strategies. *Am Fam Physician*. 2018;98(12):719-728.

Item 144**ANSWER: E**

This patient's most likely diagnosis is necrobiosis lipoidica diabetorum. Its pathophysiology is poorly understood, but lesions are commonly found on the anterior and lateral surfaces of the lower legs. Lesions may begin as a small, dusky red, elevated nodule with a sharply circumscribed border. The nodule slowly enlarges, develops an irregular outline, and flattens, and has a brownish-yellow center. A reddened border may be present. Diabetic dermopathy characteristically occurs as crops of brownish atrophic lesions. In diabetic lipoatrophy the skin is depressed but otherwise normal. Diabetic rubeosis is described as reddening of the face and extremities in patients with long-standing diabetes mellitus. Eruptive xanthomas from diabetes are typically small, reddish-yellow nodules up to 0.5 cm in diameter that appear in clusters on the extensor surfaces and buttocks.

References

Powers AC, Stafford JM, Rickels MR. Diabetes mellitus: complications. In: Loscalzo J, Fauci AS, Kasper DL, Hauser SL, Longo DL, Jameson JL, eds. *Harrison's Principles of Internal Medicine*. Vol 2. 21st ed. McGraw Hill; 2022:3128.

Item 145**ANSWER: D**

Asymptomatic vitamin B₁₂ deficiency has been reported in a significant proportion of patients taking metformin (SOR A). A randomized, placebo-controlled trial showed that patients taking metformin had a 19% decrease in vitamin B₁₂ levels and a 7% increase in absolute risk of vitamin B₁₂ deficiency compared to the placebo group.

Chromatopsia, which is a visual disturbance of color perception, is associated with use of phosphodiesterase-5 inhibitors, such as sildenafil. Metformin use is not associated with ileus. Although metformin use is contraindicated in patients with impaired renal function, metformin itself does not cause renal failure. Because diarrhea occurs relatively frequently with metformin therapy, metformin use is not commonly associated with weight gain.

References

Nabrdalik K, Hendel M, Irlik K, et al. Gastrointestinal adverse events of metformin treatment in patients with type 2 diabetes mellitus: a systematic review and meta-analysis with meta-regression of observational studies. *BMC Endocr Disord*. 2024;24(206).

Item 146**ANSWER: A**

The majority of cases of community-acquired pneumonia (CAP) in children are caused by viruses. If bacterial pneumonia were suspected, then antibiotics would be warranted. Given this patient's focal lung examination and x-ray findings, the most likely causative bacteria would be a typical pathogen such as *Streptococcus pneumoniae* or *Haemophilus influenzae*. The first-line choice of antibiotics would be amoxicillin for 7 days. Azithromycin is appropriate for treatment of atypical pneumonia. While cefdinir does treat the typical pathogens seen in CAP, it would not be recommended for first-line treatment due to its broader spectrum. Doxycycline is used to treat CAP but should not be used in a 3-year-old. Sulfamethoxazole/trimethoprim is not indicated for CAP treatment.

References

Smith DK, Kuckel DP, Recidoro AM. Community-acquired pneumonia in children: rapid evidence review. *Am Fam Physician*. 2021;104(6):618-625.

Item 147**ANSWER: B**

This patient has a dorsal dislocation of the proximal interphalangeal (PIP) joint of the finger, which is the most commonly dislocated joint in the body. Examination of the patient's finger reveals intact neurovascular findings. Conservative management is indicated given the limited articular surface involvement. A simple, stable dorsal PIP dislocation can be managed with in-office reduction, followed by splinting in approximately 30° of flexion for 2–4 weeks. Buddy taping is also recommended to prevent hyperextension deformities. X-rays should be repeated directly after reduction, and again in 1 week.

Neither MRI nor physical therapy are indicated. Immediate surgical referral is necessary for finger fracture-dislocations that are irreducible, unstable after reduction, or involve significant soft tissue, tendon, or neurovascular injury. Additional indications include fractures involving a large portion of the articular surface (typically >30%), significant displacement or angulation (>15°), malrotation, comminution, and persistent subluxation or dislocation on imaging. Open fractures (exposed bone or joint) and injuries with vascular compromise or loss of function also require urgent surgical evaluation.

References

Leggit JC, Meko CJ. Acute finger injuries: part II. Fractures, dislocations, and thumb injuries. *Am Fam Physician*. 2006;73(5):827-834.

Childress MA, Olivas J, Crutchfield A. Common finger fractures and dislocations. *Am Fam Physician*. 2022;105(6):631-639.

Item 148**ANSWER: E**

Giardia is a parasite that causes diarrheal illness. While found globally, it is especially common in areas where sanitary conditions are poor and water treatment facilities are limited. Transmission may be waterborne, foodborne, or fecal-oral. The incubation period is usually 1–2 weeks, and symptoms include greasy diarrhea, abdominal pain, and nausea. Fever and vomiting are uncommon. While some infections are self-limited, treatment is recommended for symptomatic individuals to improve their symptoms and prevent transmission to others. Metronidazole, nitazoxanide, or tinidazole are recommended for treatment of *Giardia* infection. Azithromycin, ciprofloxacin, and ivermectin are antimicrobials that are commonly used to treat other causes of traveler's diarrhea.

References

Pyzocha N, Cuda Am. Common intestinal parasites. *Am Fam Physician*. 2023;108(5):487-493.

Item 149**ANSWER: A**

Antidepressants including bupropion, trazodone, and tricyclic antidepressants are associated with an increased risk of seizure and should be avoided in patients with epilepsy. In general, SSRIs or SNRIs are safer choices. Mirtazapine and vortioxetine would also be preferred choices.

References

Maguire MJ, Marson AG, Nevitt SJ. Antidepressants for people with epilepsy and depression. *Cochrane Database Syst Rev*. 2021;(4):CD010682.

Item 150**ANSWER: A**

The U.S. Preventive Services Task Force gives a B recommendation for screening for hepatitis C in individuals 18–79 years of age. Cervical cancer screening is not recommended until at least age 21 (A recommendation). Evidence for screening for hyperlipidemia and skin cancer is uncertain. Current evidence is insufficient to assess the balance of the benefits and harms of visual skin examination to screen for skin cancer (I recommendation), but counseling for ultraviolet exposure is recommended for persons 6 months to 24 years of age with fair skin types (B recommendation).

References

Final recommendation statement: cervical cancer: screening. US Preventive Services Task Force. August 21, 2018.

Final recommendation statement: skin cancer: screening. US Preventive Services Task Force. Updated April 18, 2023.

Final recommendation statement: hepatitis C virus infection in adolescents and adults: screening. US Preventive Services Task Force. March 2, 2020.

Final recommendation statement: lipid disorders in children and adolescents: screening. US Preventive Services Task Force. Updated July 18, 2023.

Item 151**ANSWER: A**

Treatment of acute aortic dissection includes aggressive lowering of the systolic blood pressure to 100–120 mm Hg within 20 minutes and concomitant lowering of the pulse rate to <60 beats/min to reduce shear stress on the aortic wall, minimize the risk of propagating the dissection, and reduce the risk of other complications, including rupture and aneurysmal degeneration. The optimal treatment of acute ischemic stroke, acute renal failure, hypertensive encephalopathy, and intracerebral hemorrhage requires blood pressure to be lowered more cautiously to avoid the risk of complications that may be seen with overly rapid and aggressive blood pressure–lowering.

References

Balahura AM, Moroi ȘI, Scafa-Udriște A, et al. The management of hypertensive emergencies—is there a "magical" prescription for all? *J Clin Med*. 2022;11(11):3138.

Peixoto AJ. Acute severe hypertension. *N Engl J Med*. 2019;381(19):1843-1852.

Item 152**ANSWER: E**

The combination of obesity, loud snoring, and fatigue is suspicious for obstructive sleep apnea, and polysomnography is the definitive diagnostic study. A TSH level would be useful if hypothyroidism were suspected as the cause of this patient's fatigue, but it is less likely because he does not have any mood changes, dry skin, or constipation. Low vitamin B₁₂, folate, and iron levels are also less likely based on this patient's presentation. A comprehensive metabolic panel would be useful if kidney or liver function were suspected as the cause of the patient's fatigue, but these are less likely because he does not have shortness of breath, stomach pain, or chest pain.

References

Semelka M, Wilson J, Floyd R. Diagnosis and treatment of obstructive sleep apnea in adults. *Am Fam Physician*. 2016;94(5):355-360.

Latimer KM, Gunther A, Kopec M. Fatigue in adults: evaluation and management. *Am Fam Physician*. 2023;108(1):58-69.

Item 153**ANSWER: A**

The U.S. Preventive Services Task Force (USPSTF) recommends screening for anxiety in children and adolescents 8–18 years of age (B recommendation). The USPSTF does recommend screening for chlamydia and gonorrhea for sexually active females in this patient's age group (B recommendation); however, this patient is not sexually active. Screening adolescents for eating disorders, elevated blood pressure, and scoliosis all have insufficient evidence to assess the benefits and harms (I recommendation).

References

Final recommendation statement: anxiety in children and adolescents: screening. US Preventive Services Task Force. Updated October 11, 2022.

Final recommendation statement: chlamydia and gonorrhea: screening. US Preventive Services Task Force. Updated September 14, 2021.

Final recommendation statement: adolescent idiopathic scoliosis: screening. US Preventive Services Task Force. Updated January 9, 2018.

Final recommendation statement: eating disorders in adolescents and adults: screening. US Preventive Services Task Force. Updated March 15, 2022.

Final recommendation statement: high blood pressure in children and adolescents: screening. US Preventive Services Task Force. Updated November 10, 2020.

Item 154**ANSWER: D**

The hallmark finding of adhesive capsulitis, also known as frozen shoulder, is pain associated with restricted passive range of motion, particularly in external rotation, flexion, and abduction. Osteoarthritis of the glenohumeral joint also commonly causes pain and stiffness with limited passive range of motion; however, radiographs would show evidence of osteoarthritis, with joint space narrowing, sclerosis, osteophyte formation, and subchondral cystic change. With adhesive capsulitis, radiographs are typically normal, although characteristic findings are seen on MRI. Limited passive range of motion occurs with adhesive capsulitis and osteoarthritis because both conditions involve structural changes within the intra-articular space of the glenohumeral joint, thus resulting in mechanical restriction of the range of motion.

Apprehension in a position of abduction and external rotation would suggest a diagnosis of shoulder instability. Crepitus on passive range of motion would suggest a mechanical abnormality within the shoulder girdle (either intra- or extra-articular) and is nonspecific to any one condition. Limited active range of motion in abduction with preserved passive range of motion is common to extra-articular sources of shoulder pain, such as rotator cuff tendinitis and impingement, subacromial bursitis, biceps tendinitis, and partial or complete rotator cuff tears. Weakness on manual muscle testing in abduction and external rotation would raise concern for a high-grade or full-thickness rotator cuff tear.

References

Ramirez J. Adhesive capsulitis: diagnosis and management. *Am Fam Physician*. 2019;99(5):297-300.

Leafblad N, Mizels J, Tashjian R, Chalmers P. Adhesive capsulitis. *Phys Med Rehabil Clin N Am*. 2023;34(2):453-468.

Item 155**ANSWER: D**

In a patient with diabetes mellitus and suspected osteomyelitis of the foot, initial evaluation should include a probe-to-bone test, a serum inflammatory marker such as an erythrocyte sedimentation rate (ESR), and plain radiography. Plain radiography is the recommended initial imaging modality in the diagnosis of osteomyelitis due to its low cost, safety, and availability. If a plain radiograph suggests osteomyelitis, advanced imaging may not be necessary.

While an elevated WBC count can be used to grade the severity of a diabetes-related foot infection, it can be normal in about half of patients with a severe diabetes-related foot infection and is not helpful in the diagnosis of osteomyelitis. A highly elevated ESR, however, is suggestive of an osteomyelitis infection. Superficial wound cultures are poorly sensitive and specific for identifying diabetes-related wound organisms compared to deep tissue cultures and are not used in the assessment of diabetes-related foot infections or osteomyelitis. While evaluation of peripheral vasculature and oxygenation may be warranted in the management of diabetes-related foot infections and osteomyelitis, it does not contribute to the diagnosis of osteomyelitis.

References

Matheson EM, Bragg SW, Blackwelder RS. Diabetes-related foot infections: diagnosis and treatment. *Am Fam Physician*. 2021;104(4):386-394.

Lipsky BA, Senneville É, Abbas ZG, et al. Guidelines on the diagnosis and treatment of foot infection in persons with diabetes (IWGDF 2019 update). *Diabetes Metab Res Rev*. 2020;36(suppl 1):e3280.

Item 156

ANSWER: A

Adverse medication effects should always be considered when evaluating new symptoms, such as the association between estrogen use and peripheral edema. This patient's age suggests that she is likely remote from menopause, which provides a reason to consider discontinuing estrogen therapy. Peripheral edema is not associated with ACE inhibitor use. In fact, combining an ACE inhibitor with a calcium channel blocker (CCB) may decrease the edema attributable to CCB use.

Ankle-brachial indices are indicated for the evaluation of suspected arterial disease. Arterial disease typically presents as lower extremity pain during activity that improves with rest. Arterial disease may lead to decreased arterial pulses. Arterial disease should be ruled out in those at risk before starting compression therapy for venous disease. Venous compression ultrasonography is indicated when deep vein thrombosis (DVT) is suspected. DVT is unlikely in the presentation of chronic bilateral lower extremity edema. DVT should be considered in acute unilateral edema, and the Wells score should be used to determine if there is an intermediate or high risk. The combination of a low-risk Wells score and a negative D-dimer level safely rules out DVT. Venous duplex ultrasonography with reflux is helpful in evaluating for chronic venous insufficiency. While this patient may have chronic venous insufficiency, her current medication use could be causing her edema. This should be evaluated before pursuing additional diagnostic testing given her lack of alarm symptoms, such as signs of heart failure or lower extremity ulcerations.

References

Bates SM, Jaeschke R, Stevens SM, et al. Diagnosis of DVT: antithrombotic therapy and prevention of thrombosis, 9th ed: American College of Chest Physicians evidence-based clinical practice guidelines. *Chest*. 2012;141(2 suppl):e351S-e418S.

Patel H, Skok CJ, DeMarco A. Peripheral edema: evaluation and management in primary care. *Am Fam Physician*. 2022;106(5):557-564.

Item 157**ANSWER: D**

Parenteral penicillin G benzathine is the only therapy with documented effectiveness for treating syphilis in a pregnant person. In pregnant patients with any stage of syphilis and a reported penicillin allergy, the CDC recommends desensitization followed by treatment with penicillin G benzathine. Azithromycin and erythromycin are insufficient to cure maternal infection or treat fetal infection. There is insufficient data to recommend cephalosporins such as ceftriaxone for the treatment of maternal infections and the prevention of congenital infections. Doxycycline and tetracycline are not recommended for use in the second and third trimesters.

References

Sexually transmitted infections treatment guidelines, 2021: syphilis during pregnancy. Centers for Disease Control and Prevention. Reviewed July 22, 2021.

Item 158**ANSWER: A**

Anticipatory guidance for lifestyle management in children should incorporate multiple approaches that address nutrition, physical activity, and behaviors that contribute to possible energy imbalance. These recommendations include 60 minutes per day of moderate to vigorous physical activity for children 6–17 years of age. Calorie restriction is beneficial in reducing weight in adolescents when combined with additional strategies, such as physical activity and behavioral changes. While excessive video screen time has been associated with increased pediatric obesity, screen-based physical activity (eg, video games that require whole-body movement) can help reduce weight in children with overweight and obesity. Children should consume 5 servings of fruits and vegetables per day. The American Academy of Pediatrics recommends against the use of fruit juices as a substitute for whole fruit due to its high calorie content.

References

Hampl SE, Hassink SG, Skinner AC, et al. Clinical practice guideline for the evaluation and treatment of children and adolescents with obesity. *Pediatrics*. 2023;151(2):e2022060640.

Lobelo F, Muth ND, Hanson S, Nemeth BA; Council on Sports Medicine and Fitness; Section on Obesity. Physical activity assessment and counseling in pediatric clinical settings. *Pediatrics*. 2020;145(3):e20193992.

Item 159**ANSWER: D**

This patient demonstrates a symmetric pattern of paresthesia without red flag symptoms such as ataxia, double vision, or speech issues, which makes a lesion of the central nervous system or a nerve root less likely. Because the results of her routine laboratory studies are normal, electrodiagnostic testing would be the recommended next step. While gabapentin can be helpful in treating painful peripheral neuropathy, further workup is needed to determine the cause of this patient's symptoms. Lyme antibody testing and vitamin level testing plus supplementation may be helpful if electrodiagnostic testing exhibits an axonal pattern. Imaging of the brain or spine, such as with MRI, is not recommended in patients without central neurologic signs or asymmetry.

References

Castelli G, Desai KM, Cantone RE. Peripheral neuropathy: evaluation and differential diagnosis. *Am Fam Physician*. 2020;102(12):732-739.

Item 160**ANSWER: E**

Achilles tendinopathy is a common injury in active individuals. The diagnosis is made clinically; therefore, additional imaging is not indicated unless there is concern for an alternate diagnosis. Current recommendations for management include physical therapy for activity modification and targeted exercises. Neither complete rest nor corticosteroid injection are currently recommended for management of Achilles tendinopathy. While platelet-rich plasma injection continues to be studied in tendon injuries, it is not currently recommended as first-line management. Furthermore, studies remain mixed as to whether platelet-rich plasma injections are superior to placebo for pain control and return to normal activity.

References

Masiello F, Pati I, Veropalumbo E, Pupella S, Cruciani M, De Angelis V. Ultrasound-guided injection of platelet-rich plasma for tendinopathies: a systematic review and meta-analysis. *Blood Transfus.* 2023;21(2):119-136.

Arnold MJ, Moody AL. Common running injuries: evaluation and management. *Am Fam Physician.* 2018;97(8):510-516.

Silbernagel KG, Hanlon S, Sprague A. Current clinical concepts: conservative management of Achilles tendinopathy. *J Athl Train.* 2020;55(5):438-447.

Item 161**ANSWER: D**

Motivational interviewing is the communication strategy with the best evidence of leading to a change in behavior. Scare tactics have not been shown to be effective. Physicians should speak in language that the patient can understand rather than use jargon. Supplying too much detail may overwhelm the patient, especially if it is not tailored to the individual's goals.

References

Miller WR, Rollnick S. *Motivational Interviewing: Helping People Change and Grow*. 4th ed. Guildford Press; 2023.

Item 162

ANSWER: C

Patients with classic symptoms of diarrhea, abdominal pain, weight loss, bloating, malabsorption, and failure to thrive should undergo evaluation for celiac disease. Serologic testing for IgA tissue transglutaminase (tTG) is the preferred initial screening in all age groups due to its high sensitivity (95%) and specificity (95%). When celiac disease is strongly suspected, total IgA levels should also be obtained because 2%–3% of these patients will have IgA deficiency. If IgA tTG serology is positive, most patients will require esophagogastroduodenoscopy with small-bowel biopsy for confirmation, but it is not used as an initial test and some organizations state that a no-biopsy technique is safe for children with very high IgA tTG levels.

This patient has classic signs of malabsorption and unexplained weight loss, indicating need for further evaluation. Reassurance would be inappropriate at this time. Fecal fat testing may be useful in the workup of suspected malabsorption symptoms, but the initial test is usually a qualitative assessment of fat in a single specimen because it is less expensive and easier to perform than 72-hour testing. If high suspicion remains after negative initial testing, a 72-hour fecal fat test should be performed. Carbohydrate breath testing can be used to evaluate for suspected small intestinal bacterial overgrowth (SIBO). SIBO generally occurs in individuals with abnormal anatomy or other risk factors for abnormal intestinal mobility, including diabetes mellitus or scleroderma. The majority of patients with SIBO present with bloating and flatulence.

References

Pimentel M, Saad RJ, Long MD, Rao SSC. ACG clinical guideline: small intestinal bacterial overgrowth. *Am J Gastroenterol*. 2020;115(2):165-178.

Burgers K, Lindberg B, Bevis ZJ. Chronic diarrhea in adults: evaluation and differential diagnosis. *Am Fam Physician*. 2020;101(8):472-480.

Williams PM, Harris LM DO, Odom MR. Celiac disease: common questions and answers. *Am Fam Physician*. 2022;106(1):36-43.

Item 163**ANSWER: C**

Loss of axillary and pubic hair is one of the more specific symptoms of male hypogonadism. Memory issues have myriad etiologies; they may be secondary to attention disorders, mood disorders, sleep apnea, or other neurologic disorders. Fatigue or diminished stamina may be a symptom of male hypogonadism but also can be indicative of other endocrine, musculoskeletal, or rheumatologic disorders. Thinning hair at the temples may occur with aging and with androgenic alopecia. Thinning of the eyebrows may be related to thyroid disorders.

References

Bhasin S, Brito JP, Cunningham GR, et al. Testosterone therapy in men with hypogonadism: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab*. 2018;103(5):1715-1744.

Heidelbaugh JJ, Belakovskiy A. Testosterone replacement therapy for male hypogonadism. *Am Fam Physician*. 2024;109(6):543-549.

Item 164**ANSWER: A**

A venous hum is an innocent heart murmur that is caused by vein vibration. It is commonly heard in children and may appear between 3 and 6 years of age. It is typically audible only when the child is upright, and changes with head movement. In a child with a venous hum, reassurance of the parents is appropriate, and no follow-up is necessary.

References

Ford B, Lara S, Park J. Heart murmurs in children: evaluation and management. *Am Fam Physician*. 2022;105(3):250-261.

Item 165**ANSWER: E**

This patient experienced anaphylaxis due to her tree nut allergy. Her anaphylaxis was refractory and required 2 doses of epinephrine for treatment. Although she is feeling better, patients with refractory anaphylaxis should be transferred to a higher level of care, such as an emergency department, for further evaluation and management. Patients who have had severe or refractory anaphylaxis are at risk for a biphasic reaction and should be observed for 6–12 hours before discharge.

Although medications such as H₁- and H₂-blockers and corticosteroids can be helpful in the postanaphylaxis care of patients, they are not the mainstay of therapy for patients who have had severe anaphylaxis.

References

Pflipsen MC, Vega Colon KM. Anaphylaxis: recognition and management. *Am Fam Physician*. 2020;102(6):355-362.

Long B, Gottlieb M. Emergency medicine updates: anaphylaxis. *Am J Emerg Med*. 2021;49:35-39.

Item 166**ANSWER: E**

This patient requires a procedure with a high likelihood of bleeding and should not be anticoagulated at the time of the procedure. He also has a high likelihood of thromboembolic disease and should receive bridging with a heparin drip until his INR has normalized for the planned procedure. With his mechanical heart valve, the correct INR goal is 2.5–3.5. Reversal with vitamin K is not necessary because he is stable and not actively bleeding. He is not a candidate for enoxaparin due to his glomerular filtration rate. Heparin, 5000 units subcutaneously, is a prophylactic dose, not a treatment dose.

References

Dean CL. An overview of heparin monitoring with the anti-Xa assay. *Methods Mol Biol.* 2023;2663:343-353.

Douketis JD, Spyropoulos AC, Murad MH, et al. Perioperative management of antithrombotic therapy: an American College of Chest Physicians clinical practice guideline. *Chest.* 2022;162(5):e207-e243.

Item 167**ANSWER: B**

Buckle (torus) fractures are incomplete compression fractures that cause cortical bulging of the bone without cortical disruption. They are common in children because of the elasticity of their bones. Distal radius buckle fractures should be treated with short arm immobilization using a short arm splint, soft cast, or wrap for 3 weeks. Complete immobilization with a cast is unnecessary. According to the Choosing Wisely campaign, follow-up radiography is unnecessary unless the patient continues to experience pain or tenderness.

References

Patel DS, Statuta SM, Ahmed N. Common fractures of the radius and ulna. *Am Fam Physician*. 2021;103(6):345-354.

Item 168**ANSWER: E**

The U.S. Preventive Services Task Force (USPSTF) recommends screening for unhealthy drug use in adults ≥ 18 years of age (B recommendation). Unhealthy drug use is defined as using psychoactive medications for purposes other than as prescribed or illegally obtaining drugs for personal use. For this patient's age group, the USPSTF has determined the evidence to be insufficient to recommend for or against screening for cavities or periodontal disease, dyslipidemia, skin cancer, or suicide risk (I recommendation).

References

Final recommendation statement: unhealthy drug use: screening. US Preventive Services Task Force. Updated June 9, 2020.

Final recommendation statement: skin cancer: screening. US Preventive Services Task Force. Updated April 18, 2023.

Final recommendation statement: oral health in adults: screening and preventive interventions. US Preventive Services Task Force. November 7, 2023.

US Preventive Services Task Force. Depression and suicide risk in adults: screening. June 20, 2023.

Item 169**ANSWER: D**

In patients presenting with large-volume rice-water or milky-white stools, the most likely pathogen is *Vibrio* species. Patients with this presentation should be tested for infection regardless of exposure to undercooked shellfish or brackish waters. Guidelines from the Infectious Diseases Society of America recommend testing for infection with *Clostridioides difficile*, enteropathogenic *Escherichia coli*, enterotoxigenic *E coli*, and *Yersinia enterocolitica* in patients with diarrhea and fever, bloody or mucoid stools, signs of sepsis, or severe abdominal cramping or tenderness.

References

Shane AL, Mody RK, Crump JA, et al. 2017 Infectious Diseases Society of America clinical practice guidelines for the diagnosis and management of infectious diarrhea. *Clin Infect Dis*. 2017;65(12):1963-1973.

Item 170

ANSWER: C

In 2017, the U.S. nonprofit group Global Down Syndrome Foundation convened a workgroup to create an evidence-based guideline for clinicians and adults with Down syndrome and their supporters. This guideline was published in 2020 and includes responses to 10 questions that were created using the Population/Intervention/Comparator/Outcome (PICO) format. They address screening for conditions known to be of higher prevalence in individuals with Down syndrome.

In the United States, hypothyroidism is common in individuals with Down syndrome, with an estimated prevalence of approximately 50% in adults >30 years of age. Screening is therefore recommended with TSH testing every 1–2 years starting at age 21. This patient's TSH level was last checked 2 years ago and is due today.

Dementia occurs on average at younger ages in those with Down syndrome than those without Down syndrome; however, it is rare before the age of 40. If any concerning symptoms arise in this age group, a thorough investigation for alternative causes is recommended. Routine screening for dementia is recommended beginning at age 40 for individuals with Down syndrome. A validated tool such as the National Task Group–Early Detection Screen for Dementia should be used.

Cardiovascular disease prevention is recommended for individuals with Down syndrome per U.S. Preventive Services Task Force guidelines. This patient's LDL-cholesterol level was last checked 2 years ago and is not due for recheck today.

The prevalence of osteoporosis in individuals with Down syndrome has been estimated with a wide degree of variation (1.4%–45.1%). There is no evidence that DXA screening at ages younger than recommended for adults without Down syndrome (age 65) is of benefit.

Although atlantoaxial instability (AAI) is known to have a higher prevalence in those with Down syndrome than those without Down syndrome, no studies have assessed the use of cervical spine radiography for diagnosing AAI. Screening radiography is therefore not recommended; instead, it is recommended that clinicians remain alert for signs and symptoms of cervical myelopathy, such as incontinence, altered gait, brisk reflexes, and clonus.

References

Tsou AY, Bulova P, Capone G, et al. Medical care of adults with down syndrome: a clinical guideline. *JAMA*. 2020;324(15):1543-1556.

Item 171**ANSWER: E**

The risk for acute mountain sickness increases when sleeping above altitudes of about 8200 ft (2499 m). Symptoms can include difficulty sleeping, headaches, and decreased appetite. This patient has more severe symptoms of high-altitude pulmonary edema (HAPE), which is a medical emergency, and should begin supplemental oxygen and descend to a lower altitude if possible. Acetazolamide can be used to prevent acute mountain sickness, but it is not a treatment. Dexamethasone is a supplemental treatment for high-altitude cerebral edema. Nifedipine can be used in the prevention of HAPE and as a treatment in emergency situations in which descent or a hyperbaric chamber is not available. In this case, the patient is at a modern ski resort and descent should be feasible. COVID-19 infection is unlikely in this scenario.

References

Luks AM, Auerbach PS, Freer L, et al. Wilderness Medical Society clinical practice guidelines for the prevention and treatment of acute altitude illness: 2019 update. *Wilderness Environ Med*. 2019;30(4S):S3-S18.

Armstrong C. Acute altitude illness: updated prevention and treatment guidelines from the Wilderness Medical Society. *Am Fam Physician*. 2020;101(8):505-507.

Item 172**ANSWER: E**

Antibiotics are the most common cause of Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis, accounting for more than one-quarter of cases worldwide. Sulfonamide antibiotics, such as sulfamethoxazole/trimethoprim, are associated with the majority of antibiotic-associated SJS cases worldwide. Antibiotic stewardship is always important, and consideration for both relatively benign and life-threatening side effects should be considered with every prescription written.

References

Lee EY, Knox C, Phillips EJ. Worldwide prevalence of antibiotic-associated Stevens-Johnson syndrome and toxic epidermal necrolysis: a systematic review and meta-analysis. *JAMA Dermatol.* 2023;159(4):384-392.

Item 173**ANSWER: A**

Transvaginal ultrasonography or endometrial biopsy are both appropriate initial diagnostic tests for postmenopausal bleeding. An ultrasound showing an endometrial stripe ≤ 4 mm is associated with a >99% negative predictive value for endometrial cancer and no further evaluation is needed unless the bleeding is recurrent. If bleeding is persistent or recurrent, then a tissue sample would be appropriate. If an endometrial biopsy is not helpful and bleeding continues to persist or recur, then further evaluation with hysteroscopy and curettage would be indicated.

References

American College of Obstetricians and Gynecologists' Committee on Gynecologic Practice. ACOG Committee Opinion No. 734: the role of transvaginal ultrasonography in evaluating the endometrium of women with postmenopausal bleeding. *Obstet Gynecol.* 2018;131(5):e124-e129.

Item 174**ANSWER: D**

This patient's history of a catching or popping sensation combined with a positive anterior slide test, as described in the physical examination, is highly suggestive of a glenoid labral tear. This combination of history and the positive anterior slide test has a positive likelihood ratio of 6.0 in diagnosing a glenoid labral tear.

Acromioclavicular joint arthritis presents with superior shoulder pain and tenderness over the acromioclavicular joint, often with a positive cross-body adduction test. Adhesive capsulitis is characterized by diffuse shoulder pain and restricted passive range of motion. Glenohumeral osteoarthritis usually presents with a gradual onset of pain and loss of motion, particularly in older patients. A rotator cuff tear typically presents with pain and weakness, especially with overhead activities, and positive findings on the empty can and external rotation tests. Acromioclavicular joint arthritis, adhesive capsulitis, glenohumeral osteoarthritis, and rotator cuff tears are not associated with catching or popping sensations, and the crank and anterior slide tests are not specific for these conditions.

References

Burbank KM, Stevenson JH, Czarnecki GR, Dorfman J. Chronic shoulder pain: part I. Evaluation and diagnosis. *Am Fam Physician*. 2008;77(4):453-460.

Michener LA, Doukas WC, Murphy KP, Walsworth MK. Diagnostic accuracy of history and physical examination of superior labrum anterior-posterior lesions. *J Athl Train*. 2011;46(4):343-348.

Walsworth MK, Doukas WC, Murphy KP, Mielcarek BJ, Michener LA. Reliability and diagnostic accuracy of history and physical examination for diagnosing glenoid labral tears. *Am J Sports Med*. 2008;36(1):162-168.

Item 175

ANSWER: A

The U.S. Preventive Services Task Force recommends screening all adults for unhealthy alcohol use and implementing brief counseling interventions for those who screen positive (B recommendation). The screening, brief intervention, and referral to treatment (SBIRT) approach has been shown to decrease the number of patients with risky drinking after the intervention. Motivational interviewing is often used for brief counseling in primary care settings. Cognitive behavioral therapy, mindfulness-based therapy, and interpersonal psychotherapy are different psychotherapy approaches that are used as longer-term counseling methods and would not be appropriate for use as brief counseling interventions for adults with unhealthy alcohol use. Patients identified as having alcohol use disorder (AUD) by *DSM-5* criteria should be referred for treatment.

References

Perry C, Liberto J, Milliken C, et al. The management of substance use disorders: synopsis of the 2021 U.S. Department of Veterans Affairs and U.S. Department of Defense Clinical Practice guideline. *Ann Intern Med*. 2022;175(5):720-731.

Ford B, Smith E, Richards A. Management of substance use disorders: guidelines from the VA/DoD. *Am Fam Physician*. 2022;106(4):466-469.

Final recommendation statement: unhealthy alcohol use in adolescents and adults: screening and behavioral counseling interventions. US Preventive Services Task Force. Updated November 13, 2018.

O'Connor EA, Perdue LA, Senger CA, et al. Screening and behavioral counseling interventions to reduce unhealthy alcohol use in adolescents and adults: updated evidence report and systematic review for the US Preventive Services Task Force. *JAMA*. 2018;320(18):1910-1928.

Item 176**ANSWER: A**

For anal fissures, topical sphincter relaxants applied externally around the anus 2–4 times per day help reduce sphincter tone and promote healing. Topical calcium channel blockers such as diltiazem are best tolerated. Topical nitroglycerin is also effective but is associated with more side effects such as headaches and lightheadedness, which results in 1 in 5 patients stopping therapy. Botulinum toxin is also effective. Topical capsaicin is an irritant. Oral calcium channel blockers are not effective for treatment of anal fissures.

References

Madoff RD, Melton GB. Diseases of the anus and rectum. In: Goldman L, Cooney KA, eds. *Goldman-Cecil Medicine*. 27th ed. Elsevier; 2024:980.

Item 177**ANSWER: C**

Diverticulitis is an inflammation of diverticula caused by a protrusion through the intestinal wall. Risk factors include constipation, a low-fiber diet, increasing age, red meat consumption, smoking, obesity, lack of exercise, aspirin and NSAID use, and genetic susceptibility. Consuming a vegetarian diet may help to prevent diverticulitis. Red meat intake, long-term NSAID use, and tobacco use are risk factors for diverticulitis. After years of research, it was concluded that consumption of nuts, seeds, and popcorn is not a risk factor for diverticulitis (SOR B).

The incidence of diverticulitis is increasing, yet screening is not recommended. Tenderness located in the left lower quadrant of the abdomen is the most specific finding of diverticulitis. Evaluation includes a CBC, a basic metabolic panel, a C-reactive protein level, and urine analysis. CT with contrast is the imaging modality of choice if the diagnosis is uncertain. Treatment is based on severity and includes oral and intravenous antibiotics, bowel rest and a clear liquid diet, and surgical management for abscess formation or peritonitis as well as for prevention of recurrences.

References

Bailey J, Dattani S, Jennings A. Diverticular disease: rapid evidence review. *Am Fam Physician*. 2022;106(2):150-156.

Item 178**ANSWER: A**

Trigeminal neuralgia is characterized by severe paroxysms of pain in the distribution of 1 or more branches of cranial nerve V. Impairment of cranial nerve VII results in paralysis of facial muscles, such as Bell palsy. Impairment of cranial nerve X may cause dysphagia and dysphonia, cranial nerve XI involvement results in neck weakness, and cranial nerve XII damage leads to tongue paralysis.

References

Douglas VC, Hauser SL. Trigeminal neuralgia, Bell's palsy, and other cranial nerve disorders. In: Loscalzo J, Fauci AS, Kasper DL, Hauser SL, Longo DL, Jameson JL, eds. *Harrison's Principles of Internal Medicine*. 21st ed. Vol. 2. McGraw-Hill; 2022:3436-3445.

Zakrzewska JM, Linskey ME. Trigeminal neuralgia. *Am Fam Physician*. 2016;94(2):133-135.

Item 179**ANSWER: D**

Social determinants of health (SDOH) are the conditions in which people are born, work, live, and age as well as the wider set of forces and systems that shape the conditions of daily life. Several SDOH affect cognition and risk of Alzheimer disease and related dementias (ADRDs). For example, higher levels of social engagement have a protective effect against ADRDs. Conversely, SDOH such as decreasing BMI with older age, early-life adversity, low education level, food insecurity, midlife obesity, manual labor, and low socioeconomic status are associated with a greater risk of ADRDs.

References

Majoka MA, Schimming C. Effect of social determinants of health on cognition and risk of Alzheimer disease and related dementias. *Clin Ther*. 2021;43(6):922-929.

Item 180**ANSWER: B**

These findings are concerning for polymyalgia rheumatica (PMR), an inflammatory condition that typically affects adults ≥ 50 years of age. This patient's proximal muscle pain, fatigue, and decreased proximal muscle strength are consistent with this diagnosis, as are the elevated inflammatory markers (erythrocyte sedimentation rate and C-reactive protein level) and mild anemia. A normal creatine kinase level is expected in patients with PMR.

PMR is diagnosed clinically and treated with oral glucocorticoids. Rapid improvement with therapy is expected; if improvement does not occur, an alternate diagnosis should be explored. Giant cell arteritis, also known as temporal arteritis, is associated with PMR. It causes temporal headaches, visual changes, and constitutional symptoms. Early identification and treatment are essential to prevent permanent vision change. Fibromyalgia does not cause elevated inflammatory markers. Polymyositis and statin-induced myopathy may cause myalgias but are typically associated with elevations in creatine kinase, which this patient does not have. Almost all patients with systemic lupus erythematosus have an elevated antinuclear antibody titer, which this patient also does not have.

References

Raleigh MF, Stoddard J, Darrow HJ. Polymyalgia rheumatica and giant cell arteritis: rapid evidence review. *Am Fam Physician*. 2022;106(4):420-426.

Item 181**ANSWER: A**

The U.S. Preventive Services Task Force concluded with moderate certainty that the harms outweighed the benefits for routine, population-based serologic screening for infection with herpes simplex virus (HSV) in asymptomatic adolescents and adults, including pregnant patients (D recommendation). High false-positive rates are common in asymptomatic patients who undergo serologic testing for infection with HSV. In addition, treatment is not recommended in asymptomatic patients with a positive test.

References

Final recommendation statement: genital herpes infection: serologic screening. US Preventive Services Task Force. Updated February 14, 2023.

Item 182**ANSWER: A**

Intussusception is a condition in which a part of the intestine slides or "telescopes" into an adjacent section. It presents with a classic triad of intermittent abdominal pain, emesis, and blood-tinged mucous stools that are sometimes called "red currant jelly" stools. It most commonly occurs at the ileocolic junction and typically affects infants and toddlers. The initial evaluation begins with ultrasonography. An enema with air or contrast can both confirm the diagnosis and reverse the intussusception. If this procedure does not relieve the condition, surgical correction is necessary.

Necrotizing enterocolitis is a severe intestinal infection that typically presents with severe abdominal pain, lethargy, and unstable vital signs. Pyloric stenosis is notable for projectile vomiting, but not bloody stools. Volvulus typically involves a sudden onset of vomiting and abdominal pain and even distension, but without bloody stools.

References

Buel KL, Wilcox J, Mingo PT. Acute abdominal pain in children: evaluation and management. *Am Fam Physician*. 2024;110(6):621-631.

Kelley-Quon LI, Arthur LG, Williams RF, et al. Management of intussusception in children: a systematic review. *J Pediatr Surg*. 2021;56(3):587-596.

Item 183**ANSWER: E**

Seven pharmacotherapies are approved by the FDA for smoking cessation, including bupropion, varenicline, and 5 forms of nicotine replacement therapy (NRT). All of these have been shown to be beneficial in smoking cessation compared to placebo. Compared to NRT and bupropion, varenicline appears to be the most effective monotherapy with a number needed to treat of 7. Recent meta-analyses suggest that a combination of pharmacotherapies is more effective than monotherapy. Varenicline plus NRT is more likely to result in sustained smoking cessation compared to bupropion plus NRT.

References

Shang X, Guo K, E F, et al. Pharmacological interventions on smoking cessation: a systematic review and network meta-analysis. *Front Pharmacol*. 2022;13:1012433.

Gaddey HL, Dakkak M, Jackson NM. Smoking cessation interventions. *Am Fam Physician*. 2022;106(5):513-522.

Item 184**ANSWER: C**

Lifestyle changes such as appropriate fluid intake (2.5–3.0 L of liquid per day), a diet high in fiber and low in animal protein, lower sodium intake (<4–5 g per day), and moderate calcium intake (1200 mg per day) can help reduce recurrence of calcium stones. However, if patients continue to develop recurrent calcium stones, it is reasonable to initiate pharmacologic therapy. Appropriate medications include high-dose thiazide diuretics, allopurinol, and potassium citrate. Furosemide would not be appropriate as it has been associated with kidney stone formation. Magnesium oxide has not been shown to decrease the risk of stone development. Tamsulosin can be used to help with stone expulsion, but it does not help to prevent stones.

References

Fontenelle LF, Sarti TD. Kidney stones: treatment and prevention. *Am Fam Physician*. 2019;99(8):490-496.

Item 185**ANSWER: C**

This patient presents with symptomatic severe hyponatremia (sodium level <125 mEq/L). The most appropriate intervention at this time is a 100-mL bolus of 3% hypertonic saline (SOR B), which may be repeated up to 2 times at 10-minute intervals with the goal of increasing the sodium level by 4–6 mEq/L. Serum sodium levels should be monitored after every bolus and every 6 hours during the first 24 hours of treatment (SOR C). Avoiding rapid correction of severe hyponatremia is important to prevent osmotic demyelination syndrome. Depending on risk factors, sodium correction should be limited to 8–12 mEq/L over 24 hours or 18 mEq/L over 48 hours (SOR C).

The use of normal saline is appropriate treatment for nonsevere hypovolemic hyponatremia (a sodium level $\geq$ 125 mEq/L but <135 mEq/L). Intravenous desmopressin (DDAVP) is indicated for treatment of diabetes insipidus, which usually presents with high-normal sodium levels or hypernatremia. Desmopressin may also be given with 5% dextrose in water if overcorrection of severe hyponatremia has occurred.

References

Miller NE, Rushlow D, Stacey SK. Diagnosis and management of sodium disorders: hyponatremia and hypernatremia. *Am Fam Physician*. 2023;108(5):476-486.

Seay NW, Lehrich RW, Greenberg A. Diagnosis and management of disorders of body tonicity—hyponatremia and hypernatremia: core curriculum 2020. *Am J Kidney Dis*. 2020;75(2):272-286.

Item 186

ANSWER: B

In the absence of red flags (fracture, malignancy, epidural abscess, or cauda equina syndrome), the approach to acute lumbar radiculopathy should be conservative. Most patients improve over the first 2 weeks, and there is level B evidence that mobilization and manipulation is of benefit. Bed rest was previously recommended, but current data supports allowing the patient to be as active as the pain allows. While corticosteroids, NSAIDs, and muscle relaxants are often used, evidence to do so is lacking. In the subacute stage, neurodynamic mobilization and strength training could be added. If there is no improvement over 6 weeks, then MRI should be ordered.

References

Dvorin EL, Ebell MH. Short-term systemic corticosteroids: appropriate use in primary care. *Am Fam Physician*. 2020;101(2):89-94.

El Melhat AM, Youssef ASA, Zebdawi MR, Hafez MA, Khalil LH, Harrison DE. Non-surgical approaches to the management of lumbar disc herniation associated with radiculopathy: a narrative review. *J Clin Med*. 2024;13(4):974.

Thoomes E, Falla D, Cleland JA, Fernández-de-Las-Peñas C, Gallina A, de Graaf M. Conservative management for lumbar radiculopathy based on the stage of the disorder: a Delphi study. *Disabil Rehabil*. 2023;45(21):3539-3548.

Item 187

ANSWER: C

The vast majority of patients treated for essential hypertension will require 2 or more agents to achieve adequate blood pressure control. Combination therapy with a calcium channel blocker and an ACE inhibitor or angiotensin receptor blocker (ARB) would be most appropriate for this patient with stage 2 hypertension. Stage 2 hypertension is defined as a systolic blood pressure ≥ 160 mm Hg or a diastolic blood pressure ≥ 100 mm Hg (or a systolic blood pressure > 20 mm Hg above goal or a diastolic blood pressure > 10 mm Hg above goal).

Race-based prescribing, which advises preferential use of thiazide or calcium channel blockers for Black patients, has demonstrated insufficient blood pressure control and is thus falling out of favor. β -Blockers are not used as first-line treatment for hypertension, but they may be indicated in patients with specific comorbidities, such as coronary artery disease or heart failure. Aldosterone antagonists are also not typically used as first-line treatment for hypertension, but they may be used if multiple other drug classes cannot achieve adequate blood pressure control. Monotherapy is unlikely to produce adequate blood pressure control in this patient.

References

Holt HK, Gildengorin G, Karliner L, Fontil V, Pramanik R, Potter MB. Differences in hypertension medication prescribing for Black Americans and their association with hypertension outcomes. *J Am Board Fam Med*. 2022;35(1):26-34.

Egan BM, Yang J, Rakotz MK, et al. Self-reported antihypertensive medication class and temporal relationship to treatment guidelines. *Hypertension*. 2022;79(2):338-348.

Flack JM, Buhnerkempe MG. Race and antihypertensive drug therapy: edging closer to a new paradigm. *Hypertension*. 2022;79(2):349-351.

Clarke SL. Hypertension in adults: initial evaluation and management. *Am Fam Physician*. 2023;108(4):352-359.

Smith DK, Lennon RP, Carlsgaard PB. Managing hypertension using combination therapy. *Am Fam Physician*. 2020;101(6):341-349.

Item 188**ANSWER: D**

A potassium level of 2.6 mEq/L would indicate the need for inpatient care of this patient's anorexia nervosa. Eating disorders can often be treated in the outpatient setting with a multidisciplinary care team including physicians, therapists, and dietitians. However, certain findings, including severe hypokalemia, require higher levels of care. While hair loss, muscle atrophy, parotid gland enlargement, and elevated sodium can all be found in patients with eating disorders, severe electrolyte abnormalities can lead to life-threatening cardiac complications and require immediate treatment.

References

Klein DA, Sylvester JE, Schvey NA. Eating disorders in primary care: diagnosis and management. *Am Fam Physician*. 2021;103(1):22-32.

Item 189**ANSWER: C**

Hereditary nonpolyposis colorectal cancer (HNPCC), formerly called Lynch syndrome, is a hereditary cancer syndrome that increases the risk for several types of cancer, especially colon cancer. Guidelines recommend that patients with HNPCC should start colon cancer screening with colonoscopy between 20 and 25 years of age and repeat every 1–2 years. Patients at average risk for colon cancer should begin colon cancer screening at age 45. Fecal occult blood testing, stool DNA testing, and sigmoidoscopy are not adequate screening tests in this high-risk population. Lynch syndrome is associated with increased risk for several other cancers including endometrial, ovarian, stomach, small-bowel, and pancreatic.

References

Wilkins T, McMechan D, Talukder A, Herline A. Colorectal cancer screening and surveillance in individuals at increased risk. *Am Fam Physician*. 2018;97(2):111-116.

Final recommendation statement: colorectal cancer: screening. US Preventive Services Task Force. May 18, 2021.

Item 190

ANSWER: C

Pancreatitis is associated with an intense inflammatory response that increases caloric needs, and nutrition management is a key component of care. Acute pancreatitis is also associated with reduced perfusion of the intestinal mucosa, leading to tissue damage and increased intestinal permeability. Enteral feeding provides calories and increases splanchnic blood flow, leading to improved integrity of the bowel wall and improved gut motility. Patients with acute pancreatitis benefit from early enteral feeding within 24 hours of admission. A Cochrane review found that enteral nutrition has been associated with lower mortality, less multiple organ failure, and less systemic infection compared with parenteral nutrition. In severe acute pancreatitis, the benefits of enteral nutrition appear to be even more notable with a relative risk of death of 0.18 (95% CI 0.06 to 0.58). Liquid and soft diets have no evidence of benefit over a low-fat solid diet, and some studies found a shorter length of stay in patients started on solid diets rather than liquid diets. If patients cannot tolerate an oral diet within 72 hours, enteral nutrition should be provided via nasogastric or nasojejunal tube. Parenteral nutrition should be administered in patients who cannot tolerate tube feeding. Low-carbohydrate diets are not recommended because they are often high in fat and may worsen pancreatic inflammation.

References

- Meng WB, Li X, Li YM, Zhou WC, Zhu XL. Three initial diets for management of mild acute pancreatitis: a meta-analysis. *World J Gastroenterol*. 2011;17(37):4235-4241.
- Vivian E, Cler L, Conwell D, et al. Acute Pancreatitis Task Force on Quality: development of quality indicators for acute pancreatitis management. *Am J Gastroenterol*. 2019;114(8):1322-1342.
- Qi D, Yu B, Huang J, Peng M. Meta-analysis of early enteral nutrition provided within 24 hours of admission on clinical outcomes in acute pancreatitis. *J Parenter Enteral Nutr*. 2018;42(7):1139-1147.
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- Al-Omran M, AlBalawi ZH, Tashkandi MF, Al-Ansary LA. Enteral versus parenteral nutrition for acute pancreatitis. *Cochrane Database Syst Rev*. 2010;(1):CD002837.

Item 191**ANSWER: D**

According to the U.S. Preventive Services Task Force (USPSTF), women ≥ 65 years of age should be screened for osteoporosis using DXA bone mineral density testing, regardless of fracture risk assessment. Postmenopausal women < 65 years of age with 1 or more additional risk factors such as smoking, excessive alcohol consumption, or parental history of hip fracture should also be screened with DXA. This patient does not have any significant risk factors. Tools such as FRAX can be used to help counsel patients regarding the future risk of fracture, but they are not intended to be used to determine who should receive further screening with DXA.

References

Final recommendation statement: osteoporosis to prevent fractures: screening. US Preventive Services Task Force. January 14, 2025.

Item 192**ANSWER: B**

Hyperkalemia (potassium level >5.0 mEq/L) can occur with impaired renal excretion, transcellular shifts, and increased potassium intake. Management of severe hyperkalemia includes using diagnostic EKG to identify cardiac conditions requiring emergent treatment. EKG changes associated with hyperkalemia include peaked T waves, prolonged PR interval, loss of P waves, and prolonged QRS interval. Hyperkalemia resulting in prolonged QRS interval, bradycardia, and junctional rhythm is associated with adverse events, including ventricular fibrillation, cardiopulmonary resuscitation, and death within 6 hours of measurement. Findings associated with hypokalemia (potassium level <3.5 mEq/L) include flattened T waves, prolonged QT interval, ST segment depression, U waves, and T-wave inversion. Delta waves are not associated with potassium disorders but are present in Wolff-Parkinson-White syndrome.

References

Kim MJ, Valerio C, Knobloch GK. Potassium disorders: hypokalemia and hyperkalemia. *Am Fam Physician*. 2023;107(1):59-70.

Durfey N, Lehnhof B, Bergeson A, et al. Severe hyperkalemia: can the electrocardiogram risk stratify for short-term adverse events? *West J Emerg Med*. 2017;18(5):963-971.

Item 193**ANSWER: B**

For dislodged or fully avulsed permanent teeth, immediate reimplantation or repositioning is recommended (SOR C). Time is imperative in saving teeth. If reimplantation does not occur within 1 hour, there is little likelihood that it will be successful. If possible, replace the tooth in its anatomic position and have the patient bite down on a washcloth or gauze to keep it in place. It is not advised to remove a loose tooth, as the safest medium for the tooth is the socket. If the tooth is fully avulsed, hold it by the crown, rinse with saline, and reimplant. A tooth that cannot be reimplanted should be placed in a saline solution or milk. Antibiotic treatment is not imperative at this time. Without evidence of a dental or bony fracture, immediate imaging is not necessary as it would delay reimplantation and referral to a dentist.

References

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Stephens MB, Wiedemer JP, Kushner GM. Dental problems in primary care. *Am Fam Physician*. 2018;98(11):654-660.

Item 194**ANSWER: B**

This patient has hyperthyroidism, specifically Graves disease. Diffusely high radioactive iodine uptake differentiates Graves disease from other thyroid conditions, including toxic adenoma (high uptake concentrated in one spot), toxic multinodular goiter (high uptake in multiple spots), and thyroiditis (low uptake). Treatment options for Graves disease include antithyroid medications, radioactive iodine ablation, and surgical thyroidectomy. The most appropriate treatment is usually determined based on the risks and benefits of each option and patient preference.

Methimazole is the preferred antithyroid medication for treatment of Graves disease due to its improved safety profile compared to propylthiouracil, which has a greater risk of severe liver injury. Pregnant patients in the first trimester should be treated with propylthiouracil because it has a lower teratogenic risk than methimazole. Levothyroxine is used to treat hypothyroidism. Prednisone is used to treat severe hyperthyroidism, such as thyroid storm. A calcium channel blocker such as verapamil would be indicated if a β -blocker is contraindicated or not tolerated.

References

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Item 195**ANSWER: D**

Delirium is a syndrome characterized by an acute, fluctuating disturbance of attention, awareness, and cognition. Initial efforts should focus on identifying and addressing any underlying causes while providing redirection. However, if no reversible cause is found, managing delirium using medications becomes the most appropriate next step to ensure the patient's safety. While there are no FDA-approved medications for the management of delirium, second-generation antipsychotics such as olanzapine are preferred for this indication. Haloperidol is a less preferred option, compared to a second-generation antipsychotic. It is known to cause QT prolongation and other concerning side effects in older patients. Lithium is indicated for chronic management of bipolar disorder and does not have a role in the acute management of this patient's delirium. The side effects of benzodiazepines such as lorazepam make them undesirable for managing delirium, particularly in an older adult. Oxycodone is an opioid analgesic that would likely provide some degree of sedation but would not be appropriate for managing this patient's delirium.

References

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Jones HT, Davis DH. What you need to know about: delirium in older adults in hospital. *Br J Hosp Med (Lond)*. 2021;82(12):1-10.

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Item 196**ANSWER: C**

This patient clearly lacks the capacity to make decisions, and a surrogate decision-maker should be consulted. If designated, a durable power of attorney for health care should make decisions for this patient. Structured assessments for determining decision-making capacity or cognitive function may help in uncertain cases but are not necessary in this situation. Psychiatry consultation is rarely necessary to determine decision-making capacity and would not be appropriate in this situation. Decision-making capacity is determined by the treating clinician, not the court.

References

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Item 197**ANSWER: D**

Hypothermia is a potentially life-threatening condition that physicians and first responders should be able to promptly recognize and treat. While core temperature is useful for academic purposes or in a hospital, it is impractical to obtain and unimportant in the field. A simplified hypothermia assessment for field use evaluates the patient's level of consciousness, presence or absence of shivering, level of alertness, and spontaneous movements to categorize the severity of hypothermia. This patient is not moving or shivering, and although she is conscious she is not alert. This patient is experiencing moderate hypothermia and requires immediate intervention. A patient with moderate hypothermia should be kept horizontal, have large warming packs placed in the axillae and around the upper chest, be wrapped in an insulated vapor barrier (the burrito method), and be prepared for transport to the nearest medical facility, ideally one with an ICU.

Warm fluids and high-calorie foods can be used for a patient who is cold stressed or has mild hypothermia. Wet clothes should not be removed in the field unless shelter is available and rescue is more than 30 minutes away because this risks further lowering the patient's core temperature. Intubation is not necessary because the patient is protecting her airway and is conscious. Furthermore, it would not warm her.

References

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Item 198**ANSWER: A**

Prostate cancer screening recommendations vary, and there is little evidence that routine screening for prostate cancer improves patient-oriented outcomes. While the American Academy of Family Physicians does not recommend routine prostate-specific antigen (PSA)-based screening for prostate cancer, the American Urology Association (AUA) recommends prostate cancer screening with PSA testing for those 45–50 years of age. For those 50–69 years of age, the AUA recommends screening every 2–4 years between ages 50–69. The AUA also recommends obtaining a follow-up PSA level for individuals with newly elevated PSA before considering additional biomarkers, imaging, or biopsy. While this is based on expert opinion, the authors note that PSA may normalize in as many as 40% of individuals. The authors also suggest a higher threshold for biopsy based on age, with 4.5 ng/mL being the upper limit of normal for this patient's age group. The Choosing Wisely AUA initiative recommends against using empiric antibiotics to treat elevated PSA levels in asymptomatic individuals. A digital rectal examination is not recommended as a screening method alone or in addition to a screening PSA.

References

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Item 199**ANSWER: A**

Testing for fecal calprotectin, lactoferrin, and leukocytes is of limited value in evaluating a nontoxic patient with acute diarrhea, including infectious diarrhea. Testing for fecal occult blood and infection with *Clostridioides difficile* is also not recommended in these patients. Detecting fecal calprotectin in a patient with inflammatory bowel disease and bacterial diarrhea lacks consistency and is not recommended. Although the presence of fecal lactoferrin can be associated with a bacterial etiology, its lack of specificity makes it less useful. Fecal leukocytes disintegrate easily in stool samples, reducing test sensitivity.

Indications for obtaining stool testing for acute infection include the presence of severe abdominal pain or tenderness, bloody or mucoid stools, clinical findings of sepsis, fever $\geq 38.3^{\circ}\text{C}$ (101.0°F), immunocompromised state, recent hospitalization, recent antibiotic therapy, and diarrhea that lasts >7 days.

References

Meisenheimer ES, Epstein C, Thiel D. Acute diarrhea in adults. *Am Fam Physician*. 2022;106(1):72-80.

Item 200**ANSWER: C**

Dog bite wounds may be primarily closed if desired for cosmetic reasons such as wounds to the face. However, dog bite wounds that carry a higher risk of infection such as crush injuries, wounds to the hand, and puncture wounds, should be left to heal by secondary intention. Other factors that increase the risk of infection include delayed presentation (>6–12 hours for bites to the arm or leg and >12–24 hours for bites to the face) and bites in people who are immunosuppressed, including those with diabetes mellitus.

References

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